Personalization, Algorithmic Dependence, and Learning

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Abstract

Personalized recommendation systems are now an integral part of the digital ecosystem. However, users' increased dependence on these personalized algorithms has heightened concerns among consumer protection advocates and regulators. In this work, we bring an information-theoretic perspective to this problem and examine the underpinnings of algorithmic dependence and its downstream implications for users' preference learning and independent decision-making ability, an important construct given the growing fear of adversarial AI. We develop a utility framework where users consume experience goods and sequentially update their preference weights for product attributes. We theoretically establish regret bounds for different types of users based on their dependence on the personalized algorithm. Our theoretical results demonstrate the rationality of algorithmic dependence as the gain from following the personalized algorithm grows linearly with time periods. We then develop an empirical framework to obtain model-free measures of regret for different user types. We find that personalized algorithms generate significant welfare gains, but these gains come at the expense of users' preference learning and independent decision-making. Finally, we demonstrate that simple policies can help balance the trade-off between welfare and learning, offering insights for both users and platforms.

Keywords: personalized algorithms, recommendation system, preference learning, consumer protection, linear bandits, reinforcement learning

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1 Introduction

Personalized recommendation systems are now an integral part of the digital ecosystem. Digital platforms use massive amounts of consumer-level data to deliver personalized recommendations. One of the canonical examples of recommendation systems is the Netflix movie recommendation algorithm, which reportedly saves the company over one billion dollars annually by reducing the churn rate [Gomez-Uribe and Hunt, 2015]. Other examples include Facebook and Twitter's news feed personalization, Amazon's product recommendation, and YouTube's video recommendation algorithm.

In today's digital age, the online marketplace is saturated with many options, presenting users with the challenge of sifting through too many options to find what they truly want. Personalized recommendation systems have emerged as a solution to this problem by reducing users' search costs and simplifying decision-making. These systems are designed to effectively narrow down options in real-time and guide users towards products or services that best align with their preferences and needs. By doing so, a personalized recommendation system ensures that users can select a fitting item without the need to explore the vast digital landscape exhaustively.

However, as the adoption of and reliance on personalized recommendation systems grow, there are increasing concerns regarding users' algorithmic dependence [Buçinca et al., 2021]. Prior research has shown the pitfalls of algorithmic dependence by documenting the users' tendency even to take clearly incorrect recommendations [Spatharioti et al., 2023], risks to user well-being [Banker and Khetani, 2019], and inefficiencies in users' decision-making [McLaughlin and Spiess, 2022]. Regarding the mechanism behind algorithmic dependence, prior work in economics and marketing has suggested the presence of time-inconsistent preferences [Allcott et al., 2022] and search costs [Ursu, 2018]. However, little is known about the algorithm's information advantage and how it feeds algorithmic dependence.

In this work, we bring an information-theoretic perspective to this problem and examine the underpinnings of algorithmic dependence and its downstream implications for users' preference learning. In particular, we study contexts where users consume content sequentially and may have uncertainty about their preferences given the vast space of product features, which they resolve through experience. Personalized recommendations influence the process through which users learn their preferences by affecting the decisions they make. For example, a news reader interested in social justice topics may rarely explore other content if the algorithm correctly identifies her taste and only exposes her to this type of content. Thus, dependence on algorithmic recommendations can have important implications for user learning.

Understanding algorithmic dependence and its downstream effects on users' learning is crucial from a consumer protection perspective, as users who heavily rely on algorithms without fully learning their preferences are more vulnerable to digital manipulation by adversarial AI, a topic of growing concern among policy-makers [Kusnezov et al., 2023]. In particular, when users lack a clear understanding of their own preferences, they make poorer decisions in the absence of recommendation systems, creating a feedback loop in which they become increasingly dependent on these systems while learning less about their own preferences. In this paper, we study the interplay between personalization, algorithmic dependence, and preference learning and aim to answer the following questions:

1. How does the algorithm's information advantage translate into better recommendations? Are there information-theoretic guarantees on the regret performance of personalized recommendation systems?

2. How does the dependence on a personalized algorithm affect user learning? How can we quantify the impact?

3. What self-regulatory or consumer protection policies could generate good recommendations while helping users learn their preferences?

To answer these questions, we face several challenges. First, we need a theoretical framework that allows us to formally characterize the personalized algorithm's information advantages and disadvantages relative to individual users. In particular, we need our framework to capture learning by the algorithm and users separately from any given prior. Second, we need to theoretically compare the outcomes for users who follow personalized algorithms with those who do not. However, since algorithms and users operate under different conditions, the theoretical guarantees, such as regret bounds, often include parameters that make them inherently incomparable. Hence, we need to obtain comparable regret bounds that allow us to assess the rationality of algorithmic dependence. Third, we need an empirical framework to evaluate the performance of different algorithms without necessarily deploying those policies online. In particular, we want the evaluation process to be model-free so we do not need to rely on model-based outcome estimates.

To address our first set of challenges, we build a general linear utility framework, where users have some preference parameters over the space of product features, and products have search and experience features, consistent with Nelson [1974]. Given the high dimensionality of the feature space and the continuous introduction of new features and styles in many applied settings (e.g., movies, food), we allow users to exhibit any level of uncertainty about their own preference parameters and quantify this uncertainty using the well-known Shannon entropy measure [Shannon, 1948].[1] To characterize users' learning, we turn to the literature on Bayesian learning and model users who update the posterior distribution of their preference parameters [Ching et al., 2013]. For users' decision-making, we use the Thompson Sampling approach as it incorporates users' Bayesian learning and is behaviorally plausible [Schulz et al., 2019, Mauersberger, 2022, Ding et al., 2022]. Under Thompson Sampling algorithms, users sample from the posterior distribution of preferences to choose an item and update the posterior distribution according to their experience. We then characterize the personalized recommendation system and its decision-making process as a low-rank model that mimics the reality of personalized algorithms used by platforms and parsimoniously accounts for the platform's information advantage over a single user who only has access to search features.

For the second challenge, we turn to the literature on bandits [Lattimore and Szepesvári, 2020]. In particular, given the central role of information advantage in our study, we focus on the bandits literature that offers information-theoretic regret bounds for the performance of different adaptive learning algorithms [Russo and Van Roy, 2016]. To isolate the impact of recommendation systems on outcomes, we focus on two types of users: (1) self-exploring users who make decisions on their own as though there is no recommendation system, and (2) recommendation-system-dependent (RS-dependent henceforth) users who follow the recommendations provided by the recommendation system. We find that the regret for self-exploring users has a lower bound that scales linearly in time periods because self-exploring users can, at best, identify the first-best product based on the search features. Conversely, the RS-dependent user has an information-theoretic upper bound for regret that grows sub-linearly in time periods. Together, our theoretical findings suggest that the welfare gain from following the RS grows linearly in time periods, highlighting a mechanism for rational dependence solely through the algorithm's information advantage.

Third, to empirically examine the impact of personalized algorithms on both welfare and learning outcomes, we use MovieLens data, which is the main public data set used as a benchmark for personalized recommendation systems. To facilitate a model-free regret evaluation, we focus on a small sample of users who have provided many ratings so we can use their observed outcomes instead of estimating them from models. We then take a hold-out subset of size 20 from the movies they have rated and exclude it from the training part. This creates a subset of products for each user in the test set, which allows us to evaluate how different each model's prescription is from the observed first-best in the set. Lastly, we embed a state-of-the-art matrix factorization algorithm to simulate a personalized algorithm that captures the reality of algorithms used by the platforms

[1] It is worth emphasizing that our information-theoretic framework encompasses the classical economics assumption that users have full knowledge of their preferences as a special case, which makes our approach more general and agnostic regarding the degree of uncertainty in users' preferences.

[Cortes, 2018].

We apply our empirical framework to the MovieLens data and examine regret and learning outcomes. We first focus on the algorithm's information advantage and welfare gains from following the algorithm. We find a significantly better regret performance by the personalized algorithm. Notably, even compared to the self-exploring user who knows her own preferences, the personalized algorithm has a persistent advantage. Further, our results show that the algorithm quickly surpasses the performance of the self-exploring user with known preferences, emphasizing its efficiency due to low-rank learning. This is an important empirical finding, as we do not impose any specific rank constraint that forces the problem to be low-rank. Together, the algorithm's better long-run performance and faster learning rationalize algorithmic dependence.

Next, we focus on the implications of algorithmic dependence for learning. We first show that the absolute amount of preference learning is higher for self-exploring users than for RS-dependent users when they start from an identical prior. Motivated by the trade-off between welfare and learning, we develop a new regret measure defined as *counterfactual regret*, which helps quantify the potential welfare consequences of insufficient learning. The counterfactual regret measures the expected regret incurred by a user in each period if they make decisions independently without the help of the personalized algorithm. As such, a user with insufficient learning would make worse decisions on their own. The key advantage of this measure is that it has the same unit as our main regret measure, which offers greater interpretability and facilitates joint optimization of welfare and learning outcomes. We compute counterfactual regret for RS-dependent users and show that while these users enjoy a low regret due to following personalized recommendations, they have a higher counterfactual regret than self-exploring users because they become worse independent decision-makers in the absence of personalized algorithms. Therefore, our examination of learning outcomes suggests that although personalized algorithms help users make better decisions that increase their welfare, these algorithms can act as a barrier to user learning since these algorithms can limit the organic exploration process users engage in.

Motivated by the trade-off between welfare and learning, we consider a class of hybrid users who probabilistically alternate between following the recommendation system and exploring independently. We find that intermediate hybrid strategies push the Pareto frontier of welfare and learning: even a modest amount of independent exploration lets users learn almost as much as fully self-exploring users while sacrificing only a small amount of welfare. We further document substantial heterogeneity across users in the optimal mixing rate, suggesting that personalized policies may be valuable. These findings offer important implications for users who want to self-regulate their reliance on recommendation systems and for platforms or policymakers considering interventions that modulate RS availability. Lastly, we consider a range of other extensions and robustness checks—including alternative forms of user learning, alternative RS algorithms, and potential misalignment between user and RS objectives—to establish the robustness of our main insights and the boundary conditions of our analysis.

In summary, our paper makes several contributions to the literature. Substantively, we present a comprehensive study of the information advantage of personalized algorithms—how this advantage enhances consumer welfare while creating a cycle of algorithmic dependence that hinders consumers' preference learning and independent decision-making. Through a series of theoretical and empirical analyses, we show that the information advantage alone can rationalize algorithmic dependence, even in the absence of search costs or time-inconsistent preferences, which are often used to justify such dependence. While concerns related to privacy, fairness, and polarization have been extensively studied in the literature on personalized recommendation systems, the topic of algorithmic dependence and its downstream effects on users' learning has received less attention. Our work extends this policy debate by uncovering the underpinnings of algorithmic dependence and its negative consequences for users' independent decision-making, offering insights that are increasingly relevant given growing concerns about adversarial AI. Methodologically, we introduce a framework that allows us to establish theoretical regret bounds for users based on their dependence on personalized algorithms. A key innovation of our approach is its ability to capture the information advantage of personalized algorithms through a low-rank assumption and access to experience features unavailable to users. Additionally, we develop a counterfactual regret measure that serves as a valuable benchmark for evaluating the effects of adversarial AI.

2 Related Literature

First, our paper relates to the literature on personalization. Prior methodological work in this domain has offered a variety of methods to generate personalized policies, such as low-rank matrix factorization models for collaborative filtering and a host of causal machine learning methods [Linden et al., 2003, Mazumder et al., 2010, Athey and Imbens, 2019, Koren et al., 2021, Rafieian and Yoganarasimhan, 2023]. Related applied work in this domain has focused on different aspects of personalization, such as developing personalized algorithms tailored to specific problems [Hauser et al., 2009, Chung et al., 2009, Chung and Rao, 2012, Urban et al., 2014, Ansari et al., 2018, Liberali and Ferecatu, 2022, Rafieian, 2023, Rafieian et al., 2023, Boughanmi et al., 2025], the interplay between personalization and consumer protection policies [Johnson et al., 2020, Rafieian and Yoganarasimhan, 2021, Despotakis and Yu, 2023, Bondi et al., 2026], and the tension between content homogenization vs. content diversity due to personalization [Fleder and Hosanagar, 2009, Nguyen et al., 2014, Song et al., 2019, Holtz et al., 2020, Aridor et al., 2020, Anwar et al., 2024].

Our work extends this stream of work by bringing an information-theoretic view and focusing on the foundations of algorithmic dependence and its negative impact on users' learning. In particular, we combine the insights from the literature on matrix factorization with bandits to establish theoretical regret bounds for users as a function of their dependence.

Second, our paper relates to the literature on consumer search and personalized rankings [Jeziorski and Segal, 2015, Ursu, 2018, Dzyabura and Hauser, 2019, Yoganarasimhan, 2020, Korganbekova and Zuber, 2023, Donnelly et al., 2024]. Most papers in this literature build a sequential search model akin to Weitzman [1979] to model consumers' search behavior and estimate structural parameters such as search costs. Under this modeling framework, consumers do not learn their preferences through experience but realize the match value of each item upon a costly search. The exception is Dzyabura and Hauser [2019], which allows for learning from signals extracted through a sequential search process, but not from the sequential consumption of items, as in our setting. A common theme in the stream of work on consumer search is that personalized algorithms create value by reducing consumers' search costs. In that sense, search cost is the main driver behind algorithmic dependence. Our work differs from this stream of literature as we focus on a channel separate from search cost that drives algorithmic dependence: the algorithm's information advantage. In particular, we characterize the personalized algorithm's information advantage in the context of experience goods, which makes it a better predictor of whether the user likes the product or not than the users themselves.

Third, our paper relates to the literature on consumer learning. Understanding consumer learning dynamics has been of great interest to researchers in marketing [Roberts and Urban, 1988]. Ever since the seminal paper by Erdem and Keane [1996] who modeled forward-looking consumers who make decisions under uncertainty and engage in an exploration-exploitation trade-off, numerous studies have focused on choice contexts where dynamic learning plays an important role [Ackerberg, 2003, Crawford and Shum, 2005, Hitsch, 2006, Ching et al., 2013]. An important issue in this stream of work is computational complexity, which has motivated researchers to adopt heuristic-based strategies, which yield performance similar to the dynamic programming strategy, while having the advantage of computational and cognitive simplicity [Lin et al., 2015, Tehrani and Ching, 2023]. We extend this stream of literature by offering a Thompson Sampling approach for characterizing consumer choice and learning process, which is another cognitively simple alternative to the typical dynamic programming solution to the exploration-exploitation trade-off and has been shown to be a behaviorally plausible framework to model consumer behavior [Schulz et al., 2019, Ding et al., 2022]. Our work differs from this stream of literature as we consider a setting where users learn their preferences, rather than the product attributes. We further demonstrate how the increased flexibility offered by the Thompson Sampling approach can help researchers study settings with high-dimensional learning and establish information-theoretic regret bounds.

Fourth, our work relates to the vast literature on adaptive learning and multi-armed bandits. Prior research in this domain has offered a variety of algorithms to use [Lattimore and Szepesvári, 2020]. Although Thompson sampling has been around since the work by Thompson [1933], it has only recently gained traction after providing a remarkable empirical performance better than state-of-the-art benchmarks [Chapelle and Li, 2011]. Since then, many researchers have attempted to provide theoretical guarantees on Thompson sampling for a variety of adaptive learning problems [Agrawal and Goyal, 2012, 2013, Russo and Van Roy, 2014, 2016]. For a comprehensive review of Thompson sampling, please see Russo et al. [2018]. In marketing, a growing body of research explores the applications of Thompson Sampling across various domains [Schwartz et al., 2017, Misra et al., 2019, Jain et al., 2024, Ye et al., 2024, Rashid et al., 2025, Rombach et al., 2025]. Most of the literature in this domain focuses on a single learner that optimizes the action and updates parameters upon experience. Our work extends this single-agent framework to a setting with both a learning recommendation system and an agent, offering new insights for modeling general multi-agent platform settings with decision-making under uncertainty. We offer theoretical regret bounds for different users based on their level of dependence on the algorithm.

3 Modeling Framework

We consider a general platform setting, where the platform designs a Recommendation System (RS) that offers personalized product recommendations, and the user of the platform wants to consume products on the platform. The products available are experience products, meaning that only a subset of their features are available prior to consumption, while other features are only realized after consumption [Nelson, 1970, Villas-Boas, 2006]. There are numerous examples of such contexts, including movie recommendations, news personalization, and content recommendation on social media apps. In this section, we first characterize the user's utility model in §3.1 and then describe users' preference learning in §3.2. In §3.3, we discuss the user's choice in two different regimes with and without the personalized RS.

3.1 Users' Utility Model

We propose a general utility framework in which user i derives utility from selecting product (action) A_j from the product (action) set $\mathcal{A}$. In this context, each action corresponds to consuming a distinct experience product, represented by a d-dimensional set of attributes, i.e., $\mathcal{A} \subset \mathbb{R}^d$. Following the literature on experience goods [Nelson, 1970], we categorize the product features into two types: (1) *search features*, which are attributes known before consumption (e.g., a movie's runtime, genre), and

(2) *experience features*, which are realized only after consumption (e.g., the presence of a surprise ending). For convenience, we denote the first s features as search features and the remaining $d - s$ features as experience features. Figure 1 illustrates the distinction between these two feature types.

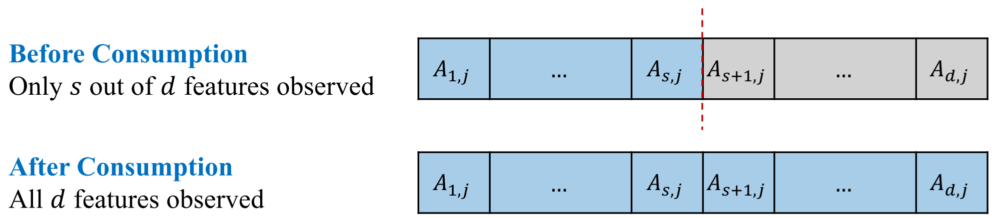


Figure 1: Illustration of search and experience features

We denote user i's utility from consuming product A_j by $u_i(A_j)$ and characterize it by a user-specific vector of preferences $\theta_i \in \mathbb{R}^d$ in a linear specification as follows:

$$u_i(A_j) = \theta_i^T A_j + \epsilon_{i,j}, \tag{1}$$

where $\epsilon_{i,j}$ is the error term, drawn from a normal distribution with mean zero and known variance σ_ϵ^2. Intuitively, the error term captures the notion of *experience variability* characterized in the consumer learning literature [Ching et al., 2013].[2] Although we assume linearity for theoretical simplicity, this assumption is not overly restrictive, as a rich set of features could be used to approximate user utility in a linear manner. To account for the distinction between search and experience features, we decompose utility into two independent pieces as follows:

$$u_i(A_j) = \underbrace{\sum_{k=1}^{s} \theta_{i,k} A_{k,j}}_{\text{utility from search features}} + \underbrace{\sum_{l=s+1}^{d} \theta_{i,l} A_{l,j}}_{\text{utility from experience features}} + \epsilon_{i,j}, \tag{2}$$

where the utility from search features is the part available to the user prior to consumption, but the utility from the experience features is only realized after consumption.

3.2 Users' Preference Learning

In high-dimensional settings with numerous product features, users are likely to experience some degree of uncertainty about their preference parameters [Branco et al., 2016, Yao et al., 2022]. This

[2]Experience variability implies that two items with identical features can generate different utility for the same user due to unobserved situational factors, product variability, inherent consumer uncertainty, and emotional volatility, among others. Consistent with the consumer learning literature, the user knows the distribution of this variability, but not its realization prior to consumption.

uncertainty becomes particularly pronounced when new product features are continually introduced, as users may be unfamiliar with these features and, therefore, uncertain about their importance [Lee et al., 2024]. In such cases, users gradually resolve their uncertainty by consuming products and learning their preferences over time. For instance, a movie viewer may be unsure about their preference for a specific sub-genre or a distinct visual style (e.g., the aesthetics of a Wes Anderson film) and develop a clearer understanding by watching multiple films within that category. Similarly, in the context of food, a user who has never tried a particular ingredient may initially be uncertain about their preference for it but gradually learn it through experience.

In this section, we extend our framework to sequential settings where users learn their preference parameters through sequential consumption of products. We adopt an information-theoretic approach that accommodates any level of uncertainty in users' preferences. A key advantage of this flexible characterization is that it encompasses the classical economic assumption of fully known preferences as a special case while also capturing more general scenarios where users face uncertainty about their preferences.

Let t denote each time period and $A_{i,t}$ the product chosen by user i in period t. For notational brevity, we define $U_{i,t} = u_i(A_{i,t})$ and let $\mathcal{H}_{i,t}$ denote the history (sequence) of products (actions) consumed and utility outcomes up until period t, that is, $\mathcal{H}_{i,t} = (A_{i,1}, U_{i,1}, A_{i,2}, U_{i,2}, \ldots, A_{i,t}, U_{i,t})$. We assume θ_i is drawn from a Normal distribution $N(\mu_{i,0}, \Sigma_{i,0})$. The user starts with a prior $\tilde{\theta}_{i,0} \sim N(\mu_{i,0}, \Sigma_{i,0})$ and updates the preference parameters at the end of each time period t given the prior sequence $\mathcal{H}_{i,t}$ according to the following rule:

$$\mu_{i,t} = \mathbb{E}[\theta_i \mid \mathcal{H}_{i,t}] \tag{3}$$

$$\Sigma_{i,t} = \mathbb{E}\left[(\theta_i - \mu_{i,t})(\theta_i - \mu_{i,t})^T \mid \mathcal{H}_{i,t}\right] \tag{4}$$

The sequential nature of learning indicates that users update their parameters in every time period in a Bayesian fashion. Given the Normal prior, we can apply the Bayes rule for linear Gaussian systems and present the user's parameter updating from $\mu_{i,t}$ and $\Sigma_{i,t}$ to $\mu_{i,t+1}$ and $\Sigma_{i,t+1}$ in Algorithm 1.[3] Algorithm 1 determines user learning given any consumption sequence $\mathcal{H}_{i,t}$. As such, for any given prior, we can examine how different consumption sequences can result in different levels of learning.

In summary, a few points are worth noting about our learning framework. First, the presence of uncertainty does not imply that users have entirely uninformative priors. For instance, a user may already have a reasonably well-calibrated belief about their enjoyment of a new visual style in

[3]The exact analytical derivation is a well-established result that can be found in most textbooks on Bayesian statistics, e.g., see Theorem 4.4.1 in Murphy [2012].

Algorithm 1 Bayesian Updating

Input: $\mu_{i,t}, \Sigma_{i,t}, A_{i,t}, U_{i,t}$

Output: $\mu_{i,t+1}, \Sigma_{i,t+1}$

1: $\Sigma_{i,t+1} \leftarrow \left(\Sigma_{i,t}^{-1} + \frac{1}{\sigma_\epsilon^2} A_{i,t} A_{i,t}^T\right)^{-1}$

2: $\mu_{i,t+1} \leftarrow \Sigma_{i,t+1} \left(\Sigma_{i,t}^{-1} \mu_{i,t} + \frac{1}{\sigma_\epsilon^2} A_{i,t} U_{i,t}\right)$

movies or a certain ingredient in food. Second, representing each product as a d-dimensional feature vector means that all products share a common underlying preference vector, so observing the realized utility from one product updates the posterior over this vector and, through it, the predicted utilities of every other product. This places our setting within the class of linear bandits, which accommodate correlated learning across products, as opposed to the standard K-armed bandit, where arms share no common structure and learning proceeds independently across them. Lastly, our learning procedure is somewhat different from the typical approach in the Bayesian learning literature in marketing and economics where consumers learn about product attributes [Erdem and Keane, 1996]. However, although preference learning may appear different from attribute learning, "the two are equivalent as long as one assumes that the utility function is linear in attributes and preference weights" (see footnote 35 in Ching et al. [2013]). Therefore, one could find an equivalence result between our analysis and one with attribute learning.

3.3 User Choice

We now discuss the user's decision-making process that determines the consumption sequence $\mathcal{H}_{i,t}$. To do so, we need to characterize the choice architecture in each period. For any product set $\mathcal{A} = \{A^{(1)}, A^{(2)}, \cdots, A^{(K)}\}$, we consider the following choice architecture when the recommendation system is available:

$$\underbrace{A^{(p)}}_{\text{recommended}}, \underbrace{A^{(1)}, A^{(2)}, \cdots, A^{(K)}}_{\text{not recommended}},$$

where one product from the set is recommended and the rest of the products are not recommended.[4] We now characterize the user's decision-making process in the definition below:

Definition 1. *Let $\mathcal{I}_{i,t}$ denote all the information available to user i at time t. The user's decision-making process is characterized by the policy $\pi(\cdot \mid \mathcal{I}_{i,t})$, which is a probability distribution over products conditional on the information and products available.*

[4]We focus on the single-item recommendation setting for theoretical simplicity, which allows us to derive tractable insights. In §6, we discuss the extent to which our qualitative insights may extend to the multi-item recommendation problem and highlight this as a promising direction for future research.

To isolate the impact of personalized algorithms on users, we consider two types of users: (1) *self-exploring user*, who makes decisions on their own without the personalized RS, and (2) *RS-dependent user* who follows the personalized recommendation in every time period.[5] In what follows, we first characterize the self-exploring user's choice in §3.3.1, and then present how the RS provides personalized recommendations and characterize the consumption sequence for the RS-dependent users in §3.3.2.

3.3.1 Self-Exploring User

In the absence of the personalized recommendation, users make decisions on their own. Given the utility framework in Equation (1), a forward-looking utility-maximizing user wants to optimize the overall utility over T periods. This naturally motivates users to learn their preference parameters through experience and balance good decision-making with proper exploration of their own preference parameters. We can define the objective function for a forward-looking user as maximizing the expected utility stream as follows:

$$\underset{\pi}{\operatorname{argmax}}\, \mathbb{E}\left[\sum_{t=0}^{T} U_{i,t} \mid \mu_{i,0}, \Sigma_{i,0}, \pi\right], \tag{5}$$

where the expectation is taken over the randomness in products $A_{i,t}$ and utilities $U_{i,t}$.[6] Typical approaches to find the optimal sequence of choices by users involve solving a dynamic programming problem, which is known to be an NP-hard problem. The lack of cognitive simplicity of dynamic programming solutions has motivated researchers to study the simpler heuristic-based strategies as the underlying learning process [Lin et al., 2015, Tehrani and Ching, 2023].

We draw inspiration from this stream of literature and assume that users employ a Thompson Sampling approach that is a cognitively simple and intuitive heuristic-based strategy consistent with Bayesian learning [Thompson, 1933]. In addition, the prior literature has documented Thompson Sampling algorithm's behavioral plausibility as a framework to model consumer choice and learning [Schulz et al., 2019, Mauersberger, 2022, Ding et al., 2022] and its excellent empirical performance in terms of welfare [Chapelle and Li, 2011].[7]

Thompson Sampling aims to find the right balance between exploration and exploitation in the

[5] We choose this stylized approach and compare these two levels of dependence and independence to offer critical and sharp insights into this problem in a tractable manner. However, as we later demonstrate in this paper, our framework can also be used to examine outcomes under more hybrid user types.

[6] In line with the bandit literature, we adopt an *undiscounted* objective (i.e., set $\delta = 1$) because we consider a finite-horizon problem. Introducing a discount factor $\delta < 1$ would downweight later periods in this setting.

[7] While these features of Thompson Sampling allow us to derive tractable insights and information-theoretic regret bounds, our qualitative findings extend to alternative near-optimal bandit algorithms and exploration strategies.

decision-making process. The algorithm starts by initializing the user's prior belief distribution about the preference weights $N(\mu_{i,0}, \Sigma_{i,0})$. It then draws $\tilde{\theta}_{i,0}$ from this distribution and computes the utility from search features for all possible products by plugging in $\tilde{\theta}_{i,0}$ for $\theta_{i,0}$ in Equation (2). It is important to note that the user cannot use experience features because those features are not available prior to consumption. In the next step, the algorithm chooses the product that maximizes the estimated utility and observes the utility $U_{i,0}$ for that instance. Finally, the algorithm applies Bayesian updating procedure in Algorithm 1 using the new instance and updates the posterior distribution of preference weights for both search and experience features, since both features are realized after consumption. The Thompson Sampling algorithm continues this process for $\mathcal{T}$ periods.

Algorithm 2 Choice and Learning for the Self-Exploring User

Input: $\mu_{i,0}, \Sigma_{i,0}, \mathcal{A}, \mathcal{T}$

Output: $\mathcal{H}_{i,\mathcal{T}}, \{\mu_{i,t}, \Sigma_{i,t}\}_{t=1}^{\mathcal{T}}$

for $t = 0 \to \mathcal{T}$ **do**

$\quad \tilde{\theta}_{i,t} \sim N(\mu_{i,t}, \Sigma_{i,t})$ ▷ Sampling preference weights

$\quad A_{i,t} \leftarrow \mathrm{argmax}_{A_j \in \mathcal{A}} \sum_{k=1}^{s} \tilde{\theta}_{i,k,t} A_{k,j}$ ▷ Selecting action based on search features

$\quad \Sigma_{i,t+1} \leftarrow \left(\Sigma_{i,t}^{-1} + \frac{1}{\sigma_\epsilon^2} A_{i,t} A_{i,t}^T\right)^{-1}$ ▷ Updating posterior variance

$\quad \mu_{i,t+1} \leftarrow \Sigma_{i,t+1} \left(\Sigma_{i,t}^{-1} \mu_{i,t} + \frac{1}{\sigma_\epsilon^2} A_{i,t} U_{i,t}\right)$ ▷ Updating posterior mean

end for

Assuming that the self-exploring user uses Thompson Sampling has several key advantages. First, it is a commonly used heuristic strategy for this dynamic problem, and early literature has shown it to be nearly optimal [Chapelle and Li, 2011]. Second, while real users may be far from both optimal and near-optimal in their behavior, focusing on an algorithm with near-optimal performance ensures that our insights are not driven by a simplistic user model. Later in §5.4.2, we extend our analysis to cases where users rely on alternative learning algorithms that are less exploratory and more greedy. Third, it is computationally light, making it advantageous in our later empirical analysis using the MovieLens data set. Fourth, due to its simplicity, it is easy to incorporate it in cases where there is a recommendation system present in the problem. We discuss this issue in the following section.

3.3.2 RS-Dependent User

We now focus on the user's choice in the presence of the recommendation system. To do so, we first introduce a personalized RS that aims to simplify the user's decision-making problem. Since we want to quantify the impact of the personalized RS on user-specific outcomes, we assume that the recommendation system's objective is the same as that of the user.[8] A natural difference between the personalized RS and a single user is the fact that the system has access to the data of all other users. To understand how the recommendation system's data advantage manifests itself in better decision-making capabilities, we first introduce some notations. Let $\Theta_{[d\times N]}$ denote the matrix of preference weights for a group of N users, i.e., $\Theta_{[d\times N]} = [\theta_1 \mid \theta_2 \mid \ldots \mid \theta_N]$. In all major platforms, N is a very large number. Similarly, let $A_{[d\times J]}$ denote the matrix of attributes for all J products ($J = |\mathcal{A}|$) where each column represents the vector of attributes for a product. We can define the matrix analog of the user's utility in Equation 1 as follows:

$$U = \Theta^T A + E, \tag{6}$$

where $U_{N\times J}$ represent the utility for each pair of user and product and $E_{N\times J}$ is a matrix of i.i.d error term drawn from a mean-zero Normal distribution with variance σ_ϵ^2. The recommendation system has access to $U^{\text{obs}}_{N\times J}$, which is an incomplete realization of matrix U as each user reveals utility for a subset of items. The main question is how the data from other users $U^{\text{obs}}_{N\times J}$ help the personalized algorithm learn about the new user i with vector of preferences θ_i.

In principle, if the prior data from users do not inform us about the new user, the recommendation system's strategy will be the same as the self-exploring user, presumably with a less informed prior because users know more about their own preferences than the RS. However, the prior empirical work on personalized recommendation systems suggests otherwise as it documents extensive similarities in user preferences [Koren et al., 2021, Dhillon and Aral, 2021]. The most common approach to characterize the similarities in user preferences is to use a factor model, which suggests that the matrix $\Theta^T A$ can be factorized into two low-rank matrices. In particular, we make the following low-rank assumption:

Assumption 1. *The expected utility matrix $\Theta^T A$ can be decomposed as follows:*

$$\Theta^T_{[d\times N]} A_{[d\times J]} = \Gamma^T_{[r\times N]} F_{[r\times J]}, \tag{7}$$

where $F_{[r\times J]} = [F_1 \mid F_2 \mid \ldots \mid F_J]$ is the matrix of product-specific factors for all J products, and $\Gamma_{[r\times N]} = [\gamma_1 \mid \gamma_2 \mid \ldots \mid \gamma_N]$ represents the matrix of user-specific factor weights for all N users.

We can now view the recommendation system's data advantage in light of Assumption 1.

[8] We adopt the objective alignment assumption to ensure that the impact of the recommendation system is not driven by misaligned objectives. Besides, there are infinitely many ways in which objectives can be misaligned whereas objective alignment is unique and well-defined. We later discuss how our main findings in our context may change under alternative scenarios of objective misalignment.

Because the recommendation system has other users' prior data, it has access to an accurate estimate of the product-specific matrix F, which captures not only search features, but also experience features. As such, the recommendation system's task of learning user i's preference parameters will turn into the task of learning user i's weights for r-dimensional products because we have: $\theta_i^T A_j = \gamma_i^T F_j$. Hence, the recommendation system's data advantage manifests in two ways: (1) learning r-dimensional weights as opposed to the self-exploring user's learning of d parameters[9], and (2) access to a low-rank embedding of product space that contains both search and experience features. Figure 2 compares the information advantage of the RS and users, emphasizing the potential for users to have more informed priors. These insights are essential for establishing theoretical guarantees for different algorithms.

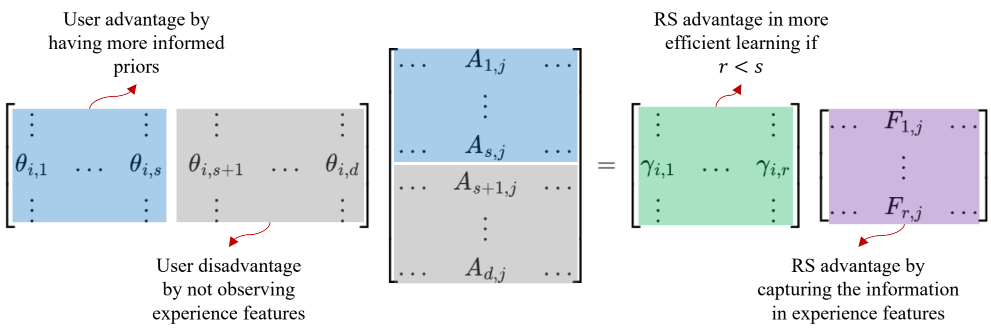


Figure 2: Comparison of information advantage by the RS and users

Before presenting the procedure that determines choice and learning for the RS-dependent user, we make the following assumption about the user's learning:

Assumption 2. *The only channel through which users learn their preferences is consumption.*

This assumption is inspired by the fact that users do not observe experience features without consuming the product. A direct implication of this assumption is that the RS-dependent user cannot learn from the recommendations beyond their own experience. In our context, this assumption is reasonable as recommendation systems are often highly complex and it is not realistic to assume that users can learn further by observing that a product is recommended.[10]

We now present the choice and learning processes for the RS-dependent user in Algorithm 3. In this setting, not only is there a user who learns her preference parameters through experience,

[9]It is worth emphasizing that this does not imply that the user's feature space is richer than that of the RS. In contrast, the r-dimensional feature space used by the RS contains richer information, as it captures experience features but in a compressed form via principal components.

[10]One could relax Assumption 2 by allowing for the possibility that a user learns from the recommendation signal itself or through social learning from other users or reviews. Later in §5.4.2, we examine these two extensions to demonstrate the robustness of our main insights.

there is also a recommendation system that learns user preferences and offers recommendations. Both players learn user i's preference parameters, but they operate in different spaces: user i learns her own parameters θ_i in the d-dimensional space, whereas the recommendation system learns user i's preference weights for factors in the r-dimensional space. To distinguish between these two learning processes, we use superscripts (θ) and (γ) to refer to the parameters of both players' prior distributions: $\mu_{i,0}^{(\theta)}$, $\Sigma_{i,0}^{(\theta)}$, $\mu_{i,0}^{(\gamma)}$, and $\Sigma_{i,0}^{(\gamma)}$.

The RS moves first in each time period. Since the recommendation system has access to F, it only wants to learn γ_i. As such, it engages in a Linear-Gaussian Thompson Sampling procedure, where it first draws $\tilde{\gamma}_{i,t}$ from the posterior distribution and then recommends the product with the highest expected utility (lines 2 and 3). The RS-dependent user always follows the recommended action (line 4). Once the utility is realized, both the user and recommendation system update parameters of their posterior distribution $\mu_{i,t+1}^{(\theta)}$, $\Sigma_{i,t+1}^{(\theta)}$, $\mu_{i,t+1}^{(\gamma)}$, and $\Sigma_{i,t+1}^{(\gamma)}$. It is worth noting that although action selection (Step 3) relies only on search features (since experience features are unobservable ex ante), the Bayesian update uses the full feature vector $A_{i,t}$ associated with the chosen item. The algorithm repeats this process for $\mathcal{T}$ periods.

Algorithm 3 Choice and Learning for the RS-Dependent User

Input: $\mu_{i,0}^{(\theta)}, \Sigma_{i,0}^{(\theta)}, \mu_{i,0}^{(\gamma)}, \Sigma_{i,0}^{(\gamma)}, F, \mathcal{A}, \mathcal{T}$

Output: $\mathcal{H}_{i,\mathcal{T}}, \{\mu_{i,t}^{(\theta)}, \Sigma_{i,t}^{(\theta)}\}_{t=1}^{\mathcal{T}}, \{\mu_{i,t}^{(\gamma)}, \Sigma_{i,t}^{(\gamma)}\}_{t=1}^{\mathcal{T}}$

1: **for** $t = 0 \to \mathcal{T}$ **do**

2: $\quad \tilde{\gamma}_{i,t} \sim N(\mu_{i,t}^{(\gamma)}, \Sigma_{i,t}^{(\gamma)})$ ▷ RS: Sampling low-dimensional weights

3: $\quad j^* \leftarrow \text{argmax}_j \sum_{k=1}^{r} \tilde{\gamma}_{i,k,t} F_{k,j}$ ▷ RS: Selecting recommendation

4: $\quad A_{i,t} \leftarrow A_{j^*}$ ▷ User: Following recommendation

5: $\quad \Sigma_{i,t+1}^{(\theta)} \leftarrow \left(\left(\Sigma_{i,t}^{(\theta)} \right)^{-1} + \frac{1}{\sigma_\epsilon^2} A_{i,t} A_{i,t}^T \right)^{-1}$ ▷ User: Updating posterior variance

6: $\quad \mu_{i,t+1}^{(\theta)} \leftarrow \Sigma_{i,t+1}^{(\theta)} \left(\left(\Sigma_{i,t}^{(\theta)} \right)^{-1} \mu_{i,t}^{(\theta)} + \frac{1}{\sigma_\epsilon^2} A_{i,t} U_{i,t} \right)$ ▷ User: Updating posterior mean

7: $\quad \Sigma_{i,t+1}^{(\gamma)} \leftarrow \left(\left(\Sigma_{i,t}^{(\gamma)} \right)^{-1} + \frac{1}{\sigma_\epsilon^2} F_{j^*} F_{j^*}^T \right)^{-1}$ ▷ RS: Updating posterior variance

8: $\quad \mu_{i,t+1}^{(\gamma)} \leftarrow \Sigma_{i,t+1}^{(\gamma)} \left(\left(\Sigma_{i,t}^{(\gamma)} \right)^{-1} \mu_{i,t}^{(\gamma)} + \frac{1}{\sigma_\epsilon^2} F_{j^*} U_{i,t} \right)$ ▷ RS: Updating posterior mean

9: **end for**

Lastly, an important point about the RS-dependent user is that the preference learning process is the same for both user types. As such, any difference in the final learning outcomes comes from the consumption sequence. We utilize this fact when we examine the learning implications of RS-dependence in our empirical framework in §5.3.

4 Theoretical Analysis

In this section, we theoretically examine the algorithms described earlier. We begin by defining our primary outcomes of interest in §4.1. We then present the regret bounds for self-exploring users who follow Algorithm 2 in §4.2. Next, in §4.3, we present the regret bounds for RS-dependent users who follow Algorithm 3. Finally, in §4.4, we theoretically compare the two regret bounds, establish bounds for welfare gains from RS dependence and examine the possibility of rational dependence.

4.1 Main Outcomes

As discussed earlier, we are interested in two key user-level outcomes: welfare and learning. In this section, we define these two key outcomes and formally present the measures we use for them. We can define user welfare for user i for T periods as the total sum of utility from actions chosen under policy π as follows:

$$\texttt{Welfare}_i(T;\pi) = \mathbb{E}\left[\sum_{t=0}^{T} u_i(A_{i,t})\right], \tag{8}$$

where $\texttt{Welfare}_i$ is a user-specific function that depends on user i's preference parameters θ_i. Another closely tied measure that is often studied in the literature on sequential decision-making and linear bandits is *expected regret*, which takes the difference between the expected utility from some notion of first-best in each period and user welfare. We formally define *expected regret* as follows:

Definition 2. *Suppose that $A_i^* \in \operatorname{argmax}_{a\in\mathcal{A}} \mathbb{E}[u_i(a) \mid \theta_i]$ is the optimal action (product) given θ_i. For the sequence of actions $\{A_{i,t}\}_{t=0}^{T}$ chosen according to policy π, the* ***expected regret*** *is given as follows:*

$$\texttt{Regret}_i(T;\pi) = \mathbb{E}\left[\sum_{t=0}^{T} \left(u_i(A_i^*) - u_i(A_{i,t})\right)\right], \tag{9}$$

where the expectation is taken over the randomness in the actions, utilities, and the prior distribution over θ_i.

This notion of expected regret is often referred to as the Bayes regret or Bayes risk. Consistent with the prior bandits literature, one advantage of using regret instead of welfare is the possibility of obtaining statistical bounds. We later use these bounds to conduct a theoretical analysis of our problem.

The second outcome we are interested in is user learning. Intuitively, the greater uncertainty the user has over her own preference parameters, the lower the degree of learning by the user. As such, we turn to the well-established concept of Shannon entropy that measures the amount of uncertainty or surprise in random variables [Shannon, 1948].

Definition 3. *Let $A^*_{i,t}$ denote the random variable corresponding to the optimal action (product) given the prior sequence $\mathcal{H}_{i,t-1}$. We measure a user's preference learning based on the* ***Shannon entropy*** *of $A^*_{i,t}$, which is defined as follows:*

$$H(A^*_{i,t}) = -\sum_{k=1}^{|\mathcal{A}|} P(A^*_i = A_k \mid \mathcal{H}_{i,t-1}) \log\left(P(A^*_i = A_k \mid \mathcal{H}_{i,t-1})\right). \tag{10}$$

According to this definition, a higher entropy means that the user is more uncertain as to what the optimal action is. For example, if the user deterministically chooses one action (maximum certainty and learning), the Shannon entropy of the optimal action will be equal to zero, i.e., $H(A_{i,t}) = 0$. On the other hand, when the user is maximally uncertain between actions, each action has an equal probability, and the Shannon entropy of optimal action will take its maximum value $H(A_{i,t}) = \log(|\mathcal{A}|)$.

A key challenge in using Shannon entropy as the primary measure of learning is that it is not directly comparable to regret, which serves as the main measure of welfare. To address this, we introduce the concept of *counterfactual regret*—the regret a user would incur at any given time period if they were to make decisions independently, without the influence of the personalized recommendation system. This measure allows us to capture the trade-off between welfare and learning: if lower regret actions come at the expense of user learning, we expect counterfactual regret to be high for that user. We formally define the *expected counterfactual regret* as follows:

Definition 4. *Let $\tilde{A}_{i,t}$ denote the counterfactual action (product), which is the random variable corresponding to the optimal action (product) given the prior sequence $\mathcal{H}_{i,t-1}$. This is the optimal action the user would choose based on her past learning through experience if the personalized algorithm was not available at period t only. For the sequence of counterfactual actions $\{\tilde{A}_{i,t}\}_{t=0}^{T}$ that would have been chosen in each period under π in the absence of RS, the expected* ***counterfactual regret*** *is given as follows:*

$$\texttt{CounterfactualRegret}_i(T;\pi) = \mathbb{E}\left[\sum_{t=0}^{T}\Big(u_i(A^*_i) - u_i(\tilde{A}_{i,t})\Big)\right], \tag{11}$$

where the expectation is taken over the randomness in the actions, utilities, and the prior distribution over θ_i.

The main benefit of using the notion of *counterfactual regret* is its direct comparability with the actual *regret*. In cases where users follow the personalized algorithms to choose the product, we expect the counterfactual product they would choose without the personalized algorithm to be different from the one offered by the algorithm. As such, there will be a discrepancy between the *counterfactual regret* and the actual *regret*. The gap between the two highlights the loss in independent decision-making ability due to RS-dependence.

4.2 Regret Bounds for Self-Exploring Users

We start by deriving the regret bounds for the self-exploring user. As shown in Algorithm 2, the user starts with a prior distribution $N(\mu_{i,0}, \Sigma_{i,0})$ and samples from the prior to select the product and then updates the posterior distribution based on the realized utility. The user repeats the procedure until convergence to the first-best action (product), which is the best product identifiable by the user. We denote the first-best for the self-exploring user by $A_i^{*,s}$ and define it as the product with the highest utility from *search* features if the preference parameters are known, that is, $A_i^{*,s} \in \operatorname{argmax}_{A_j \in \mathcal{A}} \sum_{k=1}^{s} \theta_{i,k} A_{k,j}$. As such, the first-best identifiable by the self-exploring user is different from the first-best A_i^* in the regret equation (Definition 2), which allows us to write the following decomposition:

$$
\begin{aligned}
\texttt{Regret}_i(T;\pi) &= \mathbb{E}\left[\sum_{t=0}^{T} (u_i(A_i^*) - u_i(A_{i,t}))\right] \\
&= \mathbb{E}\left[\sum_{t=0}^{T} (u_i(A_i^*) - u_i(A_i^{*,s}))\right] + \mathbb{E}\left[\sum_{t=0}^{T} (u_i(A_i^{*,s}) - u_i(A_{i,t}))\right]
\end{aligned}
\tag{12}
$$

where the first element in the equation above is a summation over a constant gap between the expected utility from the two first-bests, and the second term is another notion of regret for the self-exploring user. Specifically, we can show that the second term is the same as the linear bandit regret with an s-dimensional vector of preference weights, which allows us to use information-theoretic regret bounds for linear banidts established in the prior literature [Russo and Van Roy, 2016].[11] We use this insight to arrive at the following proposition:

Proposition 1. *Let π_{SE} denote the policy for the self-exploring user who follows the Thompson Sampling algorithm for choice and learning as in Algorithm 2. Further, let g denote the gap in expected utility between the first-best action and the first-best action based on only search features, that is, $g = \mathbb{E}\left[u_i(A_i^*) - u_i(A_i^{*,s})\right]$. The regret bound for this user is as follows:*

$$
gT \leq \texttt{Regret}_i(T;\pi_{SE}) \leq gT + \sqrt{2(\sigma_\epsilon^2 + \sigma_x^2) s H(A_{i,0}^*) T}, \tag{13}
$$

[11]We present the preliminaries from information theory and important lemmas from this paper in Web Appendix A.

where $H(A^*_{i,0})$ is the Shannon entropy of the prior distribution of optimal action for self-exploring user, defined at period 0, σ_x^2 is the variance of the utility that comes from experience features, and s is the dimensionality of the search features.

Proof. Please see Web Appendix B.3.1. □

As shown in this proposition, both lower and upper bounds contain a term linear in T. The upper bound further includes an information-theoretic bound, which is closely related to the one established in Russo and Van Roy [2016].[12] The lower bound happens in the event where the user has no uncertainty about the preferences (i.e., $H(A^*_{i,0}) = 0$).

Next, we examine the gap g, which is defined as the difference between the first-best and the first-best based on only search features. Since this gap appears in the regret lower bound, it is crucial to understand how large of a magnitude it has and the theoretical conditions under which this term converges to zero. To do so, we need to introduce new notations: let $U^s_{i,j}$ and $U^x_{i,j}$ denote the utility the user derives from the search features and experience features, respectively. We can write the following proposition to characterize the lower bound for the gap g:

Proposition 2. *Suppose that the search utility and experience utility across products are Normally distributed, such that $U^s_{i,j} \sim N(\mu_{i,s}, \sigma^2_{i,s})$ and $U^x_{i,j} \sim N(\mu_{i,x}, \sigma^2_{i,x})$. If search utility and experience utility are independent across products, the expected gap between the first-best and first-best based on the search features has the following lower bound:*

$$\mathbb{E}\left[u_i(A^*_i) - u_i(A^{*,s}_i)\right] \geq \sqrt{2\log(|\mathcal{A}|)}\left(\left(\sqrt{\sigma^2_{i,s} + \sigma^2_{i,x}} - \sigma_{i,s}\right) - O\left(\frac{\sqrt{\sigma^2_{i,s} + \sigma^2_{i,x}}}{\log(|\mathcal{A}|)}\right)\right) \tag{14}$$

Proof. Please see Web Appendix B.3.2. □

As shown in Proposition 2, the gap depends on the variance of the utility from experience features and the number of products available. In particular, this gap grows as experience features play a larger role in determining a user's utility. Although we make the independence assumption, a natural interpretation in the correlated setting is to view $\sigma^2_{i,x}$ as the residual variance of experience utility that is not explained by search features. Further, as the action space grows, the term $O\left(\sqrt{\sigma^2_{i,s} + \sigma^2_{i,x}}/\log(|\mathcal{A}|)\right)$ converges to zero, which suggests a substantially higher gap when the action space is larger.

[12] Consistent with the literature on linear bandits, our theoretical results carry through, in expectation, when we have a time-varying action set $\mathcal{A}_t$, as long as the underlying distribution of actions is stable.

4.3 Regret Bounds for RS-Dependent Users

We now turn to establishing regret bounds for the RS-dependent user. The regret analysis for the RS-dependent users is a bit more subtle. In this case, the user follows the recommendation from the personalized RS. As such, we need to find the regret bound for the RS. One could view the RS-dependent algorithm as a Thompson Sampling algorithm that operates in an r-dimensional environment, as it has access to factor information. Under Assumption 1, the first-best action that the RS could obtain is the same as the first-best action in Definition 2. Therefore, the proposition below characterizes the following regret bound for the RS-dependent user:

Proposition 3. *Let π_{RS} denote the policy for the RS-dependent user who consistently follows the personalized recommendations. Further, let $H(A_{i,0}^{RS})$ denote the Shannon entropy of the prior distribution of optimal action for the personalized RS. The regret bound for the RS-dependent user is as follows:*

$$\texttt{Regret}(T;\pi_{RS}) \leq \sqrt{2\sigma_\epsilon^2 r H(A_{i,0}^{RS})T}, \tag{15}$$

where r is the number of factors. The expected regret for the RS-dependent user is equal to the expected regret for the personalized RS.

Proof. Please see Web Appendix B.3.3. □

As shown in Proposition 3, the regret bound for the RS-dependent user does not depend on the dimensionality of the feature space but on the rank r. In many practical settings, r is much smaller than the dimensionality of the feature space (d) or search features (s) [Udell and Townsend, 2019]. On the other hand, the main challenge for the RS is the initial lack of information about user preferences. In the extreme case, one could assume that the algorithm has an uninformative prior such that the prior distribution of optimal action gives all actions the same probability, which makes the Shannon entropy of the prior distribution of optimal action equal to $\log(|\mathcal{A}|)$. However, modern RS algorithms try to overcome the cold-start problem by better exploring the side information about the user [Farias and Li, 2019]. Our information-theoretic bound accommodates these realistic scenarios where RS algorithms have a prior better than a random guess.

4.4 Welfare Gains from RS Dependence

As highlighted earlier, both the user and the RS have some information advantages. The regret bounds in Propositions 1 and 3 illustrate their dependence on factors associated with each type of information advantage and disadvantage, as depicted in Figure 2. On the one hand, the platform's algorithm benefits from access to data from other users, enabling it to efficiently identify the actual first-best in contrast to the self-exploring user, who can only determine the first-best based on search features. On the other hand, in certain scenarios, users may have greater knowledge of their own parameters than the algorithm, resulting in a very small Shannon entropy of the prior distribution of the optimal action ($H(A^*_{i,0}) \approx 0$), which is substantially lower than that of the RS-dependent user ($H(A^*_{i,0}) \ll H(A^{RS}_{i,0})$).

Our goal in this section is to compare the regret bounds for self-exploring and RS-dependent algorithms established earlier. Combining the lower bound for the self-exploring user's regret with the upper bound for the RS-dependent user's regret, we arrive at the following corollary:

Corollary 1. *The difference in regret between self-exploring users and RS-dependent users has a lower bound given by:*

$$\texttt{Regret}(T;\pi_{SE}) - \texttt{Regret}(T;\pi_{RS}) \geq gT - \sqrt{2\sigma_\epsilon^2 r H(A^{RS}_{i,0})T} \tag{16}$$

A few key insights emerge from Corollary 1. First, the difference in Equation (16) directly corresponds to the welfare gain from following the RS, as the first-best terms in both regret values cancel out. Second, we observe that the positive term grows linearly in T, while the negative term grows sub-linearly. This implies that with a nonzero gap g, the welfare gain from following the RS increases linearly over time. Third, this result does not depend on the relationship between r and s. Figure 3 highlights the region where the lower bound for welfare gain in Equation (16) is positive for different numbers of time periods, gap values and a set of initial parameter, in an instance where the error variance and rank parameters are calibrated based on the prior work on movie recommendations ($\sigma_\epsilon = 0.5$ and $r = 40$), and the Shannon entropy of the prior is fully uninformative, that is, $H(A^{\text{RS}}_{i,0}) = \log(|\mathcal{A}|)$. As shown in this figure, despite the algorithm's uninformative prior and the user's full certainty about their preferences, the region where the lower bound for the welfare gain is positive expands rapidly as the number of time periods grows, making even a small gap sufficient to make RS dependence a more utility-maximizing choice than the self-exploring alternative.

Together, our results highlight the welfare gains from the personalized algorithm created solely through its information advantage. We specifically abstract away from users' preference uncertainty and search costs to isolate the welfare gain offered by the RS purely due to its information advantage through access to other users' data. If we account for users' preference uncertainty, the region with a positive lower bound for welfare gain from the personalized RS expands, as the user must incur some exploration cost. Likewise, if the user incurs a search cost while exploring products, the illustrated region in Figure 3 will shift downward on the y-axis. Isolating the information advantage channel is important, as it suggests that even in the absence of search costs or preference uncertainty,

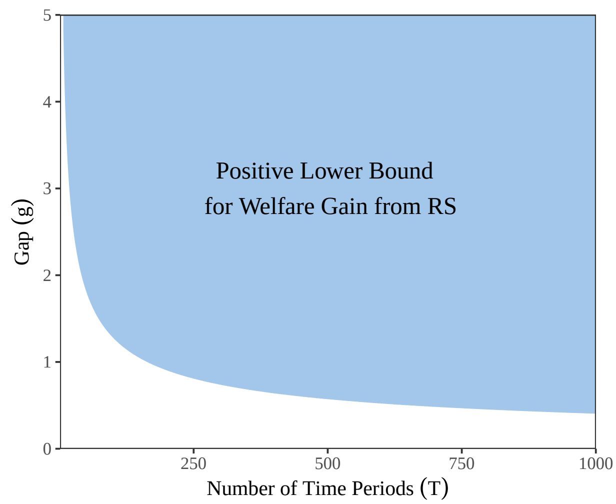


Figure 3: Region with a guaranteed welfare gain from RS dependence across model parameters

users will rationally develop a dependence on the recommendation system (RS). We summarize the regret bounds, assumptions, and implications in Web Appendix C for ease of reference.

5 Empirical Framework

In this section, we first provide a broader discussion of our theoretical analysis to identify key empirical questions that warrant further investigation. As established in Corollary 1, the welfare benefits of RS tend to outweigh its downsides as the number of periods T increases. However, in real-world settings, users do not engage with the system for an infinite number of periods. As a result, self-exploring users may achieve better welfare outcomes within finite periods when (1) the gap g is small, and/or (2) the RS initially provides poor recommendations and requires time to stabilize. The latter scenario is more likely if the effective rank r is high or if the RS starts with an uninformative prior. Therefore, assessing the net welfare impact of RS dependence in finite samples remains an empirical question, contingent on the actual values of g and r.

Second, another key outcome of interest in our study is learning. In particular, we want to know how following RS influences users' preference learning and independent decision-making ability, measured by *counterfactual regret*. In principle, if the personalized recommendations create enough variation in products to learn preference parameters, we do not expect it to affect user learning. We know that the algorithm needs to explore in the beginning, which naturally creates some variation in product features consumed. However, comparing the upper bounds in Propositions 1 and 3, we know that the RS scales the regret in the order of $\sqrt{r/s}$-fraction of the self-exploring user if they start from the same prior, that is, $H(A^*_{i,0}) = H(A^{\text{RS}}_{i,0})$. As such, the RS algorithm is likely to reach the exploitation stage quickly, potentially limiting the extent of exploration necessary for effective user learning. Specifically, if the exploitation phase primarily recommends a narrow set of products, users may experience insufficient variation to fully learn their preferences. Importantly, whether or not the exploitation stage of the algorithm generates enough variation for user learning largely depends on the distribution of user preferences and, therefore, is an empirical question.

In our empirical framework, we aim to examine the finite-sample properties of self-exploring and RS-dependent algorithms in terms of both welfare and learning measures. To accomplish these objectives, we require an empirical setting that meets the following criteria: (1) availability of preference data on user-product pairs, (2) access to a rich set of search features for products that allow us to estimate user preferences and simulate different learning patterns under various algorithms, and (3) the ability to integrate a state-of-the-art personalized recommendation system capable of learning complex user preferences and providing relevant recommendations.

To meet these three criteria, we utilize the MovieLens 1M dataset.[13] The MovieLens dataset includes over one million user ratings from 6,040 distinct users on 3,706 unique movies, which serve as our measure of user preferences. Beyond ratings, the dataset provides detailed information about the movies, including genres, themes, and an extensive collection of tags associated with each movie, which we leverage as the set of search features. Lastly, the MovieLens dataset has been widely used as a benchmark for research on personalized recommendation systems, ensuring that we can integrate state-of-the-art recommendation algorithms into our empirical framework.

In this section, we begin by outlining our empirical strategy in §5.1. We then analyze the welfare gains derived from the personalized RS due to its information advantage in §5.2. Next, in §5.3, we investigate the learning outcomes for self-exploring and RS-dependent users. Finally, in §5.4, we explore the policy implications and identify potential strategies that balance the trade-off between welfare and learning objectives.

5.1 Empirical Strategy

In order to satisfactorily answer the empirical questions, we face several challenges. Before discussing these challenges in great detail, we need to state a few assumptions upfront. The first assumption we make is about the relationship between ratings and user utility. While our theoretical framework uses the concept of user utility, we only observe user ratings. Let $Y_{[N\times J]}$ denote the rating matrix for N users and J movies. We present our assumption that links ratings to utility as follows:

Assumption 3. *User i's utility from watching movie j is well-approximated by user ratings, i.e., $U_{[N\times J]} \approx Y_{[N\times J]}$.*

[13]This dataset is publicly available and can be accessed at `https://grouplens.org/datasets/movielens/1m/`.

Our next assumption pertains to the set of available search features. Since our data contain detailed movie tags, we use these tags as the primary search features for each movie. These tags include attributes such as originality, great ending, and good soundtrack–features that are visible to users before consumption, as they appear in the movie profile on the website (for a complete list of movie tags, see Web Appendix D.1). Naturally, one could use all tags to characterize the dimensionality of user preferences. However, it is reasonable to assume that only a subset of these tags are important for user utility. Specifically, we assume that the top k most frequently used tags capture the search features. In our analysis, we set $k = 50$ and formalize this assumption as follows:

Assumption 4. *The matrix of a movie's search features is defined as $A_{[51\times J]} = [A_1 \mid A_2 \mid \cdots \mid A_J]$, where A_j represents features of movie j in terms of the aggregated rating and top 50 tags selected from the data.*

We emphasize that the choice of k is flexible. While we use $k = 50$ for our main analysis for modeling reasons, all qualitative insights hold for other characterizations. The final assumption we make is the following about the stability of true user preferences:

Assumption 5. *For any user i, the true user vector of preferences is represented as θ_i for all time periods, independent of the movies watched.*

It is important to note that, in our setting, the user can still learn their preferences through experience. While the actual preferences remain stable, the user's belief about these preferences can evolve over time. With all the assumptions clearly defined, we discuss these challenges along with their corresponding empirical strategy in the following sections.

5.1.1 Data Sparsity

For each user in our data, we only observe ratings for a subset of products, which makes it challenging to estimate user-level parameters θ_i. To overcome this challenge, we select a test sample of 400 users who have rated over 420 movies.[14] Having a rich set of ratings for a user allows us to estimate parameters at the individual-level, by simply regressing the outcomes on movie features. As stated in Assumption 4, we use 51 features in our analysis that contain search features that users could see on the TMDB website, including the aggregated rating for each movie as well as 50 other tag information.

[14]We acknowledge that this sample is not necessarily representative and was selected out of empirical necessity. However, our goal is to provide a proof of concept for the key insights regarding information advantage, algorithmic dependence, and user learning, for which sample representativeness is not a primary requirement.

5.1.2 Model-free Regret Measurement

Empirically measuring regret for algorithms different from the one used in the data is fundamentally challenging. One approach is to impute ratings for unrated movies, but this requires highly accurate model-based estimates and risks information leakage between the imputation model and the algorithms being evaluated. Alternatively, restricting analysis to the set of movies each user has already rated avoids this issue but limits the ability of algorithms to identify strong recommendations outside the observed choices. Moreover, the movie recommendation setting differs slightly from the theoretical framework, where the action set $\mathcal{A}$ is fixed. In reality, once content is consumed, users may not derive the same utility from repeated consumption. This makes the set of available products for each policy different, making an apples-to-apples comparison between algorithms impossible.

To address these challenges, we randomly select a hold-out set of 20 movies for each user in our test set, ensuring that these movies have been rated by the user so that we have access to the actual outcomes and the first-best option among them. We then exclude these 20 movies from the user's training set, preventing the algorithm from accessing any information about the user's ratings for these movies. Notably, while the hold-out set remains fixed for each user, it varies across users in our test data. This setup allows us to train any algorithm without the 20 hold-out movies and evaluate its regret and counterfactual regret in a model-free manner by directly using the user's observed ratings for these hold-out movies.

5.1.3 Integrating the Personalized RS

To evaluate outcomes for RS-dependent users, we require a personalized RS that continuously updates its recommendations as it acquires more information about each user. As such, this has to be a personalized RS that can handle the cold-start problem. For each user i in the test set, we use the existing data of 5,040 other users in the training set and combine it with the information about user i available at the beginning of each time period t. Following Cortes [2018], we use a matrix factorization algorithm for our personalized recommendation system, which is widely used in the industry.[15] We then train this recommendation system and predict ratings for a hold-out set of 20 movies. Based on these predictions, the RS recommends the movie with the highest predicted rating, which the RS-dependent user subsequently consumes.

[15]To better align with practice, we use a state-of-the-art matrix factorization algorithm in our main empirical analysis, which differs from Algorithm 3 based on Thompson Sampling. Later in §5.4.3, we show that our results are robust to a range of bandit-based personalized algorithms, including Thompson Sampling in Algorithm 3, as well as alternative approaches such as Bayes-UCB and Bayes-EXP4.

5.2 Algorithm's Information Advantage and Regret Performance

We now turn to examining the algorithm's information advantage and the welfare gains from following the RS. This empirical analysis complements the theoretical results from §4 by using real data to assess the finite-sample performance of different algorithms. Specifically, we aim to address two key questions: (1) how quickly the RS learns to make high-quality recommendations on the hold-out set of 20 movies and (2) whether these recommendations outperform those that users could identify based on observable search features. The first question pertains to the efficiency of low-dimensional learning through the factor model, while the second captures the inherent gap g-the difference between the first-best option identified by the RS and the best choice the user could make based on their search features.

As in our theoretical analysis, we focus on two user types: self-exploring users and RS-dependent users. In both cases, the user begins with a prior distribution over preferences that reflects average taste: the initial weight on the aggregate rating feature is set to one, all other feature weights are set to zero, and the prior variance for every parameter is initialized at one. We calibrate the experience-noise parameter $\hat{\sigma}_\epsilon = 0.86$ directly from the data, using the RMSE between actual ratings and the imputed ratings produced by the matrix factorization algorithm of Cortes [2018]; this captures the inherent variability of experience features in our setting. While we rely on these specific initializations, we stress that they do not drive our main qualitative insights, which continue to hold under a wide range of alternative choices.

In each round, the user selects a movie from the available set, consumes it, observes the realized utility, and updates their preferences accordingly. The key difference between the two user types lies in the sequence of movies they consume, which in turn drives differences in learning and regret on a hold-out test set.

To operationalize this, for each user we randomly select 20 rated movies to form a hold-out test set, with the remaining movies constituting the training set. We then simulate the consumption process separately for each user type as follows:

- **Self-Exploring User:** The self-exploring user selects movies from the training set using a Thompson Sampling algorithm. We use the actual rating of each movie as its realized utility, and once a movie is consumed, it is removed from the training set. At each period, we evaluate the movie the user would choose from the hold-out set given their current beliefs. This allows us to compute hold-out regret by comparing the chosen movie to the best available option based on the user's actual ratings. Details are provided in Algorithm 4 in Web Appendix D.2.
- **RS-Dependent User:** The RS-dependent user follows the recommendations of the algorithm at each step. The recommender system can be any standard approach; in our main analysis, we

follow Cortes [2018], and later consider alternative algorithms in §5.4.3. As before, we use the actual rating as the realized utility for each movie, and consumed movies are removed from the training set. At each period, we also evaluate the recommendation from the hold-out set, which allows us to compute model-free regret based on the user's observed ratings. Details are provided in Algorithm 5 in Web Appendix D.2.

In addition to the dynamic cases above where either the user or the RS learns, we consider two static benchmarks to contextualize the performance of these algorithms:

- **Aggregate Rating:** This benchmark selects, in each period, the movie with the highest aggregate rating from the hold-out set. It serves as a simple baseline with no user-specific learning. As a result, its performance remains constant over time.
- **User with Known Preferences:** This benchmark assumes that the user's preference parameters are fully known from the outset and remain fixed over time. Specifically, we use the preference parameters obtained by the Self-Exploring User after consuming all movies in the training set. This represents the best performance achievable by a self-exploring user.[16] Any gap between this benchmark and the personalized RS reflects the gap factor g characterized in our theoretical analysis.

Figure 4 shows the performance of both user types relative to the static benchmarks over 400 periods, aggregated across 400 users and 100 simulations per user. For each user, a simulation corresponds to a randomized hold-out set of 20 movies. Several key insights emerge from this figure. First, the RS outperforms all three benchmarks on the hold-out test set, demonstrating its ability to identify and leverage patterns in user behavior. Second, there is a persistent gap between the regret under *RS-Dependent User* and the *Self-Exploring User*, highlighting the intrinsic information advantage of the RS and the gap between the first-best and the first-best based on search features. Third, the RS quickly surpasses the user with known preferences, underscoring its efficiency due to low-rank learning. Importantly, this result is not driven by an imposed rank restriction (e.g., $r < 51$); instead, the personalized RS algorithm determines the effective rank in a data-driven manner. In Web Appendix D.3, we provide additional performance metrics to further compare the RS with these benchmarks. We also document the heterogeneity in our findings across different levels of nicheness in user preferences in Web Appendix D.4.

In summary, our findings highlight the information advantage of the personalized RS algorithm, offering a rational explanation for why users become dependent on algorithms. This is significant because prior literature has often cited factors such as time-inconsistent preferences and search costs to justify algorithmic dependence [Allcott et al., 2022, Ursu, 2018]. However, as our analysis

[16]Later in §5.4.2, we perform robustness checks with a richer feature set.

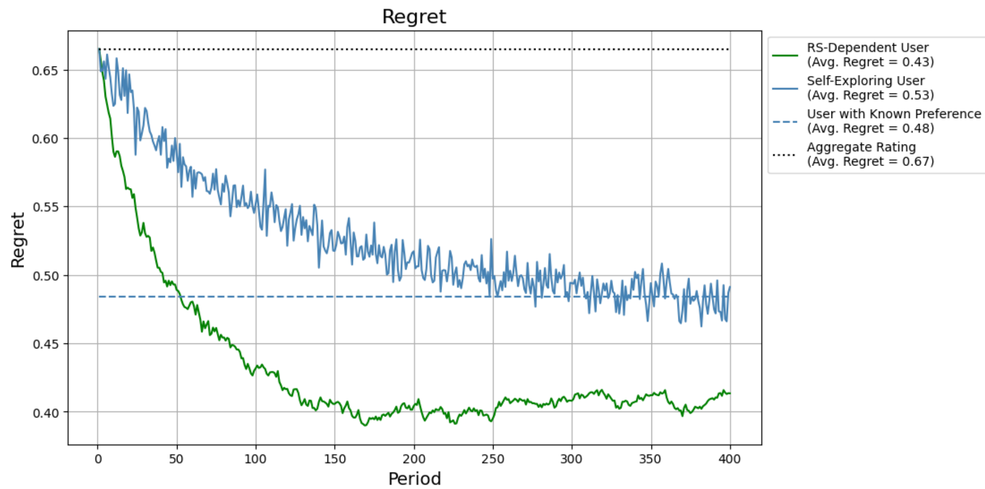


Figure 4: The regret performance of different algorithms on the hold-out set

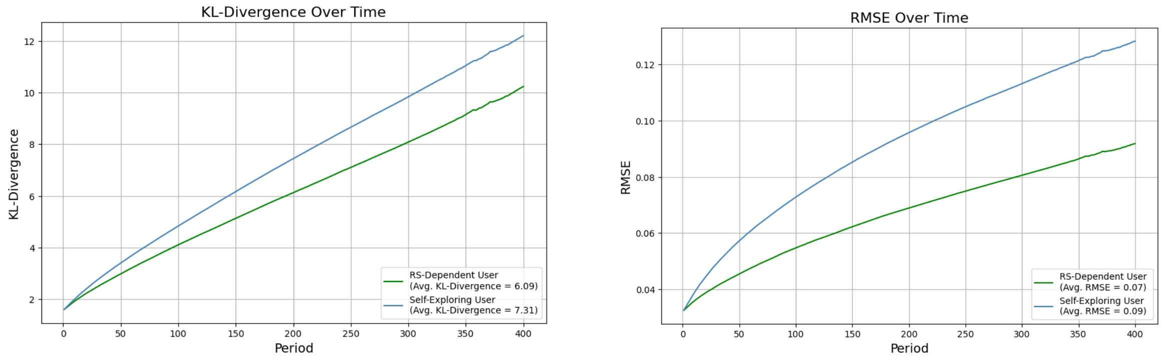
suggests, there is a clear limitation for users in the context of experience goods, which rationally forces them to rely on personalized RS, even absent search costs or time-inconsistent preferences.

5.3 Implications for Users' Preference Learning

As shown in our theoretical and empirical analysis so far, RS-dependence brings substantial welfare gains for users, making it a rational strategy. However, this dependence also has implications for users' preference learning, as it alters the prior experience from which they resolve uncertainty about their preferences. In this section, we examine the implications of RS-dependence for user learning, first using statistical measures in §5.3.1, and then using economic measures that capture users' ability to make independent decisions without the assistance of the RS in §5.3.2.

5.3.1 Statistical Measures of Learning

We present the results from the exercise in §5.2 using statistical measures of preference learning for the two user types: (1) the Self-Exploring User and (2) the RS-Dependent User. Both users start from the same prior distribution of preferences reflecting average taste. Our goal is to assess how much learning occurs in each case. To quantify this difference, we use two key measures: (1) Kullback-Leibler (KL) divergence, which is commonly used to assess the difference between two probability distributions and measures the divergence between the posterior distribution at each point from the prior distribution, and (2) Root Mean Square Error (RMSE), which calculates the square root of the average squared differences between the vector of mean preference parameters at each point and the vector of prior weights. These measures help evaluate how much the user has learned over time.



(a) KL Divergence (b) RMSE

Figure 5: User learning under different algorithms

Figure 5 illustrates the learning outcomes for both self-exploring and RS-dependent users, measured using KL divergence and RMSE. As both figures demonstrate, the difference from the prior increases at a faster rate for the self-exploring user compared to the RS-dependent user. This suggests that the self-exploring user learns more efficiently over time. It is important to note that in this analysis, both user types begin with the same prior preferences, so the observed differences reflect the relative learning rates of the two approaches. We further validate the robustness of these findings in Web Appendix D.5, where we consider scenarios in which users begin with more informed priors. Across all cases, the self-exploring user consistently exhibits greater learning. This suggests that actively exploring a diverse range of movies accelerates preference learning, whereas the RS-dependent user, despite benefiting from personalized recommendations, experiences a slower learning trajectory. Thus, our findings highlight that following personalized recommendations comes at the expense of the user's ability to learn.

5.3.2 Economic Measure of Learning: Counterfactual Regret

We now turn to our economic measure of learning and examine whether the reduced statistical learning illustrated in Figure 5 translates into meaningful economic losses. To quantify the impact of this trade-off by a metric comparable to welfare, we use the measure of *counterfactual regret*, as outlined in Equation (11) and Definition 4. This measure allows us to assess how the RS algorithm influences the user's independent decision-making ability by comparing their performance with different strategies.

To measure counterfactual regret, we use the updated preference distribution for the RS-Dependent User at each time period and measure their regret on the hold-out set of movies without the algorithm's help. We add this line as RS-Dependent User Counterfactual to Figure 4 and visualize the patterns in Figure 6. If RS-dependence truly acts as a barrier to preference learning, we should observe that the RS-Dependent User Counterfactual experiences higher regret compared to the Self-Exploring User and the RS-Dependent User. This would suggest that although RS-dependence provides welfare gains by offering better recommendations, it limits the user's ability to independently make decisions, thereby hindering their learning.

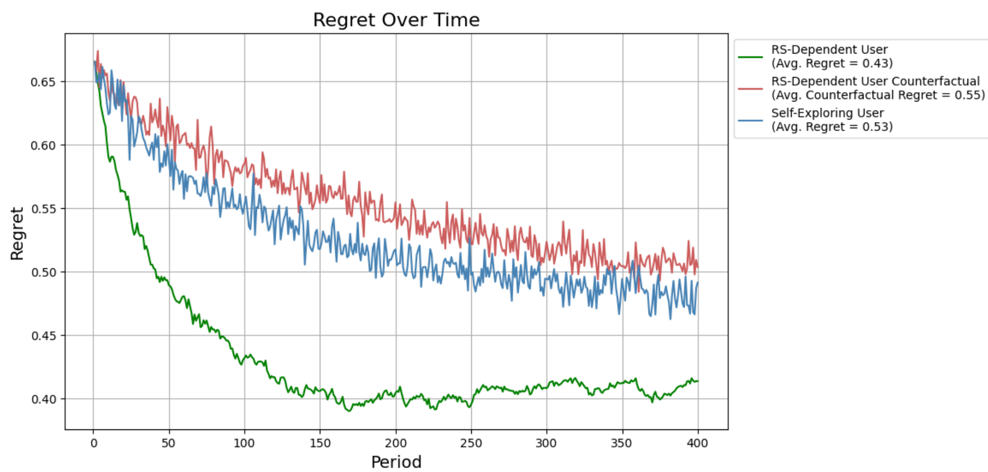


Figure 6: Comparison of regret counterfactual regret for different algorithms

The results presented in Figure 6 underscore the trade-off between the benefits of following personalized recommendations and the cost in terms of independent decision-making ability. The red line, representing counterfactual regret, illustrates the amount of regret an RS-Dependent User incurs when making decisions independently (i.e., without relying on the personalized recommendation system). This line shows a persistent gap between the expected regret (from following the RS) and the counterfactual regret (from independent decision-making). Importantly, the expected counterfactual regret for the RS-Dependent User is consistently higher than the expected regret for the Self-Exploring User.[17]

In summary, we find that although following personalized recommendations reduces expected regret, it comes at the expense of users' independent decision-making ability. This finding is important for several reasons. First, it highlights the trade-off between welfare gains and learning ability. Second, it raises concerns about dependency on personalized RS systems, particularly in contexts where independent decision-making is critical (e.g., to avoid adversarial AI attacks or to maintain a high level of user autonomy). Thus, our analysis of learning outcomes emphasizes the need to carefully balance the benefits of personalized algorithms with the potential risks of over-dependence.

[17]It is worth clarifying that the counterfactual regret for the self-exploring user is the same as the expected regret.

5.4 Extensions and Robustness Checks

5.4.1 Hybrid Users

Our analysis so far has focused on two benchmark behaviors: RS-dependent users who always follow the recommendation system, and self-exploring users who always act independently. These represent two extremes. In practice, however, users may adopt hybrid strategies that combine recommendation-based choices with independent exploration. This raises the question: can such mixed strategies balance the trade-off between welfare and learning identified in §5.2 and §5.3?

To study this, we consider a class of hybrid strategies in which users probabilistically alternate between following the recommendation system (RS) and making independent decisions. Specifically, in each period, the user follows the RS with probability $1 - p$ and explores independently with probability p. This parameter p captures the degree of self-exploration. When $p = 0$, the user is fully RS-dependent; when $p = 1$, the user is fully self-exploring. Intermediate values of p correspond to hybrid behaviors that mix the two modes. This formulation is equivalent to introducing stochastic availability of the RS, where the system is unavailable with probability p. See Web Appendix E.1 for implementation details.

As in §5.2 and §5.3, we evaluate performance using regret and counterfactual regret on a hold-out set of 20 movies per user. For each value of p, we simulate consumption sequences under the corresponding hybrid strategy and compute both metrics. Figure 7 presents the results, showing a clear trade-off between regret and counterfactual regret, closely related to the exploration–exploitation dilemma. We plot the Pareto frontier across different values of p and find that intermediate strategies outperform the extremes: certain hybrid strategies Pareto dominate the fully self-exploring benchmark by achieving both lower expected and counterfactual regret. For example, a policy with $p = 0.2$ attains counterfactual regret comparable to the self-exploring user while maintaining expected regret close to the RS-dependent benchmark

We further complement our analysis by allowing for user heterogeneity in their hybrid policy, that is, different p for different users. We present the results from this exercise in Web Appendix E.2. Our results indicate that moderate levels of mixing bring regret and counterfactual regret closer to each other, improving learning without a substantial increase in regret. Next, in Web Appendix E.3, we consider a simple equally weighted multi-objective optimization that balances regret and counterfactual regret, and identify the optimal p_i for each user. We document substantial heterogeneity in the optimal hybrid policy across users in this exercise, suggesting that users differ in both the benefits they derive from the RS and the losses they incur from reduced independent learning.

Together, our findings suggest that relatively simple hybrid strategies can effectively balance

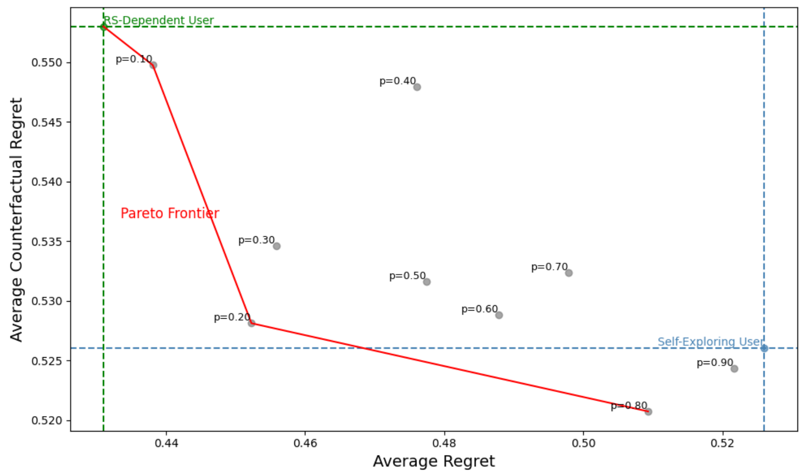


Figure 7: The regret and counterfactual regret performance of hybrid (random availability) strategies

welfare and learning. From a policy perspective, this interpretation maps naturally to interventions that limit or regulate the availability of recommendation systems. Such policies can be implemented uniformly (imposing a common p across users) or in a personalized manner, where p is tailored to individual users. In this way, a policymaker can use controlled randomness to balance welfare and learning across the population.

5.4.2 Alternative Forms of User Learning

In our main analysis, we assume that users are Bayesian learners who learn only from experience signals and update their preference parameters over a certain set of search features. In this section, we relax different parts of these assumptions to demonstrate the robustness of our main insights and establish boundary conditions. In particular, we consider the following extensions:

- **ϵ-Greedy Self-Exploring User:** In our main analysis, the self-exploring user follows a Thompson Sampling (TS) linear bandit framework. However, the posterior sampling process in TS may be perceived as relatively sophisticated for users to adopt. In practice, some users may behave in a more greedy and less exploratory manner. As such, we consider an ϵ-greedy self-exploring user who continues to update preferences in a Bayesian fashion but does not engage in posterior sampling. Instead, the user selects the option with the highest posterior mean utility with probability $1 - \epsilon$, and chooses randomly with probability ϵ to allow for learning. Details of this alternative learning model are provided in Web Appendix F.1. As shown in our results, the gap between RS-dependent and self-exploring users is even more pronounced when the self-exploring user adopts alternative learning rules such as ϵ-greedy.
- **Social Learning:** In our main analysis, users have no access to information about experience

features. In practice, however, they may partially compensate for this lack of information through written reviews, experts, friends, and user-generated content. To capture this, we allow for social learning by giving users access to the aggregated ratings of their nearest neighbors, which provide indirect information about experience-based utility. Details of this approach are provided in Web Appendix F.2. Our results show that while social learning improves outcomes for self-exploring users, the gap between RS-dependent and self-exploring users remains substantial, reinforcing the robustness of our main qualitative insights.

- **Learning from the Recommendation Signal:** Under Assumption 2, users learn only through consumption. In practice, however, a more sophisticated self-exploring user may also learn from the recommendation signal itself, even without consuming the recommended item. To capture this, we conduct an exercise in which we introduce a feature indicating whether an item has ever been recommended by the algorithm. This allows users to infer potential preference alignment from the recommendation signal and adjust their beliefs accordingly. Details are provided in Web Appendix F.3. Our results show that while learning from the recommendation signal reduces regret for the self-exploring user, the gap between RS-dependent and self-exploring users remains substantial.
- **Learning from a Richer Feature Set:** In our main analysis, we use top 50 tags and the aggregate rating as features for a movie. We perform two robustness checks in Web Appendix F.4, where we expand the feature set to 100 dimensions and also consider a dense PCA-based feature representation. Our qualitative results remain unchanged across these specifications.
- **Correlation Between Search and Experience Features:** Our theoretical analysis assumes independence between search and experience features. In practice, however, some search features may be informative about experience features (e.g., the thriller genre may increase the likelihood of a surprise ending). To account for this, we conduct two simulation exercises that introduce correlation between these features in both low- and high-dimensional settings (see Web Appendix F.5). Our results show that the magnitude of the gap depends on the extent to which search features capture the utility derived from experience features.

In summary, these extensions and robustness checks demonstrate that our main insights are robust and not driven by the specific assumptions about user learning in the model.

5.4.3 Alternative RS Algorithms

In our main analysis, we use the matrix factorization method proposed by Cortes [2018], primarily to align with common practice and ensure that our empirical results are not driven by the imposed Thompson Sampling structure of the RS algorithm. In this section, we extend our analysis to three alternative bandit-based RS algorithms built on low-rank factorization: (1) Thompson Sampling, (2) Bayes-UCB, and (3) Bayes-EXP4. The first is analogous to Algorithm 3, while the latter two incorporate adaptive decision-making using UCB and EXP4 approaches. Details of these algorithms are provided in Web Appendix G. Our results show that all key insights from the main analysis remain qualitatively unchanged.

5.4.4 Objective Misalignment

In our main analysis, we assume alignment between the user's objective and that of the RS. More generally, however, adverse effects on users naturally arise when these objectives are misaligned [Kleinberg et al., 2024]. The purpose of the alignment assumption in our baseline is to show that even when the RS seeks to enhance user welfare, it may still come at the expense of user learning. In Web Appendix H, we extend the analysis to allow for objective misalignment. We find that although the information advantage of the RS shrinks, the negative impact on users' preference learning becomes more pronounced.

6 Discussion and Conclusion

Personalized recommendation systems have become a cornerstone of the digital ecosystem. However, growing user dependence on these algorithms has raised concerns among consumer protection advocates and regulators. In this work, we take an information-theoretic approach to examine the foundations of algorithmic dependence and its implications for users' preference learning and independent decision-making—an increasingly critical issue given rising concerns about adversarial AI. We develop a utility framework in which users consume experience goods and sequentially learn their preferences, allowing us to analyze how personalized algorithms influence the learning process. Our theoretical results establish regret bounds for different user types based on their level of dependence on personalized recommendations. We show that algorithmic dependence is rational, as the welfare gains from following the personalized algorithm relative to self-exploration grow linearly over time. To complement our theoretical findings, we introduce an empirical framework that provides model-free measures of regret across different user types. We find that personalized algorithms generate significant welfare gains, but these gains come at the cost of users' preference learning and independent decision-making. Finally, we demonstrate that simple policies can help balance the trade-off between welfare and learning, offering insights for both platforms and users.

In summary, our study makes several important contributions to the literature. From a substantive standpoint, we provide a comprehensive analysis of how personalized recommendation systems influence both welfare and learning outcomes, combining theoretical and empirical models. While these algorithms improve welfare by offering better product recommendations, they can also hinder users' learning and independent decision-making ability. Therefore, our work contributes to the policy debate around personalized algorithms by uncovering the foundations of algorithmic dependence and emphasizing its negative effects on users' ability to make independent decisions, offering insights that are particularly pertinent in light of the growing concerns surrounding adversarial AI.

Methodologically, we develop a theoretical framework to study algorithmic dependence and establish theoretical regret bounds based on users' reliance on personalized algorithms. A key innovation of our approach is its ability to capture the information advantage of these algorithms through a low-rank assumption and access to experience features unavailable to users. Additionally, we introduce a counterfactual regret measure that serves as a valuable benchmark for assessing the impact of adversarial AI. From a policy perspective, we demonstrate that hybrid strategies can effectively balance the trade-off between welfare and learning. Our findings provide valuable insights for both managers and consumers. For managers, we propose that achieving a balance between performance and user learning is feasible at a low business cost. Similarly, our results offer self-regulation insights for consumers, enabling them to manage their own dependence on recommendation systems by taking simple steps.

Nevertheless, our paper has certain limitations that open avenues for future research. First, our findings are largely based on the established theories on user learning. One could design a long-run randomized experiment and verify these findings in the field. Second, our paper focuses on experiential preference learning. Future research can extend our work to different forms of learning (e.g., learning how to perform a task) and study the impact of algorithms on those learning outcomes. Third, our work provides one explanation for how an algorithm's access to richer information can lead to better recommendations and faster learning, even when users behave as rational decision-makers. Future research could explore alternative channels through which the algorithm's advantage may arise, particularly those stemming from users' bounded rationality or limited ability to process rich information. Finally, for simplicity, our paper focuses on a single-item recommendation setting. Multi-item recommendation settings are more complex, as they require a combinatorial optimization framework [Ariannezhad et al., 2023]. While our key high-level insights regarding information advantage, algorithmic dependence, and user learning are likely to extend to the multi-item setting, future work could examine this problem more thoroughly in that context and identify where the two settings diverge.

Competing Interests Declaration

Author(s) have no competing interests to declare.

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Web Appendix

A Preliminaries from Information Theory

Since we work with information-theoretic bounds, we need to provide some basic definitions for the concepts we use in the proofs. We start with Shannon entropy that appears in our regret analysis and is one of the most fundamental concepts in information theory:

Definition 5. *For variable X, the Shannon entropy is defined as:*

$$H(X) = -\sum_{x \in \mathcal{X}} P(X = x) \log\left(P(X = x)\right) \tag{17}$$

Naturally, Shannon entropy quantifies the uncertainty or information content in a probability distribution. It measures how unpredictable or surprising an outcome is when drawn from a given set of probabilities. Another important definition is the Kullback-Leibler divergence, which is defined for any two probability measures as follows:

Definition 6. *The Kullback-Leibler (KL) divergence between two probability measures P and Q, where P is absolutely continuous with respect to Q, is defined as:*

$$D_{KL}(P \| Q) = \int \log\left(\frac{dP}{dQ}\right) dP. \tag{18}$$

Here, $\frac{dP}{dQ}$ is the Radon–Nikodym derivative of P with respect to Q, which represents the density of P relative to Q.

KL divergence measures how much one probability distribution differs from another. Intuitively, it quantifies the amount of extra information required to describe data sampled from one distribution (P) when using a different distribution (Q) as a reference. is not symmetric, meaning switching P and Q gives a different result. This reflects the fact that using Q to approximate P may lead to more or less information loss than vice versa. As expected, $D_{KL}(P \| Q)$ is zero if and only if $P = Q$ everywhere, meaning there is no difference between the two distributions.

Lastly, we define the concept of mutual information between two variables as follows:

Definition 7. *The mutual information between two random variables X and Y is defined as:*

$$I(X; Y) = D_{KL}(P(X, Y) \| P(X)P(Y)) \tag{19}$$

Alternatively, it can be expressed in terms of entropy:

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) \tag{20}$$

Intuitively, mutual information (MI) measures the amount of information that one random variable provides about another. It quantifies how much knowing X reduces uncertainty about Y (or vice versa). Unlike KL divergence, the mutual information is symmetric. The independence properties nicely transfer to the concept of mutual properties. For example, if Z is independent of both X and Y, we have:

$$I(X;Y \mid Z) = I(X;Y) \tag{21}$$

Further, the probability chain rule appears in additive terms as follows:

$$I(X;(Y,Z)) = I(X;Y) + I(X;Z \mid Y) \tag{22}$$

The chain rule offers great convenience in settings with a large number of variables. Lastly, we present a KL divergence form of mutual information as follows:

$$I(X;Y) = \sum_{x \in \mathcal{X}} P(X=x) D_{KL}(P(Y \mid X=x) \| P(Y)) \tag{23}$$

For the proof of this equation, please refer to Russo and Van Roy [2016].

B Proofs

In this section, we present the proofs for the propositions in the main text. We start with some useful lemmas in Web Appendix B.1 that later allow us to prove the propositions in the main text. We then present a proof for the general case of a linear reward function in Web Appendix B.2. We present the proofs for propositions in the main text in Web Appendix B.3.

B.1 Useful Lemmas for a Generic Linear Reward Case

We present a general case of a reward function R, which is defined as a linear function of action characteristics through parameters β as follows:

$$R(A_j) = \beta^T A_j + \nu_j, \tag{24}$$

where A_j is an action with d-dimensional features such that $A_j \in \mathcal{A} \subset \mathbb{R}^d$ and $\beta \in \mathbb{R}^d$ is the weights and ν_j is the idiosyncratic term with known variance σ_ν. One may notice the similarity between this reward characterization and our utility specification. The reason we define this more generally is to use it as a reference in the analysis of regret in both cases. As such, we refer to reward when discussing the general case, and utility in our main models that are specific to the context of experience products. We further drop index i for brevity. One could easily add individual-specific indices for reward parameters.

Let A^* denote the first-best action, that is, $A^* = \operatorname{argmax}_{A_j \in \mathcal{A}} \mathbb{E}[R(A_j) \mid \beta]$, which allows us to define expected regret the same way as in Definition 2. Further, let A^{TS} denote the action chosen by the Thompson Sampling algorithm given the prior information set $\mathcal{H}$. A key property of the Thompson Sampling algorithm is its probability matching, which is reflected in the following equation:

$$P(A^{\text{TS}} = A_j \mid \mathcal{H}) = P(A^* = A_j \mid \mathcal{H}) \tag{25}$$

Intuitively, it means that the probability that the first-best is equal to each action is the same as the probability the algorithm chooses that action, given the information available. The feature above is why the Thompson Sampling is often referred to as the *probability matching* algorithm, and makes the application of this algorithm appealing in real settings [Chapelle and Li, 2011]. We use this feature in establishing the regret bounds as both the self-exploring and RS-dependent users in our context use different version of the Thompson Sampling algorithm.

We define an important matrix $M \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{A}|}$, where each element in this matrix is defined as follows:

$$M_{j,k} = \sqrt{P(A^* = A_j)}\sqrt{P(A^* = A_k)}\left(\mathbb{E}\left[R(A_j) \mid A^* = A_k\right] - \mathbb{E}\left[R(A_j)\right]\right), \tag{26}$$

where the two terms under the square root are the probabilities of the optimal action being equal to actions corresponding to the row and column of the matrix, and the last term is the expected reward from action A_j given the information that action A_k is first-best minus the unconditional expected reward for action A_j. Though it may not be clear why we define such a matrix, its properties allow us to obtain bounds that later help us in establishing the information-theoretic regret bounds. Before we present a few important lemmas, we stress that the matrix M is not to be confused with our utility matrix as it is only defined for product pairs. Lastly, it is worth emphasizing that the matrix M can be defined for any prior history $\mathcal{H}$. We drop that from the notation for simplicity.

We start by a simple fact for any matrix and write the following lemma:

Lemma 1. *For any matrix $Q \in \mathbb{R}^{k\times k}$, the following property holds:*

$$\textit{tr}(Q) \leq \|Q\|_F \sqrt{\textit{Rank(Q)}}, \tag{27}$$

where $\textit{tr}(Q)$ is the trace of the matrix ($\textit{tr}(Q) = \sum_{i=1}^{k} Q_{i,i}$), and $\|Q\|_F$ is the Frobenius norm of the matrix, that is, $\|Q\|_F = \sqrt{\sum_{i=1}^{k}\sum_{j=1}^{k} Q_{i,j}^2}$.

Proof. Let $\tilde{r}$ denote the rank of matrix $Q_{[k\times k]}$ where $\tilde{r}$ singular values are denoted by $\tilde{\sigma}_1, \tilde{\sigma}_2, \cdots, \tilde{\sigma}_{\tilde{r}}$. Further, let $\|Q\|_*$ denote the nuclear norm of matrix Q such that $\|Q\|_* = \sum_{i=1}^{\tilde{r}} \tilde{\sigma}_i$. On the one hand, we can write:

$$\text{tr}(Q) = \text{tr}\left(\frac{1}{2}Q + \frac{1}{2}Q^T\right) \overset{(1)}{\leq} \|\frac{1}{2}Q + \frac{1}{2}Q^T\|_* \overset{(2)}{\leq} \frac{1}{2}\|Q\|_* + \frac{1}{2}\|Q^T\|_* \overset{(3)}{=} \|Q\|_*, \tag{28}$$

where inequality (1) is a result of Von Neumann's trace inequality, (2) applies the triangle inequality, and (3) uses the fact that $\|Q\|_* = \|Q^T\|_*$. On the other hand, we can write the following inequality for the nuclear norm:

$$\|Q\|_* = \sum_{i=1}^{\tilde{r}} \tilde{\sigma}_i \overset{(1)}{\leq} \sqrt{\tilde{r}}\sqrt{\sum_{i=1}^{\tilde{r}} \tilde{\sigma}_i^2} = \sqrt{\tilde{r}}\|Q\|_F, \tag{29}$$

where (1) is an application of Cauchy-Schwarz inequality. Combining Equations (28) and (29), we arrive at the following inequality and complete the proof:

$$\text{tr}(Q) \leq \sqrt{\tilde{r}}\|Q\|_F \tag{30}$$

□

We can apply Lemma 1 to matrix M and arrive at the following inequality:

$$\text{tr}(M) \leq \sqrt{\text{Rank}(M)}\|M\|_F \tag{31}$$

We now attemp to further simplify all three pieces in the equation above: (1) trace of matrix M, (2) rank of matrix M, and (3) Frobenius norm of matrix M. We do so in the following sections:

B.1.1 Trace of Matrix M

We start with the trace of the matrix. We can write the following lemma:

Lemma 2. *For the matrix M defined in Equation* (26)*, the following equality holds:*

$$\textit{tr}(M) = \mathbb{E}\left[R\left(A^*\right)\right] - \mathbb{E}\left[R\left(A^{TS}\right)\right] \tag{32}$$

Proof. The proof of this lemma uses the probability matching feature of the Thompson Sampling algorithm that yields $P(A^{\text{TS}} = A_j) = P(A^* = A_j)$. We can write:

$$\begin{aligned}
\text{tr}(M) &= \sum_{A_j \in \mathcal{A}} P(A^* = A_j) \left(\mathbb{E}\left[R(A_j) \mid A^* = A_k\right] - \mathbb{E}\left[R(A_j)\right]\right) \\
&= \sum_{A_j \in \mathcal{A}} P(A^* = A_j) \mathbb{E}\left[R(A_j) \mid A^* = A_k\right] - \sum_{A_j \in \mathcal{A}} P(A^* = A_j) \mathbb{E}\left[R(A_j)\right] \\
&= \sum_{A_j \in \mathcal{A}} P(A^* = A_j) \mathbb{E}\left[R(A_j) \mid A^* = A_k\right] - \sum_{A_j \in \mathcal{A}} P(A^{\text{TS}} = A_j) \mathbb{E}\left[R(A_j) \mid A^{\text{TS}} = A_k\right] \\
&= \mathbb{E}\left[R\left(A^*\right)\right] - \mathbb{E}\left[R\left(A^{\text{TS}}\right)\right]
\end{aligned} \tag{33}$$

□

B.1.2 Rank of Matrix M

We now focus on the second element in Equation (31): rank of matrix M. The following lemma characterizes the upper bound for the rank of this matrix as follows:

Lemma 3. *For the matrix M defined in Equation* (26)*, the rank is bounded as follows:*

$$\textit{Rank}(M) \leq d \tag{34}$$

Proof. We can rewrite $M_{j,k}$ as follows:

$$\begin{aligned}
M_{j,k} &= \sqrt{P(A^* = A_j)} \sqrt{P(A^* = A_k} \left(\mathbb{E}\left[R(A_j) \mid A^* = A_k\right] - \mathbb{E}\left[R(A_j)\right]\right) \\
&= \sqrt{P(A^* = A_j)} \sqrt{P(A^* = A_k} \left(\mathbb{E}\left[\beta^T \mid A^* = A_k\right] A_j - \mathbb{E}\left[\beta^T\right] A_j\right) \\
&= \sqrt{P(A^* = A_j)} \sqrt{P(A^* = A_k} \left(\mathbb{E}\left[\beta^T \mid A^* = A_k\right] - \mathbb{E}\left[\beta^T\right]\right) A_j
\end{aligned} \tag{35}$$

Since both $\mathbb{E}\left[\beta^T \mid A^* = A_k\right]$ and A_j are d-dimensional, we show that the rank of matrix M is upper bounded by d. Therefore, Rank$(M) \leq d$. □

B.1.3 Frobenius Norm of Matrix M

We now focus on the third piece of the inequality in Equation (31): the Frobenius norm of matrix M. This is one of the pieces where the information-theoretic aspect of our problem appear. We first show a simple lemma about the mutual information:

Lemma 4. *For the three variables A^*, A^{TS}, and $R(A^{TS})$, the following relationship holds:*

$$I\left(A^*; \left(A^{TS}, R\left(A^{TS}\right)\right)\right) = \sum_{A_j, A_k \in \mathcal{A}} p_j^* p_k^* D_{KL}\left(P\left(R(A_j) \mid A^* = A_k\right) \| P(R(A_j)\right) \tag{36}$$

where $p_j^* = P(A^* = A_j)$ *and* $p_k^* = P(A^* = A_k)$.

Proof. We start by rewriting $I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right)$ using a few simple facts about mutual information:

$$\begin{aligned}
I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right) &= I(A^*; A^{\text{TS}}) + I(A^*; R(A^{\text{TS}}) \mid A^{\text{TS}}) \\
&= I(A^*; R(A^{\text{TS}}) \mid A^{\text{TS}}) \\
&= \sum_{A_j \in \mathcal{A}} P(A^{\text{TS}} = A_j) I(A^*; R(A^{\text{TS}}) \mid A^{\text{TS}} = A_j) \\
&= \sum_{A_j \in \mathcal{A}} P(A^{\text{TS}} = A_j) I(A^*; R(A_j)) \\
&= \sum_{A_j \in \mathcal{A}} p_j^* \left(\sum_{A_k \in \mathcal{A}} p_k^* D_{KL}\left(P\left(R(A_j) \mid A^* = A_k\right) \| P(R(A_j)\right) \right) \\
&= \sum_{A_j, A_k \in \mathcal{A}} p_j^* p_k^* D_{KL}\left(P\left(R(A_j) \mid A^* = A_k\right) \| P(R(A_j)\right),
\end{aligned} \tag{37}$$

where the first line is the application of chain rule, the second line uses the fact that A^* is independent of A^{TS}, the third line just expands, the fourth line uses the independence of A^{TS} from both A^* and $R(A^{\text{TS}})$, and the fifth line applies Equation (23). $\square$

The right-hand side of Equation (37) presents the mutual information in terms of KL divergence, which can be further simplified using Pinsker's inequality through the following lemma:

Lemma 5. *Suppose that the error term in the reward function in Equation* (24) *comes from* $N(0, \sigma_\nu^2)$. *The following inequality holds:*

$$D_{KL}\left(P\left(R(A_j) \mid A^* = A_k\right) \| P(R(A_j)\right) \geq \frac{\left[\left(\mathbb{E}\left[R(A_j) \mid A^* = A_k\right] - \mathbb{E}\left[R(A_j)\right]\right)\right]^2}{2\sigma_\nu^2} \tag{38}$$

Proof. Please see the proof for Fact 9 and Lemma 3 in Russo and Van Roy [2016]. $\square$

We can now combine Lemma 5 with Lemma 4 to write the following lemma about the Frenobius norm of matrix M:

Lemma 6. *The following inequality holds for the Frobenius norm of matrix* M *in Equation* (26)*:*

$$\|M\|_F \leq 2\sigma_\nu^2 I\left(A^*; \left(A^{TS}, R\left(A^{TS}\right)\right)\right) \tag{39}$$

Proof. We can expand the Frobenius norm of matrix M as follows:

$$\begin{aligned}\|M\|_F &= \sum_{A_j, A_k \in \mathcal{A}} P(A^* = A_j) P(A^* = A_k) \left(\mathbb{E}\left[R(A_j) \mid A^* = A_k\right] - \mathbb{E}\left[R(A_j)\right]\right)^2 \\ &\leq \sum_{A_j, A_k \in \mathcal{A}} P(A^* = A_j) P(A^* = A_k)(2\sigma_\nu^2) D_{KL}\left(P\left(R(A_j) \mid A^* = A_k\right) \| P(R(A_j)\right) \\ &= 2\sigma_\nu^2 I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right),\end{aligned} \tag{40}$$

where the inequality in the second line comes from Lemma 5. □

B.1.4 Revisiting the Inequality for Matrix M

We now revisit the inequality for matrix M in Equation (31), using all the lemmas above that help us simplify a relationship the mutual information and the per-period regret. In particular, Lemma 2 connects the per-period regret term to the trace of matrix M, and Lemma 6 connects the mutual information to the Frobenius norm of matrix M. We can write the following lemma for the relationship between the mutual information and the regret:

Lemma 7. *The ratio of the squared per-period regret to the mutual information between A^* and A^{TS} has the following upper bound:*

$$\frac{\left[\mathbb{E}\left[R\left(A^*\right)\right] - \mathbb{E}\left[R\left(A^{TS}\right)\right]\right]^2}{I\left(A^*; \left(A^{TS}, R\left(A^{TS}\right)\right)\right)} \leq 2\sigma_\nu^2 d \tag{41}$$

Proof. We can write:

$$\begin{aligned}\frac{\left[\mathbb{E}\left[R\left(A^*\right)\right] - \mathbb{E}\left[R\left(A^{\text{TS}}\right)\right]\right]^2}{I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right)} &= \frac{\text{tr}(M)}{I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right)} \\ &\leq \frac{\text{Rank}(M)\|M\|_F}{I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right)} \\ &\leq \frac{d 2\sigma_\nu^2 I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right)}{I\left(A^*; \left(A^{\text{TS}}, R\left(A^{\text{TS}}\right)\right)\right)} \\ &= 2\sigma_\nu^2 d,\end{aligned} \tag{42}$$

where the first line applies Lemma 2, the second line applies the matrix inequality in Equation (31), and the third line applies Lemma 6. □

B.2 Regret Bounds for the General Case of Linear Reward

We now establish the regret bound for the general case with linear rewards and d-dimensional action space as follows:

Lemma 8. *Suppose that the reward function is linear in action features as in Equation* (24)*, where the error term comes from $N(0, \sigma_\nu^2)$ and the algorithm has access to the information about all* *features when making the decision. Further, assume that $H(A_0^*)$ is the Shannon entropy of the optimal action given the prior. The following regret bound holds for the Thompson Sampling algorithm with policy π^{TS} for any T:*

$$\texttt{Regret}(T, \pi^{TS}) = \sqrt{2\sigma_\nu^2 d H(A_0^*) T} \tag{43}$$

Proof. The proof uses the bound established in Lemma 7 and other inequalities as follows:

$$\begin{aligned}
\texttt{Regret}(T, \pi^{\mathrm{TS}}) &= \mathbb{E}\left[\sum_{t=0}^{T}\left(R\left(A^*\right) - R\left(A_t^{\mathrm{TS}}\right)\right)\right] \\
&= \mathbb{E}\left[\sum_{t=0}^{T}\mathbb{E}\left[\left(R\left(A^*\right) - R\left(A_t^{\mathrm{TS}}\right)\right) \mid \mathcal{H}_t\right]\right] \\
&= \mathbb{E}\left[\sum_{t=0}^{T}\frac{\mathbb{E}\left[\left(R\left(A^*\right) - R\left(A_t^{\mathrm{TS}}\right)\right) \mid \mathcal{H}_t\right]}{\sqrt{I\left(A^*; \left(A_t^{\mathrm{TS}}, R\left(A_t^{\mathrm{TS}}\right)\right) \mid \mathcal{H}_t\right)}}\sqrt{I\left(A^*; \left(A_t^{\mathrm{TS}}, R\left(A_t^{\mathrm{TS}}\right)\right) \mid \mathcal{H}_t\right)}\right] \\
&\leq \sqrt{2\sigma_\nu^2 d}\ \mathbb{E}\left[\sum_{t=0}^{T}\sqrt{I\left(A^*; \left(A_t^{\mathrm{TS}}, R\left(A_t^{\mathrm{TS}}\right)\right) \mid \mathcal{H}_t\right)}\right] \\
&\leq \sqrt{2\sigma_\nu^2 d}\ \sqrt{T\mathbb{E}\left[\sum_{t=0}^{T} I\left(A^*; \left(A_t^{\mathrm{TS}}, R\left(A_t^{\mathrm{TS}}\right)\right) \mid \mathcal{H}_t\right)\right]} \\
&= \sqrt{2\sigma_\nu^2 d}\ \sqrt{T\left[\sum_{t=0}^{T} I\left(A^*; \left(A_t^{\mathrm{TS}}, R\left(A_t^{\mathrm{TS}}\right)\right) \mid \{A_\tau^{\mathrm{TS}}, R\left(A_\tau^{\mathrm{TS}}\right)\}_{\tau=0}^{t-1}\right)\right)\right]} \\
&= \sqrt{2\sigma_\nu^2 d}\ \sqrt{TI\left(A^*; \{A_\tau^{\mathrm{TS}}, R\left(A_\tau^{\mathrm{TS}}\right)\}_{\tau=0}^{t}\right)} \\
&= \sqrt{2\sigma_\nu^2 d}\ \sqrt{T\left[H\left(A^*\right) - H\left(A^* \mid \{A_\tau^{\mathrm{TS}}, R\left(A_\tau^{\mathrm{TS}}\right)\}_{\tau=0}^{t}\right)\right]} \\
&\leq \sqrt{2\sigma_\nu^2 d}\ \sqrt{TH\left(A^*\right)},
\end{aligned} \tag{44}$$

where the first line simply writes the equation for regret, the second line uses the law of iterated expectation to account for per-period information $\mathcal{H}_t$ that is available, the third line simply includes a mutual information term in the numerator and denominator that is shown to be positive in Lemma 6, the fourth line uses Lemma 7, the fifth line applies the Cauchy-Schwarz inequality to the summation, the sixth line expands the mutual information term, the seventh line applies the chain rule for mutual information, the eighth line rewrites the mutual information in terms of Shannon entropy, and the ninth line uses the fact that the Shannon entropy is always non-negative. □

The lemma above, along with the ones earlier, all heavily borrow from Russo and Van Roy [2016]. We present them here in the web appendix of our paper, so the reader can refer to the proofs here. In light of Lemma 8, we can prove the propositions in our main text.

B.3 Proof for Propositions

B.3.1 Proof for Proposition 1

Proof. The proof for Proposition 1 follows the decomposition presented in the main text of the paper. We can first expand that decomposition as follows:

$$\begin{aligned}
\texttt{Regret}_i(T;\pi) &= \mathbb{E}\left[\sum_{t=0}^{T}\left(u_i(A_i^*) - u_i(A_{i,t})\right)\right] \\
&= \mathbb{E}\left[\sum_{t=0}^{T}\left(u_i(A_i^*) - u_i(A_i^{*,s})\right)\right] + \mathbb{E}\left[\sum_{t=0}^{T}\left(u_i(A_i^{*,s}) - u_i(A_{i,t})\right)\right] \\
&= gT + \mathbb{E}\left[\sum_{t=0}^{T}\left(u_i(A_i^{*,s}) - u_i(A_{i,t})\right)\right],
\end{aligned} \tag{45}$$

where $A_i^{*,s}$ is the first-best product given the search features. We apply Lemma 8 to the second term. We use the fact that the utility function by the self-exploring user can is a reward function with s-dimensional parameters, where the variation of the utility from experience features goes into the error term. We denote the variance of this error term by σ^2_{SE}. As such, applying Lemma 8 gives us the upper bound of $\sqrt{2\sigma^2_{\text{SE}} sH(A_0^*)T}$. We further note that the variance σ^2_{SE} is upper bounded as follows:

$$\sigma^2_{\text{SE}} \leq \sigma^2_{\epsilon} + \sigma^2_{x}, \tag{46}$$

where the equality holds if there is no mutual information between the experience and search features. This completes the proof. □

B.3.2 Proof for Proposition 2

Proof. This proof uses a simple lemma that finds upper and lower bounds for the expected maximum of k draws from a Normally distributed variable. We state the lemma as follows:

Lemma 9. *Let $X_1, X_2, \ldots, X_k \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d. normal random variables with mean μ and standard deviation σ. Define the maximum:*

$$M_k = \max(X_1, X_2, \ldots, X_k).$$

Then, the expectation $\mathbb{E}[M_k]$ satisfies the following bounds for any $k \geq 2$:

$$\mu + \sigma\left(\sqrt{2\log k} - \frac{\log\log k + \log(4\pi)}{\sqrt{8\log k}}\right) \leq \mathbb{E}[M_k] \leq \mu + \sigma\left(\sqrt{2\log k}\right) \tag{47}$$

Proof. Let $X_1, X_2, \ldots, X_k \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d. normal random variables. Define the standardized variables:

$$Z_i = \frac{X_i - \mu}{\sigma}, \quad \text{so that } Z_i \sim \mathcal{N}(0, 1).$$

Let $M_k = \max(X_1, X_2, \ldots, X_k)$ and define the corresponding standardized maximum:

$$Z_{(k)} = \max(Z_1, Z_2, \ldots, Z_k).$$

To find the upper bound, we apply a series of inequalities and definitions for some $t > 0$ as follows:

$$\begin{aligned} \exp\left(t\mathbb{E}[Z_{(k)}]\right) &\leq \mathbb{E}\left[\exp\left(tZ_{(k)}\right)\right] \\ &= \mathbb{E}\left[\max_i \exp\left(tZ_i\right)\right] \\ &\leq \sum_{i=1}^{k} \mathbb{E}\left[\exp\left(tZ_i\right)\right] \\ &= k\exp\left(\frac{t^2}{2}\right), \end{aligned} \tag{48}$$

where the first line applies the Jensen's inequality, the second line uses the definition of $Z_{(k)}$, the third line applies the union bound, and the fourth line applies the moment generating function. Taking logs from both sides of Equation (48), we have $t\mathbb{E}[Z_{(k)}] \leq \log(k) + t^2/2$, which we can simplify as follows:

$$\mathbb{E}[Z_{(k)}] \leq \frac{\log(k)}{t} + \frac{t}{2} \tag{49}$$

To find a tight upper bound, we choose t that minimizes the RHS of Equation (49), which is $t = \sqrt{2\log(k)}$. This gives us the following upper bound:

$$\mathbb{E}[Z_{(k)}] \leq \sqrt{2\log(k)}. \tag{50}$$

To obtain the lower bound, we seek to approximate $\mathbb{E}[Z_{(k)}]$, which can be related to extreme value approximations. The cumulative distribution function (CDF) of $Z_{(k)}$ is given by:

$$F_{Z_{(k)}}(z) = P(Z_{(k)} \leq z) = P(Z_1 \leq z, \ldots, Z_k \leq z) = \left(\Phi(z)\right)^k.$$

The probability density function (PDF) is then:

$$f_{Z_{(k)}}(z) = k\left(\Phi(z)\right)^{k-1}\phi(z),$$

where $\Phi(z)$ and $\phi(z)$ denote the standard normal CDF and PDF, respectively. A well-known result in extreme value theory provides an approximation for $\mathbb{E}[Z_{(k)}]$:

$$\mathbb{E}[Z_{(k)}] \geq \sqrt{2\log(k)} - \frac{\log(\log(k)) + \log(4\pi)}{\sqrt{8\log(k)}}. \tag{51}$$

Combining Equations (50) and (51), we get the following result:

$$\sqrt{2\log(k)} - \frac{\log(\log(k)) + \log(4\pi)}{\sqrt{8\log(k)}} \leq \mathbb{E}[Z_{(k)}] \leq \sqrt{2\log k}, \quad \text{for } k \geq 2.$$

By scaling back to the original normal variables:

$$\mathbb{E}[M_k] = \mu + \sigma\mathbb{E}[Z_{(k)}],$$

we obtain the desired bounds:

$$\mu + \sigma\left(\sqrt{2\log(k)} - \frac{\log(\log(k)) + \log(4\pi)}{\sqrt{8\log(k)}}\right) \leq \mathbb{E}[M_k] \leq \mu + \sigma\left(\sqrt{2\log k}\right).$$

□

We now use this lemma for the expected maximum when it comes from $N(\mu_{i,s}+\mu_{i,x}, \sigma_{i,s}^2+\sigma_{i,x}^2)$ (actual first-best) and when it comes from $N(\mu_{i,s}, \sigma_{i,s}^2)$ (search first-best). We first apply the lower bound in Lemma 9 to get the following result:

$$\begin{aligned}\mathbb{E}\left[u_i(A_i^*)\right] &\geq (\mu_{i,x}+\mu_{i,s}) + \sqrt{\sigma_{i,x}^2+\sigma_{i,s}^2}\left(\sqrt{2\log(|\mathcal{A}|)} - \frac{\log(\log(|\mathcal{A}|)) + \log(4\pi)}{\sqrt{8\log(|\mathcal{A}|)}}\right) \\ &= (\mu_{i,x}+\mu_{i,s}) + \sqrt{2\log(|\mathcal{A}|)}\sqrt{\sigma_{i,x}^2+\sigma_{i,s}^2}\left(1 - \frac{\log(\log(|\mathcal{A}|)) + \log(4\pi)}{4\log(|\mathcal{A}|)}\right)\end{aligned} \tag{52}$$

We now apply the upper bound in Lemma 9 to the utility from the search first-best, which results in the following result:

$$\begin{aligned}\mathbb{E}\left[u_i(A_i^*)\right] &= \mathbb{E}\left[\sum_{k=1}^{s}\theta_{i,k}A_{k,i}^*\right] + \mathbb{E}\left[\sum_{l=s+1}^{d}\theta_{i,l}A_{l,i}^*\right] \\ &= \mathbb{E}[\max_j U_{i,j}^s] + \mu_{i,x} \\ &\leq \mu_{i,s} + \sigma_{i,s}\left(\sqrt{2\log(|\mathcal{A}|)}\right) + \mu_{i,x},\end{aligned} \tag{53}$$

where the first equality only splits the search and experience features, the second line uses the independence between search and experience utility across products, and the third line applies Lemma 9.

Combining Equations (52) and (53), we get the following result and complete the proof:

$$\mathbb{E}\left[u_i(A_i^*) - u_i(A_i^{*,s})\right] \geq \sqrt{2\log(|\mathcal{A}|)}\left(\left(\sqrt{\sigma_{i,s}^2+\sigma_{i,x}^2} - \sigma_{i,s}\right) - O\left(\frac{\sqrt{\sigma_{i,s}^2+\sigma_{i,x}^2}}{\log(|\mathcal{A}|)}\right)\right) \tag{54}$$

□

B.3.3 Proof for Proposition 3

Proof. The proof for Proposition 3 directly applies Lemma 8 to the case of an r-dimensional action space. In this case, the variance of the error term for the reward function is σ_ϵ^2. Therefore, the upper bound for regret can be written as follows:

$$\texttt{Regret}(T; \pi_{\text{RS}}) \leq \sqrt{2\sigma_\epsilon^2 r H(A_{i,0}^{\text{RS}}) T},$$

□

C Summary of Theoretical Results

To synthesize the results in Section 4, Table A1 summarizes all the theoretical findings along with the assumption they require and their implications. Before presenting the table, we present a general assumption based on the modeling framework that all our theoretical results require as follows:

Assumption A1 (General)**.** *The general modeling framework has the following conditions:*

1. *Users have a linear utility over the product features with an i.i.d error term with known variance.*

2. *Product features are search and experience features, experience features are not known ex ante, and the search and experience utilities are independent across products.*

Table A1: Summary of Theoretical Results

Quantity	**Bound**	**Assumptions**	**Implications**
Self-Exploring User's Regret (Proposition 1)	LB: gT UB: $gT + \sqrt{2(\sigma_\epsilon^2 + \sigma_x^2)\, s\, H(A_{i,0}^*)\, T}$	A1 for general statement; 2 needed if RS is present	Both bounds linear in T; the LB is attained when the user has no preference uncertainty, i.e., $H(A_{i,0}^*) = 0$.
Expected Gap (Proposition 2)	LB: $\sqrt{2\log\lvert\mathcal{A}\rvert}\big(\sqrt{\sigma_{i,s}^2 + \sigma_{i,x}^2} - \sigma_{i,s}\big)$ (plus a lower-order $O(\cdot)$ term)	A1 + Normality of $U_{i,j}^s, U_{i,j}^x$; 2 needed if RS is present	Gap grows with experience-feature variance $\sigma_{i,x}^2$ and with $\lvert\mathcal{A}\rvert$; the $O(\cdot)$ term vanishes as $\lvert\mathcal{A}\rvert$ grows.
RS-Dependent User's Regret (Proposition 3)	UB: $\sqrt{2\sigma_\epsilon^2\, r\, H(A_{i,0}^{RS})\, T}$	A1, 1, and 2	Bound depends on rank r, not on d or s; Cold-start is the main challenge for the RS captured by $H(A_{i,0}^{RS})$.
Welfare Gain from RS (Corollary 1)	LB: $gT - \sqrt{2\sigma_\epsilon^2\, r\, H(A_{i,0}^{RS})\, T}$	A1, 1, and 2	Linear gain in T dominates the sub-linear loss; welfare gain grows linearly over time. Independent of the relationship between r and s.

Notes. "LB" and "UB" denote lower and upper bounds, respectively. "General" refers to the baseline modeling assumptions stated in Assumption A1. $H(\cdot)$: Shannon entropy of the prior over the optimal action; s: number of search features; r: rank of the utility matrix; σ_ϵ^2: noise variance; σ_x^2: experience variance; g: expected utility gap between global first-best and search-only first-best.

D Details of the Empirical Framework

D.1 Details on the Movie Attributes

Table A2 presents the top tags in the MovieLens dataset.

Table A2: Top 100 Movie Lens Data Tags

Top 1-34	Top 35-68	Top 69-100
original	pg-13	unlikely friendships
mentor	cinematography	passionate
great ending	redemption	very interesting
catastrophe	light	dramatic
dialogue	intense	relationships
good	family	so bad it's funny
great	corruption	independent film
chase	not funny	murder
runaway	unusual plot structure	sexy
good soundtrack	twists & turns	drinking
storytelling	entirely dialogue	childhood
vengeance	suprisingly clever	complex
story	pornography	creativity
weird	transformation	lone hero
drama	cult film	atmospheric
greed	adapted from:book	based on book
great acting	happy ending	first contact
imdb top 250	very funny	entertaining
culture clash	death	narrated
brutality	life & death	friendship
fun movie	social commentary	obsession
adaptation	stylized	based on a book
criterion	interesting	loneliness
life philosophy	enigmatic	sexualized violence
suspense	fight scenes	oscar (best supporting actress)
melancholic	harsh	very good
predictable	police investigation	gunfight
visually appealing	revenge	stereotypes
talky	justice	underrated
great movie	quirky	secrets
oscar (best directing)	excellent script	nudity (full frontal - brief)
clever	feel-good	
destiny	gangsters	
fantasy world	violence	

D.2 Details of Self-Exploring and RS-Dependent Algorithms in Empirical Analysis

Suppose user i in the data has rated M different movies. We define the set of movies rated by this user as $\mathcal{A}_i = \{A_{i,m}\}_{m=1}^{M}$, where each element corresponds to a movie rated by user i characterized by its features $A_{i,m} \in \mathbb{R}^{51}$. Corresponding to this set, we have the data of actual ratings by the user denoted by $\mathcal{Y}_i = \{Y_i(a)\}_{a \in \mathcal{A}_i}$. We select 20 of these movies at random and form a hold-out test set $\mathcal{A}_i^{\text{test}}$, which constructs our training set $\mathcal{A}_i^{\text{train}} = \mathcal{A}_i \setminus \mathcal{A}_i^{\text{test}}$. We now present the algorithm for simulating the results for the self-exploring user in our empirical analysis as follows:

Algorithm 4 Choice and Learning for the Self-Exploring User in Empirical Analysis

Input: $\mu_{i,0} = e_1 \in \mathbb{R}^{51}, \Sigma_{i,0} = I_{[51\times 51]}, \mathcal{A}_i^{\text{train}}, \mathcal{A}_i^{\text{test}}, \mathcal{Y}_i, \mathcal{T} = 400, \hat{\sigma}_\epsilon = 0.86$
Output: $\mathcal{H}_{i,\mathcal{T}}^{\text{train}}, \mathcal{H}_{i,\mathcal{T}}^{\text{test}}, \{\mu_{i,t}, \Sigma_{i,t}\}_{t=1}^{\mathcal{T}}$

```
1:  A^train_{i,0} ← A^train_i
2:  for t = 0 → T do
3:      θ̃_{i,t} ~ N(μ_{i,t}, Σ_{i,t})                          ▷ Sampling preference weights
4:      A_{i,t} ← argmax_{A_j ∈ A^train_{i,t}} θ̃^T_{i,t} A_j     ▷ Selecting action based on 51 search features
5:      U_{i,t} ← Y_i(A_{i,t})                                  ▷ Entering the observed rating as the realized utility
6:      A^test_{i,t} ← argmax_{A_j ∈ A^test_i} θ̃^T_{i,t} A_j     ▷ Selecting action in the hold-out set
7:      U^test_{i,t} ← Y_i(A^test_{i,t})                        ▷ Realizing the utility from hold-out choice
8:      Σ_{i,t+1} ← (Σ^{-1}_{i,t} + (1/σ̂²_ε) A_{i,t} A^T_{i,t})^{-1}   ▷ Updating posterior variance
9:      μ_{i,t+1} ← Σ_{i,t+1} (Σ^{-1}_{i,t} μ_{i,t} + (1/σ̂²_ε) A_{i,t} U_{i,t})   ▷ Updating posterior mean
10:     A^train_{i,t+1} = A^train_{i,t} \ A_{i,t}
11: end for
```

Since the output returns $\mathcal{H}_{i,\mathcal{T}}^{\text{test}}$ and the distribution of preference parameters at each time period, we can measure regret and other metrics for self-exploring users.

To present the algorithm for RS-dependent users, the only step that changes is the action selection. Let $\texttt{RS}_t(\cdot)$ denote the personalized recommendation system function that applies to a set of movies and recommends one. Unlike Algorithm 3 in our theoretical analysis, here we do not impose any structure on the algorithm to necessarily follow Thompson Sampling, in order to allow benchmarking the performance of any state-of-the-art personalized algorithm. In our main text, we follow Cortes [2018] for that purpose, but later in Web Appendix G, we consider a host of alternative RS algorithms.

Algorithm 5 presents the procedure for simulating the results for the RS-dependent user in our empirical analysis as follows. As shown in this algorithm, the output allows us to quantify regret and other metrics of interest.

D.3 Alternative Performance Metrics

In §5.2, we use regret as our main performance metric. In this subsection, we report two additional ranking-based measures that evaluate the quality of broader recommendation lists rather than only the single chosen item. The first is *F1-Score@10*, which compares the top 10 recommendations from the model with the actual top 10 items in the hold-out set [Chang et al., 2015]. The second is *Normalized Discounted Cumulative Gain (nDCG@10)*, which evaluates the quality of the ranked top-10 list by placing greater weight on correctly ranking the very best items near the top [Järvelin and Kekäläinen, 2002]. Both measures are standard in the recommendation-systems literature [Koren et al., 2021].

Algorithm 5 Choice and Learning for the RS-Dependent User in Empirical Analysis

Input: $\mu_{i,0} = e_1 \in \mathbb{R}^{51}, \Sigma_{i,0} = I_{[51\times 51]}, \mathcal{A}_i^{\text{train}}, \mathcal{A}_i^{\text{test}}, \mathcal{Y}_i, \mathcal{T} = 400, \hat{\sigma}_\epsilon = 0.86, \texttt{RS}_0(\cdot)$
Output: $\mathcal{H}_{i,\mathcal{T}}^{\text{train}}, \mathcal{H}_{i,\mathcal{T}}^{\text{test}}, \{\mu_{i,t}, \Sigma_{i,t}\}_{t=1}^{\mathcal{T}}$

1: $\mathcal{A}_{i,0}^{\text{train}} \leftarrow \mathcal{A}_i^{\text{train}}$
2: **for** $t = 0 \to \mathcal{T}$ **do**
3: $\quad \tilde{\theta}_{i,t} \sim N(\mu_{i,t}, \Sigma_{i,t})$ ▷ Sampling preference weights
4: $\quad A_{i,t} \leftarrow \texttt{RS}_t(\mathcal{A}_{i,t}^{\text{train}})$ ▷ Selecting action based on the RS recommendation
5: $\quad U_{i,t} \leftarrow Y_i(A_{i,t})$ ▷ Entering the observed rating as the realized utility
6: $\quad A_{i,t}^{\text{test}} \leftarrow \texttt{RS}_t(\mathcal{A}_{i,t}^{\text{test}})$ ▷ Realizing the utility from hold-out choice
7: $\quad U_{i,t}^{\text{test}} \leftarrow Y_i(A_{i,t}^{\text{test}})$ ▷ Same as line 5 in the hold-out set
8: $\quad \Sigma_{i,t+1} \leftarrow \left(\Sigma_{i,t}^{-1} + \frac{1}{\hat{\sigma}_\epsilon^2} A_{i,t} A_{i,t}^T\right)^{-1}$ ▷ Updating posterior variance
9: $\quad \mu_{i,t+1} \leftarrow \Sigma_{i,t+1}\left(\Sigma_{i,t}^{-1}\mu_{i,t} + \frac{1}{\hat{\sigma}_\epsilon^2} A_{i,t} U_{i,t}\right)$ ▷ Updating posterior mean
10: $\quad \mathcal{A}_{i,t+1}^{\text{train}} = \mathcal{A}_{i,t}^{\text{train}} \setminus A_{i,t}$ ▷ Updating the set of remaining movies
11: $\quad \texttt{RS}_{t+1} = g(\texttt{RS}_t, A_{i,t}, U_{i,t})$ ▷ Updating the RS
12: **end for**

Figure A1 shows that the personalized recommender system continues to outperform the main user-side benchmarks under both broader ranking metrics. In panel (a), the RS achieves the highest F1-Score@10, with an average of about 0.716, compared with about 0.689 for the self-exploring user, 0.703 for the User with Known Preference, and 0.667 for the Aggregate Rating benchmark. In panel (b), the same ordering largely holds for nDCG@10: the RS again performs best, with an average of about 0.936, compared with about 0.917 for the self-exploring user, 0.921 for the User with Known Preference, and 0.902 for Aggregate Rating. These results confirm that the recommender system's information advantage extends beyond regret and remains strong when performance is evaluated using the full top-10 ranking.

At the same time, the dynamics of the learning user differ somewhat across the two ranking metrics. In F1-Score@10, the self-exploring user improves steadily over time but remains below the User with Known Preference benchmark throughout most of the horizon. This suggests that, although sequential learning improves the user's ability to place relevant items into the top 10, it is still difficult to match the static benchmark in terms of recovering the full top-10 set. In nDCG@10, however, the gap is narrower, and the self-exploring user eventually approaches and slightly exceeds the User with Known Preference line at long horizons. This indicates that dynamic Bayesian updating can be especially useful for refining the relative ordering of highly valued items near the top of the list, even if it does not fully recover the same set of top-10 items on average.

Overall, the evidence from F1-Score@10 and nDCG@10 reinforces the main message of the paper. Users who learn sequentially can substantially improve ranking quality over time, and in some dimensions they can nearly match or even locally surpass a static known-preference benchmark. Nevertheless, the personalized recommender system remains clearly dominant throughout, implying that its advantage comes not only from selecting a single good recommendation, but also from maintaining a superior ordering of multiple high-value items.

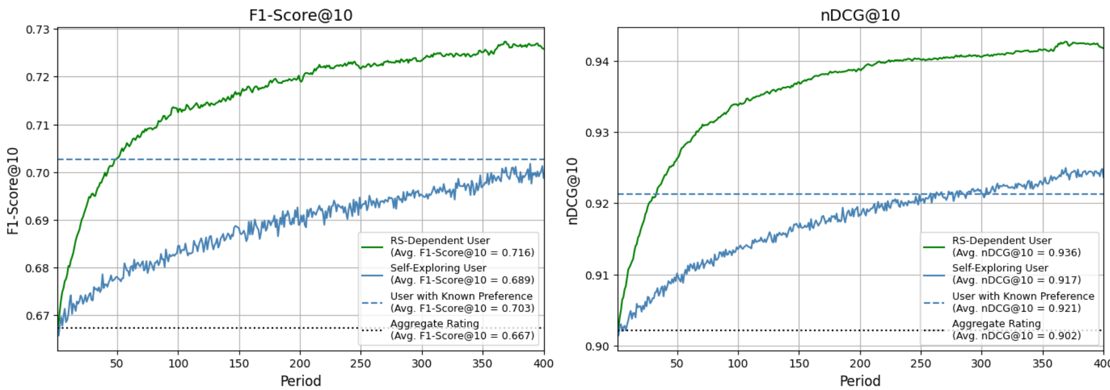


Figure A1: Performance of different algorithms using F1-Score and nDCG.

D.4 Heterogeneity in Welfare Gains from RS Based on Nicheness of User Preferences

The main analysis reports average welfare gains from the recommender system across users. In this section, we examine how these gains vary with the nicheness of user preferences. This heterogeneity is important because the RS may be especially valuable for users whose preferences are difficult to infer from aggregate popularity signals. At the same time, extremely niche users may also be difficult for the RS itself to predict if there are few similar users in the data.

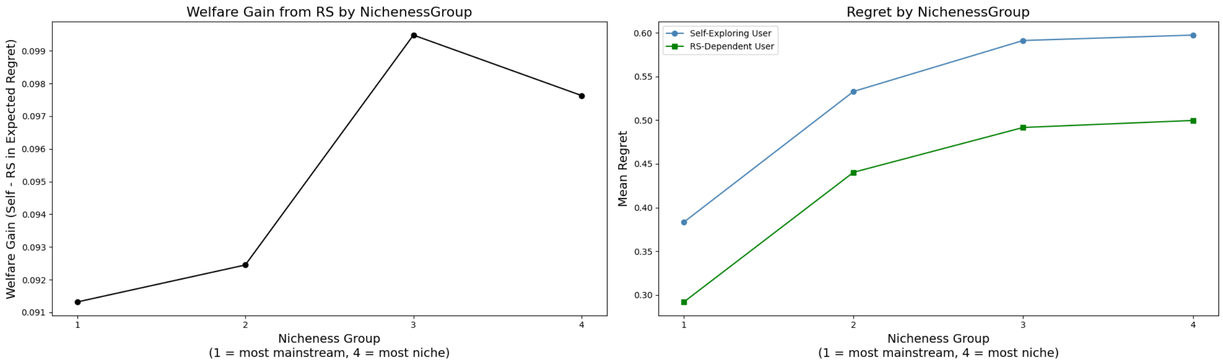
We measure user i's nicheness by the correlation between the aggregate rating of movie A_j in the training sample and user i's own rating of that movie. Intuitively, if user i has mainstream preferences, the aggregate rating is a good proxy for user i's evaluation, and this correlation is high. If user i has niche preferences, the aggregate rating is a poor predictor of user i's evaluation, and the correlation is low. We then divide users into four nicheness groups based on this measure. Group 1 contains the most mainstream users, while Group 4 contains the most niche users.

For each group, we compare the regret of the RS with the regret of the *Self-Exploring User* benchmark. We define the welfare gain from the RS as the reduction in regret relative to the *Self-Exploring User* benchmark:

$$\text{Welfare Gain} = \text{Regret}^{\text{SE}} - \text{Regret}^{\text{RS}} .$$

This measure is positive when the RS lowers regret relative to self-directed learning.

Figure A2 reports both the welfare gains and the underlying regret levels across nicheness groups. Regret increases as users become more niche for both the RS and the *Self-Exploring User*: mean regret rises from about 0.38 in Group 1 to about 0.60 in Group 4 for the Self-Exploring User, and from about 0.29 to about 0.50 for the RS. Thus, niche users are harder to predict for both approaches. The RS welfare gain remains positive for all groups, but it is not strictly monotone: it is about 0.091 in Group 1, 0.092 in Group 2, 0.099 in Group 3, and 0.098 in Group 4. This pattern suggests two opposing forces. As users become more niche, aggregate ratings become less informative and the value of personalization rises; however, for the most niche users, collaborative information also becomes less reliable because there are fewer similar users in the data. Hence, recommender systems provide broad benefits across the preference distribution, but the largest marginal gains arise for users with intermediate-to-high nicheness.



(a) Welfare gain.

(b) Regret.

Figure A2: Welfare gain and regret from the RS and the *Self-Exploring User* benchmark across nicheness groups.

D.5 User's Prior Information Compared to RS

In §5.3, we assume both user types start from a prior that assigns weight one to the aggregate rating and zero weight to all other features. Although this aggregate prior is already relatively informative, users may have even more informed priors. To study this, we construct priors from the posterior distribution of the self–exploring user in §5.3 after she has watched 20, 40, 60, or 80 movies. We then use each of these posteriors as the starting prior for both the self–exploring and the RS–dependent user and simulate their subsequent learning.

Figure A3 plots the resulting KL divergence over time for each prior length (four panels corresponding to priors at periods 20, 40, 60, and 80). The KL divergence is always computed relative to the diffuse prior used in §5.3 (prior mean one for the aggregate rating, prior mean zero for all other parameters, and prior variance one for all parameters). Across all four prior settings, the self-exploring user consistently attains a higher KL divergence than the RS–dependent user at a given period, indicating more learning for the self-exploring user. As the prior becomes more informed (longer prior learning phase), both users start further from the diffuse prior and the incremental growth in KL over subsequent periods becomes smaller, reflecting that an already well–informed prior leaves less room for additional learning.

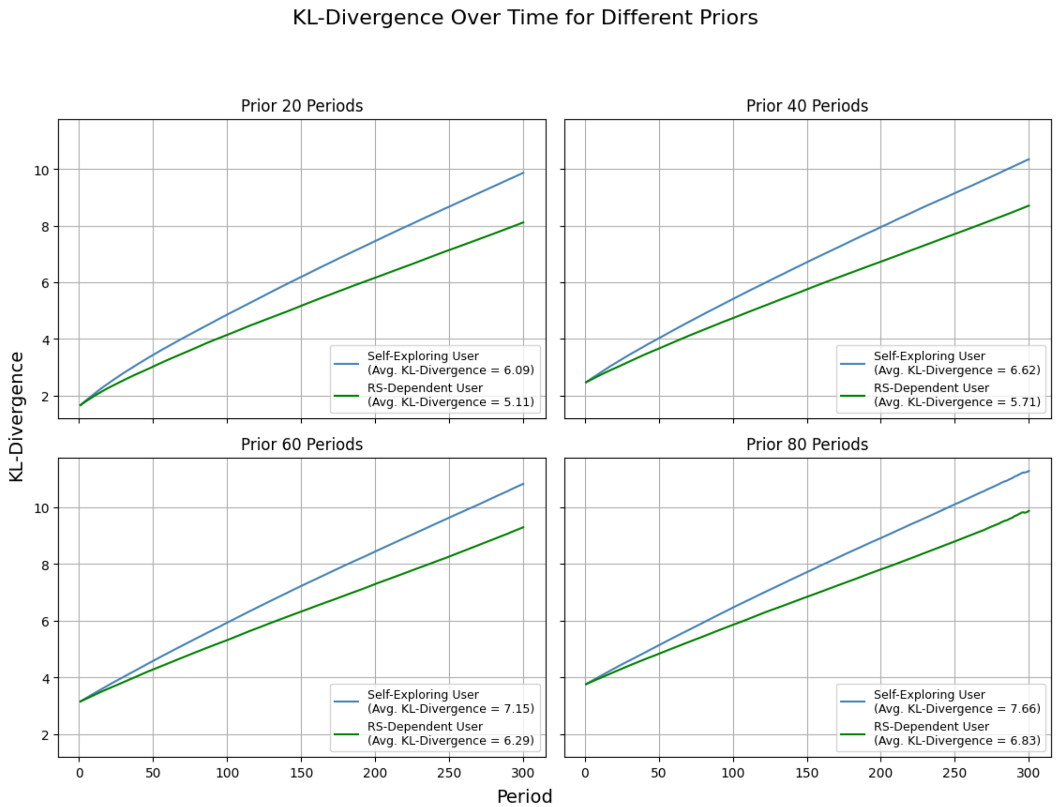


Figure A3: User Learning under Different Algorithms (with User's Prior Information)

E Hybrid Users

E.1 Algorithm for Hybrid Users

In Section 5.4.1, we propose hybrid policies with random use of the RS through a parameter p that determines the probability that the RS is not followed. We show the detailed pseudo-code in Algorithm 6. Most parts of this algorithm are similar to Algorithm 3. The main difference is that the algorithmic recommendation is only followed with a probability p.

Algorithm 6 Choice and Learning for the Hybrid User

Input: $\mu_{i,0} = e_1 \in \mathbb{R}^{51}, \Sigma_{i,0} = I_{[51\times 51]}, \mathcal{A}_i^{\text{train}}, \mathcal{A}_i^{\text{test}}, \mathcal{Y}_i, \mathcal{T} = 400, \hat{\sigma}_\epsilon = 0.86, \mathtt{RS}_0(\cdot)$
Output: $\mathcal{H}_{i,\mathcal{T}}^{\text{train}}, \mathcal{H}_{i,\mathcal{T}}^{\text{test}}, \{\mu_{i,t}, \Sigma_{i,t}\}_{t=1}^{\mathcal{T}}$

```
$\mathcal{A}_{i,0}^{\text{train}} \leftarrow \mathcal{A}_i^{\text{train}}$
for $t = 0 \to \mathcal{T}$ do
    $\xi_{i,t} \sim \text{Bernoulli}(p)$                                          ▷ Determine RS Availability
    if $\xi_{i,t} = 1$ then
        $\tilde{\theta}_{i,t} \sim N(\mu_{i,t}, \Sigma_{i,t})$                          ▷ Sampling preference weights
        $A_{i,t} \leftarrow \text{argmax}_{A_j \in \mathcal{A}_{i,t}^{\text{train}}} \tilde{\theta}_{i,t}^T A_j$      ▷ Selecting action based on 51 search features
        $U_{i,t} \leftarrow Y_i(A_{i,t})$                                    ▷ Entering the observed rating as the realized utility
        $A_{i,t}^{\text{test}} \leftarrow \text{argmax}_{A_j \in \mathcal{A}_i^{\text{test}}} \tilde{\theta}_{i,t}^T A_j$       ▷ Selecting action in the hold-out set
        $U_{i,t}^{\text{test}} \leftarrow Y_i(A_{i,t}^{\text{test}})$                          ▷ Realizing the utility from hold-out choice
    else
        $A_{i,t} \leftarrow \mathtt{RS}_t(\mathcal{A}_{i,t}^{\text{train}})$                     ▷ Selecting action based on the RS recommendation
        $U_{i,t} \leftarrow Y_i(A_{i,t})$                                    ▷ Entering the observed rating as the realized utility
        $A_{i,t}^{\text{test}} \leftarrow \mathtt{RS}_t(\mathcal{A}_{i,t}^{\text{test}})$                    ▷ Selecting action in the hold-out set
        $U_{i,t}^{\text{test}} \leftarrow Y_i(A_{i,t}^{\text{test}})$                          ▷ Realizing the utility from hold-out choice
    end if
    $\Sigma_{i,t+1} \leftarrow \left(\Sigma_{i,t}^{-1} + \frac{1}{\hat{\sigma}_\epsilon^2} A_{i,t} A_{i,t}^T\right)^{-1}$          ▷ Updating posterior variance
    $\mu_{i,t+1} \leftarrow \Sigma_{i,t+1}\left(\Sigma_{i,t}^{-1}\mu_{i,t} + \frac{1}{\hat{\sigma}_\epsilon^2} A_{i,t} U_{i,t}\right)$        ▷ Updating posterior mean
    $\mathtt{RS}_{t+1} = g(\mathtt{RS}_t, A_{i,t}, U_{i,t})$                           ▷ Updating the RS
    $\mathcal{A}_{i,t+1}^{\text{train}} = \mathcal{A}_{i,t}^{\text{train}} \setminus A_{i,t}$                       ▷ Updating the set of remaining movies
end for
```

E.2 Hybrid User and User Heterogeneity

In §5.4.1 in the main text of the paper, we consider a uniform hybrid/mixed strategy for each user. We now extend this hybrid policy to be heterogeneous across users. To bring user heterogeneity and mixed decision strategies into the main text, we incorporate the Bernoulli hybrid user in Algorithm 6 directly into the simulation and draw $p \sim \text{Unif}[0, 1]$ once per user and keep it fixed across periods and simulations for that user.

We simulate 400 users and draw, for each user i, a propensity $p_i \sim \text{Unif}[0, 1]$ independently.

We then *cluster users by* p_i into four bins:

$$p_i \in [0, 0.25), \qquad p_i \in [0.25, 0.5), \qquad p_i \in [0.5, 0.75), \qquad p_i \in [0.75, 1].$$

For each of these four bins, we present a figure illustrating the following user types and their corresponding regret on the hold-out set of movies:

- **Hybrid User:** For users in the bin, at each period the user follows the RS with probability $1 - p_i$ and makes an independent decision otherwise.
- **Hybrid User Counterfactual:** This user follows the Hybrid User policy for all periods before, but in each period of the counterfactual evaluation, they make their own choices (without the algorithm's help, regardless of the Bernoulli draw).
- **Self-Exploring User:** This user chooses movies on their own and updates their preferences based on their realized utility.
- **RS-Dependent User:** This user relies on the personalized RS for their movie choices and updates their preferences upon consuming the algorithm's recommendations.
- **RS-Dependent User Counterfactual:** This user follows the personalized RS recommendations in all prior periods, but in each period of the counterfactual evaluation, they make their own choices (without the algorithm's help).

The last three types are identical to those in the main text of the paper. Figure A4 shows the results by different bins. Compared to our results in §5.3, the two new lines are for Hybrid User and Hybrid User Counterfactual, shown in black and gray respectively. When p_i is closer to zero, the Hybrid User exhibit similar behavior to the RS-Dependent user in both regret and counterfactual regret. In contrast, when p_i is closer to one, the behavior is much closer to the Self-Exploring user. It is only in the moderate levels of p_i that we see improvement in Hybrid User Counterfactual, without too much compromise in Hybrid User's regret.

In summary, the results in Figure A4 show a qualitative pattern similar to our results in the main text. Adding some randomness to following the RS leads to greater learning at a low cost to actual regret. In addition, the analysis with user heterogeneity suggests that the level of exploration can be tailored to each user.

E.3 User Heterogeneity in the Optimal Hybrid Policy

Our results in §5.4.1 and Web Appendix E.2 show that a mixed strategy can improve the trade-off between welfare and learning. In this section, we consider a simple equally weighted multi-objective optimization that balances regret and counterfactual regret, and identify the optimal p_i for each user. If users differ in how much they benefit from the RS and how much they lose from reduced independent learning, the optimal degree of self-exploration need not be the same across users. Let π_p denote the hybrid policy in which the user explores independently with probability p and follows the RS with probability $1 - p$. We can write the objective as follows:

$$\mathcal{L}_i(\pi_p) = \frac{1}{2}\texttt{Regret}_i(\pi_p; T) + \frac{1}{2}\texttt{CounterfactualRegret}_i(\pi_p; T), \tag{55}$$

where the first term captures the welfare cost from worse recommendations, while the second term captures the learning loss that appears when the user later needs to choose independently. We then

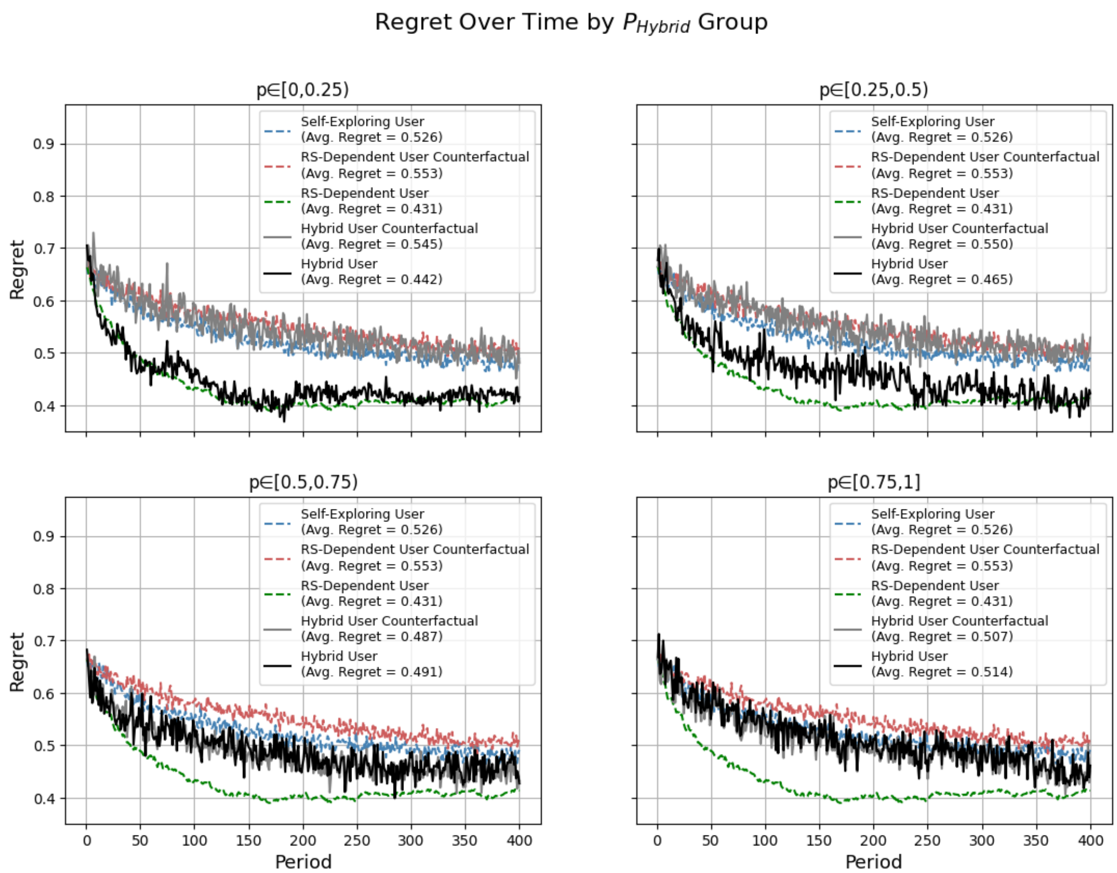


Figure A4: Hybrid user heterogeneity

define the optimal hybrid-policy parameter for user i as:

$$p_i^* = \arg\min_{p \in \mathcal{P}} \mathcal{L}_i(p), \tag{56}$$

where $\mathcal{P}$ is the grid of candidate values.

Figure A5 plots the histogram of $\{p_i^*\}_i$ across users. The main pattern is that the distribution is dispersed rather than concentrated at a single value, although it is tilted toward relatively low and moderate values of p. This indicates substantial heterogeneity in how users trade off the welfare gains from following the recommender system against the benefits of independent learning. For some users, a relatively low RS-off probability is optimal, meaning that they benefit more from remaining closely guided by the RS. For other users, a substantially higher RS-off probability is optimal, implying that more frequent periods of independent choice improve their overall objective by reducing learning loss. In this sense, the figure complements the main-body result: moderate mixing is often beneficial, but the precise degree of mixing varies meaningfully across users.

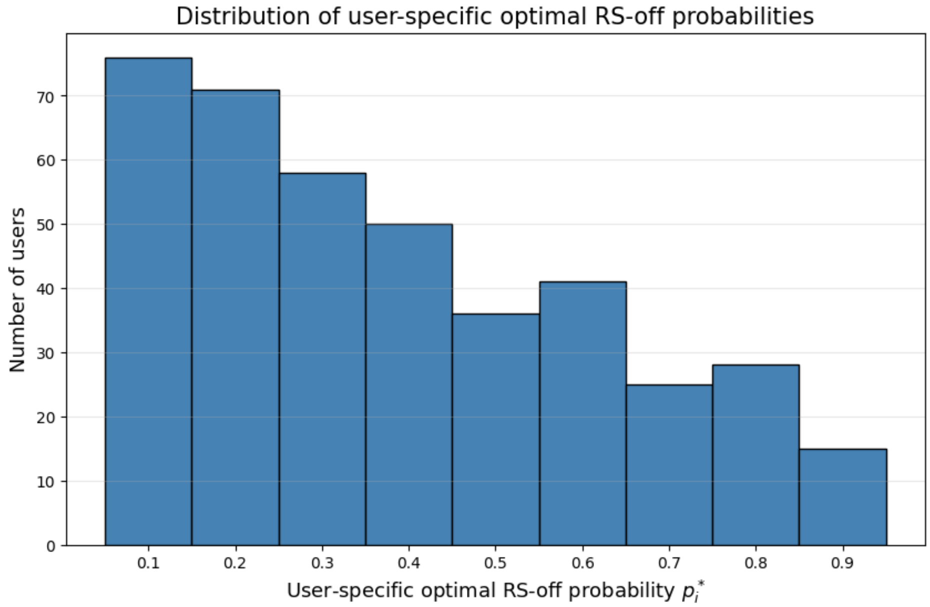


Figure A5: Distribution of user-specific optimal RS-off probabilities p_i^* under the hybrid policy.

This result strengthens the interpretation of the hybrid policy as a practical way to balance welfare and learning. In the main text, we show that intermediate values of p can bring regret and counterfactual regret closer together. The web appendix shows that this balancing point is not universal. A common policy can improve outcomes on average, but it will generally be too restrictive for some users and too permissive for others. By contrast, a personalized hybrid policy allows the platform or policymaker to choose the level of RS availability that best fits each user's own welfare-learning trade-off.

From a policy perspective, this interpretation maps naturally to interventions that limit or regulate the availability of recommendation systems. Such policies can be implemented uniformly, by imposing a common value of p across users, or in a personalized manner, by tailoring p to individual users. The dispersion in Figure A5 suggests that the personalized approach is more promising. By controlling how often users rely on the RS versus explore independently, a policymaker or platform can better balance immediate welfare and long-run learning across the population. More broadly, the figure shows that the cost of consumer learning loss is heterogeneous, which strengthens the case for individualized hybrid policies rather than one-size-fits-all interventions.

F Alternative Forms of User Learning

F.1 Alternative Self-Exploring User: ϵ-Greedy with Bayesian Updates

In the main text, the self-exploring user learns using Thompson Sampling (TS) within a linear bandit framework. The user updates beliefs about 51 movie features, and selects actions by sampling from the posterior. In this section, we retain the same Bayesian updating structure but replace TS with an ϵ-greedy choice rule.

Under the ϵ-greedy policy, at each round t, the user evaluates each movie using the posterior mean utility using the mean of preference weights $\mu_{i,t}$. This is in contrast with the TS algorithm that draws probabilistically from the posterior distribution, thereby taking into account the posterior uncertainty of the preferences. The user then selects the movie with the highest posterior mean with probability $1-\epsilon$ (exploitation), and chooses uniformly at random from the action set $\mathcal{A}$ with probability ϵ (exploration). Thus, ϵ directly governs the exploration rate. Algorithm 7 summarizes how this algorithm works.

Algorithm 7 Choice and Learning for the ϵ-Greedy Self-Exploring User

Input: $\mu_{i,0} = e_1 \in \mathbb{R}^{51}, \Sigma_{i,0} = I_{[51\times 51]}, \mathcal{A}_i^{\text{train}}, \mathcal{A}_i^{\text{test}}, \mathcal{Y}_i, \mathcal{T} = 400, \hat{\sigma}_\epsilon = 0.86, \texttt{RS}_0(\cdot)$
Output: $\mathcal{H}_{i,\mathcal{T}}^{\text{train}}, \mathcal{H}_{i,\mathcal{T}}^{\text{test}}, \{\mu_{i,t}, \Sigma_{i,t}\}_{t=1}^{\mathcal{T}}$

```
A^train_{i,0} ← A^train_i
for t = 0 → T do
    ξ_{i,t} ~ Bernoulli(ε)                                    ▷ Determine RS Availability
    if ξ_{i,t} = 1 then
        A_{i,t} ← argmax_{A_j ∈ A^train_{i,t}} Σ_{k=1}^s μ_{i,k,t} A_{k,j}   ▷ Selecting action based on mean utility
        U_{i,t} ← Y_i(A_{i,t})                                ▷ Entering the observed rating as the realized utility
        A^test_{i,t} ← argmax_{A_j ∈ A^test_{i,t}} Σ_{k=1}^s μ_{i,k,t} A_{k,j}   ▷ Selecting action in the hold-out test set
        U^test_{i,t} ← Y_i(A^test_{i,t})                      ▷ Realizing the utility from hold-out choice
    else
        A_{i,t} ~ Uniform(A^train_{i,t})                      ▷ Selecting action uniformly at random
        U_{i,t} ← Y_i(A_{i,t})                                ▷ Entering the observed rating as the realized utility
        A^test_{i,t} ~ Uniform(A^test_{i,t})                  ▷ Selecting action in the hold-out set
        U^test_{i,t} ← Y_i(A^test_{i,t})                      ▷ Realizing the utility from hold-out choice
    end if
    Σ_{i,t+1} ← (Σ^{-1}_{i,t} + (1/σ̂²_ε) A_{i,t} A^T_{i,t})^{-1}          ▷ Updating posterior variance
    μ_{i,t+1} ← Σ_{i,t+1} (Σ^{-1}_{i,t} μ_{i,t} + (1/σ̂²_ε) A_{i,t} U_{i,t})   ▷ Updating posterior mean
    A^train_{i,t+1} = A^train_{i,t} \ A_{i,t}                 ▷ Updating the set of remaining movies
end for
```

As in the main text, we consider three different policies as follows:

- **ϵ-Greedy Self-Exploring User:** As presented in Algorithm 7, for the self-exploring user, decisions are made based on the ϵ draw and mean preference weights. In both the training set and the hold-out set in our empirical exercise, the user follows the ϵ-greedy rule: with probability

$1 - \epsilon$, selecting the posterior-best movie, and with probability ϵ, making a random exploratory choice. After each consumption, the user updates their posterior beliefs based on the realized utility. Period-t regret is defined as the difference between the utility of the best available movie in the hold-out set and that of the chosen movie.

- **RS-Dependent User:** For the RS-Dependent User, movie choices in the training set are determined by the recommender system. The user consumes the recommended movies and updates their beliefs accordingly, but does not control exploration directly. As a result, regret is computed based on the RS-selected actions and is independent of ϵ.
- **RS-Dependent User Counterfactual:** The RS-Dependent User Counterfactual follows the personalized RS recommendations in *all prior training periods*, and updates their posterior beliefs based on the realized utility from these RS-selected movies. Thus, their belief trajectory is entirely shaped by RS-guided exploration. However, in each period of counterfactual evaluation, the user makes their *own* choices without the help of the algorithm. This captures how the user would perform if their learning were shaped entirely by the RS, but their final decisions were made independently. Specifically, holding the RS-induced posterior beliefs fixed, the user applies the same ϵ-greedy rule as the self-exploring user on the holdout set: with probability $1 - \epsilon$, choosing the posterior-best movie, and with probability ϵ, exploring randomly.

We report results for four exploration rates, $\epsilon \in 0.01, 0.02, 0.08, 0.09$. Figure A6 shows the regret for the self-exploring user (blue), the RS-dependent user (green), and the RS-dependent counterfactual (red). Most importantly, compared to our main analysis with a self-exploring user using Thompson Sampling in Figure 6, the gap between the RS-dependent and self-exploring users widens. This reflects the greedy nature of Algorithm 7, which limits effective exploration and learning by ignoring posterior uncertainty. Moreover, the self-exploring user's regret is not monotonic in ϵ: moderate exploration reduces regret by improving learning, whereas excessive exploration leads to too many random choices and increases regret. This pattern reflects the standard exploration–exploitation trade-off, where intermediate values of ϵ balance faster learning with fewer short-run mistakes.

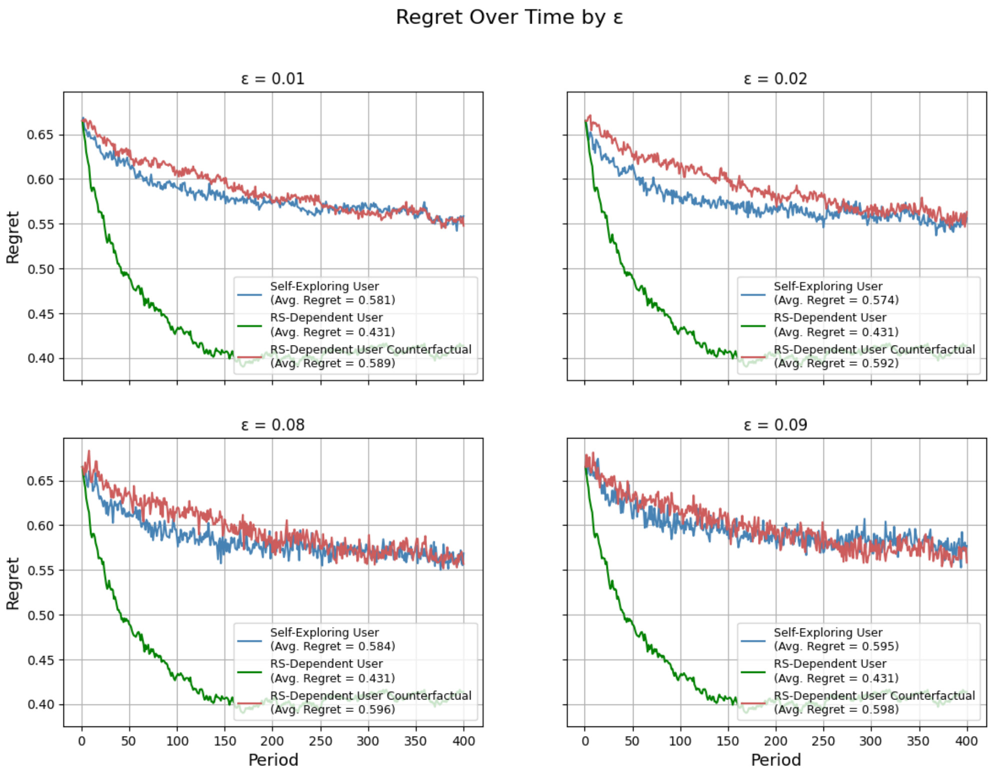


Figure A6: Regret and counterfactual regret across different ϵ-greedy self-exploring policies

In contrast, counterfactual regret increases with ϵ. However, this is a mechanical effect, as the ϵ does not influence the consumption sequence and learning for the RS-dependent user, but worsens the quality of their independent decision-making by adding more randomness in measuring the counterfactual regret. Across all scenarios, the amount of learning for the RS-dependent user is kept unchanged because of the independence of RS from ϵ. We illustrate this point in Figure A7, where the learning for the RS-dependent user is unchanged across scenarios. However, as expected, increasing ϵ improves the learning of the self-exploring user.

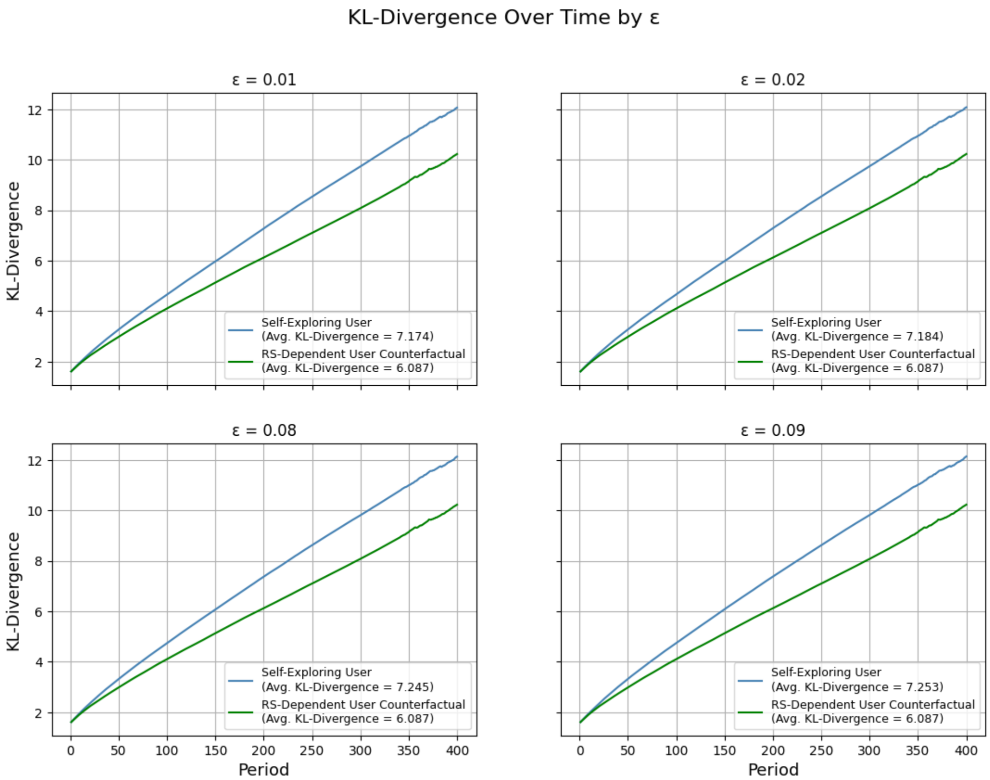


Figure A7: User learning across different ϵ-greedy self-exploring policies

F.2 Social Learning via Nearest-Neighbor Information

In the main text, the *Self-Exploring User* updates preferences using the global average rating and the top 50 tag features. In this subsection, we examine whether enriching the user's information environment reduces the recommender system's informational advantage. In particular, we allow users to incorporate an additional source of outside information that proxies for social learning, public reviews, or correlated consumption patterns.

The exercise proceeds in three steps. First, we construct a nearest-neighbor rating signal for each user–movie pair. Second, we add this signal to the user's feature space and let the user learn over the augmented representation. Third, we examine how performance changes when the social signal becomes noisier.

Construction of the Nearest-neighbor Signal. We operationalize outside information through a nearest-neighbor rating signal. The idea is that a user may learn not only from global average ratings, but also from the ratings of other users with similar historical tastes. We therefore augment the baseline learning problem with one additional nearest-neighbor average rating feature. The learning user now updates beliefs over a 52-dimensional feature space: the global average rating, the top 50 tag features, and one user-specific neighbor-based rating feature.

To construct this signal, let $Y \in \mathbb{R}^{U \times J}$ denote the sparse user–item rating matrix, and let Y_i

denote row i. We compute cosine similarity between users using their observed ratings:

$$\mathrm{sim}(i, \ell) = \frac{Y_i^\top Y_\ell}{\|Y_i\|_2 \, \|Y_\ell\|_2},$$

excluding user i itself. For each simulated user i, we select the top $K = 100$ nearest neighbors according to $\mathrm{sim}(i, \cdot)$.

Using these neighbors, we construct a user-specific movie-level social signal. Let $\bar{m}(j)$ denote the global mean rating of movie j, averaged over all users who rated it. For user i and movie j, define the neighbor-average rating as

$$\widehat{m}_i(j) = \begin{cases} \dfrac{1}{|\mathcal{N}_i(j)|} \sum_{\ell \in \mathcal{N}_i(j)} Y_{\ell j}, & \text{if } \mathcal{N}_i(j) \neq \varnothing, \\ \bar{m}(j), & \text{if no neighbor rated } j, \end{cases}$$

where $\mathcal{N}_i(j)$ is the subset of user i's K nearest neighbors who rated movie j. Thus, $\widehat{m}_i(j)$ summarizes how socially proximate users evaluated movie j, while falling back on the global average rating when no neighbor information is available.

We do not treat this nearest-neighbor signal as a replacement for the global average rating. Instead, we include it as an additional feature in the user's Bayesian learning problem. We define the augmented movie feature vector as

$$A_{i,j} = \big(\bar{m}(j),\ A_j^{\text{tag}},\ \widehat{m}_i(j)\big) \in \mathbb{R}^{52},$$

where $A_j^{\text{tag}} \in \mathbb{R}^{50}$ denotes the vector of the top-50 tag features. Relative to the baseline feature vector, $A_{i,j}$ adds one user-specific social-learning feature. Hence, unlike the baseline movie feature vector A_j, the augmented vector $A_{i,j}$ depends on both the movie and the user because $\widehat{m}_i(j)$ is constructed from user i's nearest-neighbor set.

Let the user's preference vector in this augmented feature space be

$$\theta_{i,t} = \big(\theta_{i,t}^{\text{global}}, \theta_{i,t}^{\text{tag}}, \theta_{i,t}^{\text{neighbor}}\big) \in \mathbb{R}^{52}.$$

where $\theta_{i,t}^{\text{neighbor}}$ is the new element added compared to our main exercise, and captures social learning. The user's predicted utility for movie j in period t is then

$$\widehat{u}_{i,j,t} = \theta_{i,t}^\top A_{i,j},$$

We initialize the user's prior mean so that the coefficient on the global average rating is equal to one, while all other coefficients are initialized at zero:

$$\mu_{i,0} = (1, 0, \ldots, 0)^\top \in \mathbb{R}^{52}, \qquad \Sigma_{i,0} = I_{52}.$$

After consuming a movie and observing realized utility, the user updates the posterior over the augmented preference vector using the same Bayesian updating rule as in Algorithm 1. As in the main analysis, we compare the learning user to the recommender system and to two benchmarks:

- **Aggregate Rating baseline.** This is the non-personalized benchmark based on average movie ratings.
- **User with Known Preference benchmark.** In this exercise, the known-preference benchmark is also enriched. Instead of being estimated only from the global average rating and tags, it is estimated from the same expanded information set as the learning user: the global average rating, the 50 tag features, and the nearest-neighbor rating feature.
- **RS-Dependent User.** The RS benchmark is unchanged and continues to use its personalized recommendation procedure.

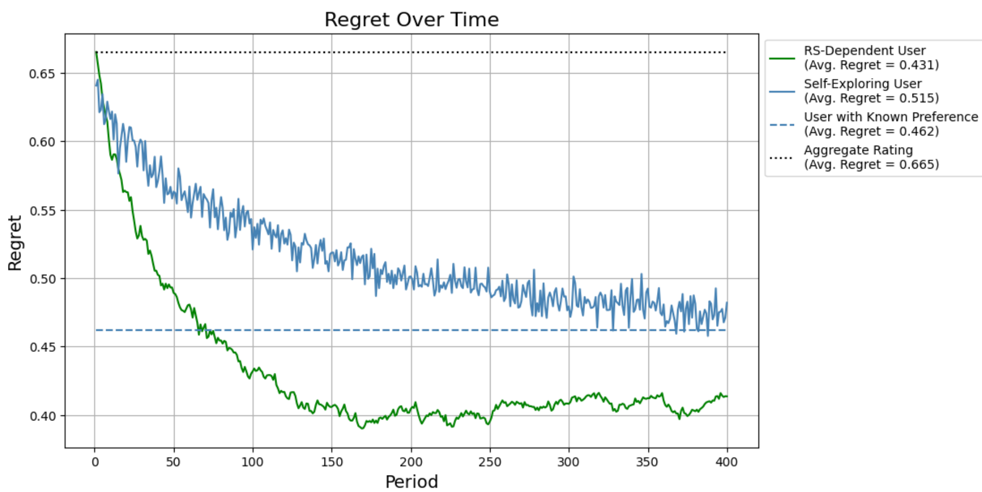


Figure A8: Regret performance with nearest-neighbor information added as an extra feature.

Figure A8 compares four trajectories: the RS-Dependent User, the enriched Self-Exploring User, the enriched User with Known Preference benchmark, and the Aggregate Rating baseline. Adding nearest-neighbor information improves user learning: the Self-Exploring User now attains an average regret of approximately 0.515, compared with about 0.526 in the baseline model. The enriched User with Known Preference benchmark also improves, with average regret falling to approximately 0.462, compared with 0.484 in the baseline model. Nevertheless, the RS-Dependent User still performs best, with average regret of approximately 0.431. Thus, the richer social-learning information environment narrows the gap between the Self-Exploring User and the RS-Dependent User to about 0.084, but it does not eliminate the RS's informational advantage.

Noisy Social Information. We next examine whether the quality of the social signal matters. The nearest-neighbor construction above assumes that the user observes information from the most similar users. In practice, however, outside information may be noisy: users may rely on broad public reviews, weakly related peers, or social signals that are only imperfectly correlated with their own preferences. To capture this possibility, we introduce random users into the neighbor set.

For each user i, let ρ denote the share of random users in the neighbor set. Define

$$K_{\text{top}} = \lfloor (1-\rho)K \rfloor, \qquad K_{\text{rand}} = K - K_{\text{top}}.$$

We then take the K_{top} most similar users by cosine similarity and augment them with K_{rand} users sampled uniformly without replacement from the complement set

$$\{1, \ldots, U\} \setminus \left(\{i\} \cup \mathcal{N}_i^{\text{top}}\right).$$

The resulting mixed neighbor set is denoted by $\mathcal{N}_i^{(\rho)}$. For each value of ρ, we rebuild the neighbor-average feature $\widehat{m}_i(j)$ using $\mathcal{N}_i^{(\rho)}$ and re-run the same 52-feature learning exercise. Increasing ρ therefore makes the neighbor-based signal less informative by replacing socially proximate users with unrelated users.

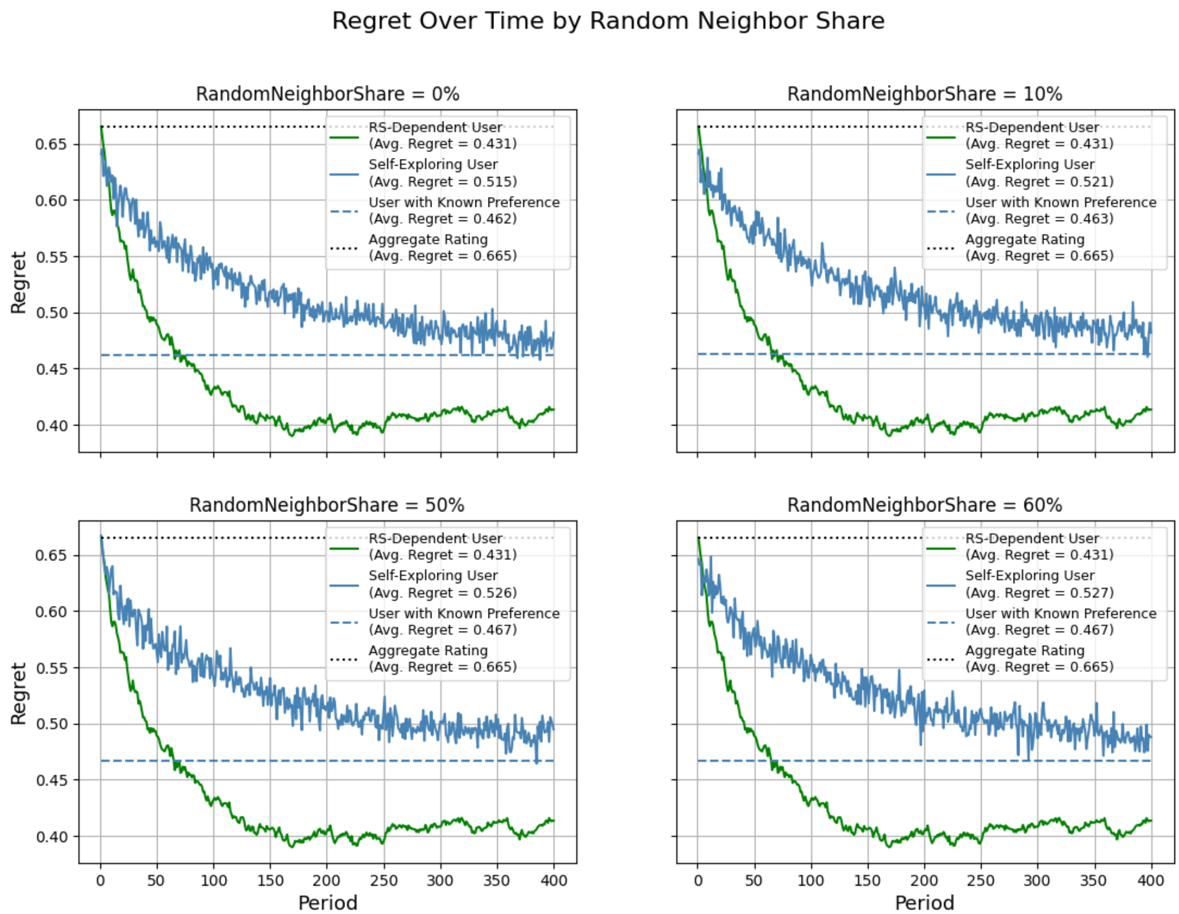


Figure A9: Regret performance with nearest-neighbor information under different shares of random neighbors.

Figure A9 shows that social information is valuable only when it is sufficiently informative. When the neighbor signal is relatively clean, it helps the Self-Exploring User: average regret is about 0.515 with 0% random neighbors and rises only slightly to about 0.521 with 10% random neighbors. As the signal becomes noisier, performance deteriorates: average regret increases to about 0.526 with 50% random neighbors and to about 0.527 with 60% random neighbors. The User with Known Preference benchmark follows the same pattern, with average regret rising from about 0.462 at 0% random neighbors to about 0.467 at 50%–60% random neighbors. By contrast, the RS-Dependent User is unaffected by the random-neighbor share, with average regret remaining around 0.431 across panels. Thus, as social information becomes noisier, the value of social learning declines and the gap between the RS-Dependent User and the Self-Exploring User widens.

F.3 Learning from RS Recommendations as an Observable Feature

In the baseline model, users learn only from their own consumption experience. A direct implication of Assumption 2 is that the RS-dependent user cannot learn from the recommendations beyond their own experience. A natural objection is that users may also learn from the recommender itself. If the RS is optimizing a similar objective, then the act of recommending a title may already reveal useful information about the item's likely match quality. In this subsection, we examine this possibility by allowing the recommendation event to enter the user's learning problem as an additional observable feature.

We implement this idea by augmenting the user's feature space with one binary indicator for whether an item is recommended by the RS in the past. We define the RS recommendation indicator as $s_{i,j,t}$, such that $s_{i,j,t} = 1$ if item j is the item that the RS would recommend to user i in period t, and $s_{i,j,t} = 0$ otherwise.

Augmented User-side Feature Representation. On the user side, we retain the usual 51-dimensional feature vector and append the RS recommendation indicator as an additional learnable feature. Let

$$A_j^{\mathrm{base}} \in \mathbb{R}^{51}$$

denote the baseline user-side movie feature vector, consisting of the aggregate rating and the top 50 tag features. The augmented feature vector is

$$A_{i,j,t} = \begin{bmatrix} A_j^{\mathrm{base}} \\ s_{i,j,t} \end{bmatrix} \in \mathbb{R}^{52}.$$

The notation $A_{i,j,t}$ emphasizes that this augmented feature vector varies not only across movies, but also across users and periods, because the RS recommendation indicator depends on user i's RS posterior at time t.

Let the user's preference vector in this augmented feature space be

$$\theta_{i,t} = \begin{bmatrix} \theta_{i,t}^{\mathrm{base}} \\ \theta_{i,t}^{\mathrm{RS}} \end{bmatrix} \in \mathbb{R}^{52}.$$

The final component, $\theta_{i,t}^{\mathrm{RS}}$, captures how much weight user i places on the RS recommendation signal. The user's predicted utility for movie j in period t is

$$\widehat{u}_{i,j,t} = \theta_{i,t}^{\top} A_{i,j,t}.$$

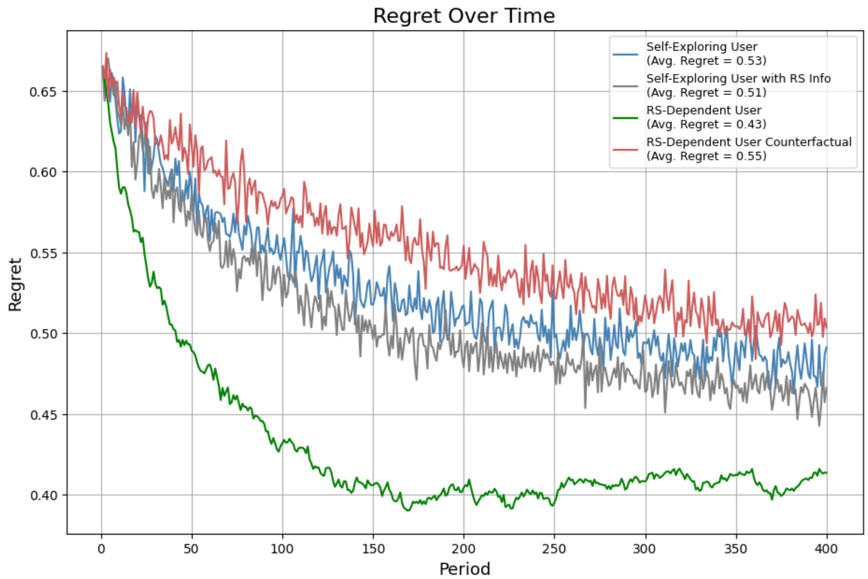


Figure A10: Regret performance when RS recommendations enter the user's learning problem as an observable feature.

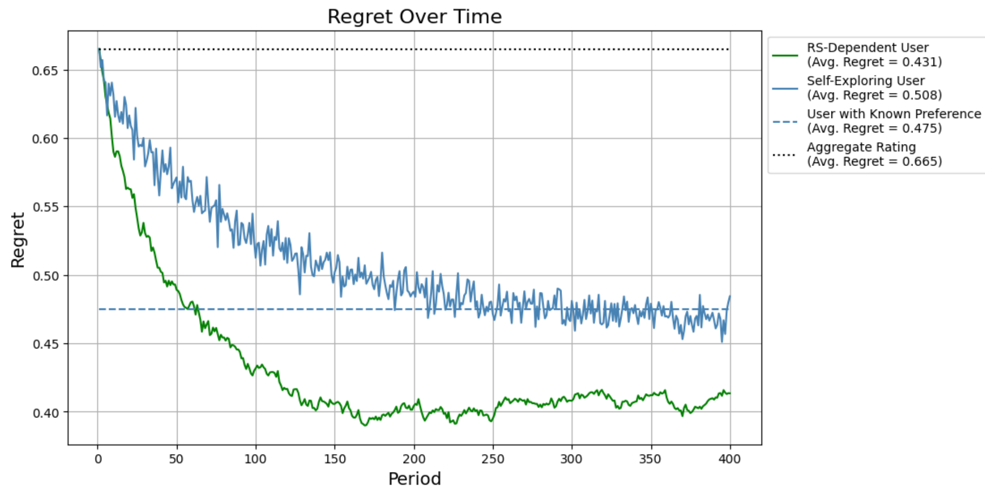
Figure A10 reports the regret trajectories. Adding the RS recommendation indicator helps user learning: the Self-Exploring User with RS information attains an average regret of about 0.51, compared with about 0.53 for the baseline Self-Exploring User without the signal. The improvement is most visible in the early periods, which is consistent with the idea that recommendation events provide useful guidance when the user has limited personal feedback. However, the RS-Dependent User still performs best, with average regret of about 0.43. Thus, allowing users to learn from the RS indicator narrows the user–RS gap from roughly 0.10 to about 0.08, but it does not eliminate the recommender system's informational advantage.

F.4 Alternative Feature Representations for User Learning

In the main text, the self-exploring user updates preferences using the top 50 most frequently mentioned tags together with the aggregate rating. A natural concern, however, is that our results may depend on this particular feature construction. In this subsection, we examine whether the recommender system's informational advantage persists under alternative user-side feature representations.

We consider two robustness checks. The first replaces the top 50 tag features with a lower-dimensional principal-component representation. Specifically, we construct the first 20 principal components from the full set of movie tags and combine them with the aggregate rating. These 20 components capture approximately 80% of the total variation in item features.

The second robustness check expands the tag-based feature space. Instead of using the top 50 most frequently mentioned tags, we allow users to learn from the top 100 tags, again together with the aggregate rating. This representation gives users a richer and more detailed set of observable product attributes.



(a) Regret performance when users learn from the top 20 PCA features.

(b) Regret performance when users learn from the top 100 tags.

Figure A11: Regret under alternative feature representations for user learning.

Figure A11 reports the regret trajectories under the two alternative feature representations. In Panel (a), replacing the top 50 tags with the top 20 PCA features improves the Self-Exploring User's performance relative to the baseline: average regret falls from about 0.515 to about 0.508. In Panel (b), expanding the feature space to the top 100 tags produces qualitatively similar results, with the Self-Exploring User attaining average regret of about 0.518. In both panels, the User with Known Preference benchmark also performs well, with average regret of about 0.475 under PCA and 0.471 under the 100-tag representation. However, the personalized RS continues to achieve the lowest regret, with average regret of about 0.431.

Overall, these robustness checks show that the main conclusion is not driven by the specific choice of the top 50 tags as the user-side feature space. PCA features appear somewhat more useful than simply adding more raw tags, suggesting that a compact representation can summarize item variation more efficiently. Nevertheless, richer or more efficient observable features only partially improve user learning and do not eliminate the RS's informational advantage.

F.5 Inference from Search to Experience (2D $\rightarrow$ 100D)

In our theoretical analysis, we assume independence between search and experience utility across products. However, it is possible that a search feature carries information about an experience feature—for example, the thriller genre may suggest a higher prior on a "surprise ending" (an experience feature) than the documentary genre. We discuss how to allow for this correlation by interpreting $\sigma^2_{i,x}$ as the residual variance of experience utility, i.e., the part not explained by search features. The same logic applies in our empirical analysis: since we observe only search features, the coefficients users estimate capture the collective relationship between these features and realized utility, including any indirect contribution from correlated experience features.

In this section, we run two illustrative exercises where we have access to experience features for the purpose of the exercise, and there is correlation between search and experience features. First, we introduce a simple two-dimensional environment with one search feature and one experience feature. This setting clarifies the inference mechanism and includes an Oracle RS benchmark that directly observes both features. Second, we scale the exercise to a high-dimensional environment with 50 search features and 50 experience features, and compare user learning with a matrix-factorization recommender system trained only on ratings.

Two-dimensional setup. In the two-dimensional setup, each movie j has one observable search feature s_j and one latent experience feature x_j. We write the movie feature vector as

$$A_j = \begin{bmatrix} s_j \\ x_j \end{bmatrix} \in \mathbb{R}^2.$$

The two features are generated with controlled correlation:

$$s_j \sim \mathcal{N}(0,1), \qquad x_j = \rho s_j + \sqrt{1-\rho^2}\,\varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0,1),$$

where $\rho \in [0,1]$ governs how informative the search feature is about the experience feature.

User i has preference vector

$$\theta_i = \begin{bmatrix} \theta_{i,s} \\ \theta_{i,x} \end{bmatrix} \in \mathbb{R}^2, \qquad \theta_{i,s}, \theta_{i,x} \sim \mathcal{N}(0,1),$$

and receives utility

$$u_i(A_j) = \theta_i^\top A_j + \epsilon_{i,j} = \theta_{i,s} s_j + \theta_{i,x} x_j + \epsilon_{i,j}, \qquad \epsilon_{i,j} \sim \mathcal{N}(0, \sigma^2_\epsilon).$$

Before choosing among the holdout set, the user observes only the search feature s_j. The experience feature x_j is revealed only after consumption. Thus, before making a choice, the user must decide whether to use the statistical relationship between s_j and x_j.

Each user maintains a Gaussian posterior over the preference vector:

$$\theta_i \mid \mathcal{H}_{i,t} \sim \mathcal{N}(\mu_{i,t}, \Sigma_{i,t}).$$

At period t, the user draws

$$\tilde{\theta}_{i,t} = \begin{bmatrix} \tilde{\theta}_{i,s,t} \\ \tilde{\theta}_{i,x,t} \end{bmatrix} \sim \mathcal{N}(\mu_{i,t}, \Sigma_{i,t})$$

and chooses through Thompson sampling. We compare three decision rules:

- **Correct-inference user.** The correct-inference user recognizes the statistical relationship between search and experience features. Since

  $$\mathbb{E}[x_j \mid s_j] = \rho s_j,$$

  the user evaluates movie j according to

  $$\widehat{u}_{i,j,t}^{\text{corr}} = \tilde{\theta}_{i,s,t} s_j + \tilde{\theta}_{i,x,t} \mathbb{E}[x_j \mid s_j] = \big(\tilde{\theta}_{i,s,t} + \rho \tilde{\theta}_{i,x,t}\big) s_j.$$

  This rule captures a user who can correctly infer expected experience utility from the observable search feature.

- **No-inference user.** The no-inference user ignores the relationship between search and experience attributes. Because x_j is not observed before choice, this user evaluates movie j using only the observable search component:

  $$\widehat{u}_{i,j,t}^{\text{no-inf}} = \tilde{\theta}_{i,s,t} s_j.$$

  Comparing this rule with the correct-inference rule isolates the value of using search attributes to infer latent experience utility.

- **Oracle RS.** The Oracle RS observes both s_j and x_j before choice. It therefore selects the movie with the highest true systematic utility:

  $$j_{i,t}^{\text{Oracle}} = \arg\max_{j \in H_i} \theta_i^\top A_j.$$

  By construction, the Oracle RS has zero regret in every period. It provides a full-information lower bound and helps interpret how far the user-side inference policies remain from the best achievable outcome.

Panel (a) of Figure A12 reports the two-dimensional exercise for $\rho = 0.2$ and $\rho = 0.8$. Regret falls sharply in the early periods for both user policies and then flattens, indicating that most learning in this simple two-feature environment occurs early. Correct inference improves over no inference in both cases, but the gain is much larger when the search feature is more informative about the experience feature: at $\rho = 0.2$, average regret is about 0.89 for the correct-inference user and 0.95 for the no-inference user, while at $\rho = 0.8$, the corresponding values are about 0.36 and 0.69. The

Oracle RS remains at zero regret because it observes both search and experience features before choice, so the gap between the correct-inference user and the Oracle RS captures the residual value of directly observing experience features.

High-dimensional extension. We next scale the design to a high-dimensional setting. Each movie now has 50 observable search features and 50 latent experience features. Let

$$A_j = \begin{bmatrix} A_j^s \\ A_j^x \end{bmatrix} \in \mathbb{R}^{100},$$

where $A_j^s \in \mathbb{R}^{50}$ denotes the search-feature vector and $A_j^x \in \mathbb{R}^{50}$ denotes the experience-feature vector.

The two parts of the feature vector are generated from low-dimensional subspaces. Let $r_s, r_x \ll 50$, and draw latent factors

$$z_j^{(s)} \sim \mathcal{N}(0, I_{r_s}), \qquad z_j^{(x)} \sim \mathcal{N}(0, I_{r_x}),$$

together with fixed random basis matrices

$$B_s \in \mathbb{R}^{50\times r_s}, \qquad B_x \in \mathbb{R}^{50\times r_x}.$$

We define

$$A_j^s = B_s z_j^{(s)}, \qquad A_j^x = \rho A_j^s + \sqrt{1-\rho^2}\, B_x z_j^{(x)},$$

plus a small isotropic noise term. The effective item dimension is therefore approximately

$$r_s + (1-\rho^2) r_x.$$

As ρ increases, the search and experience spaces become more aligned, and the effective dimensionality of the latent experience component declines.

User preferences are given by

$$\theta_i = \begin{bmatrix} \theta_i^s \\ \theta_i^x \end{bmatrix} \in \mathbb{R}^{100}, \qquad \theta_i^s, \theta_i^x \sim \mathcal{N}(0, I_{50}),$$

and systematic utility is

$$\theta_i^\top A_j = (\theta_i^s)^\top A_j^s + (\theta_i^x)^\top A_j^x.$$

Users behave as in the two-dimensional case. We compare three policies:

- **Correct-inference user.** The correct-inference user uses the statistical relationship between search and experience features to impute latent experience utility before making holdout choices. In particular, the user uses the conditional expectation of the experience-feature vector given the observable search-feature vector.

- **No-inference user.** The no-inference user does not use the relationship between search and experience features. This user relies only on observable search attributes when making holdout choices.

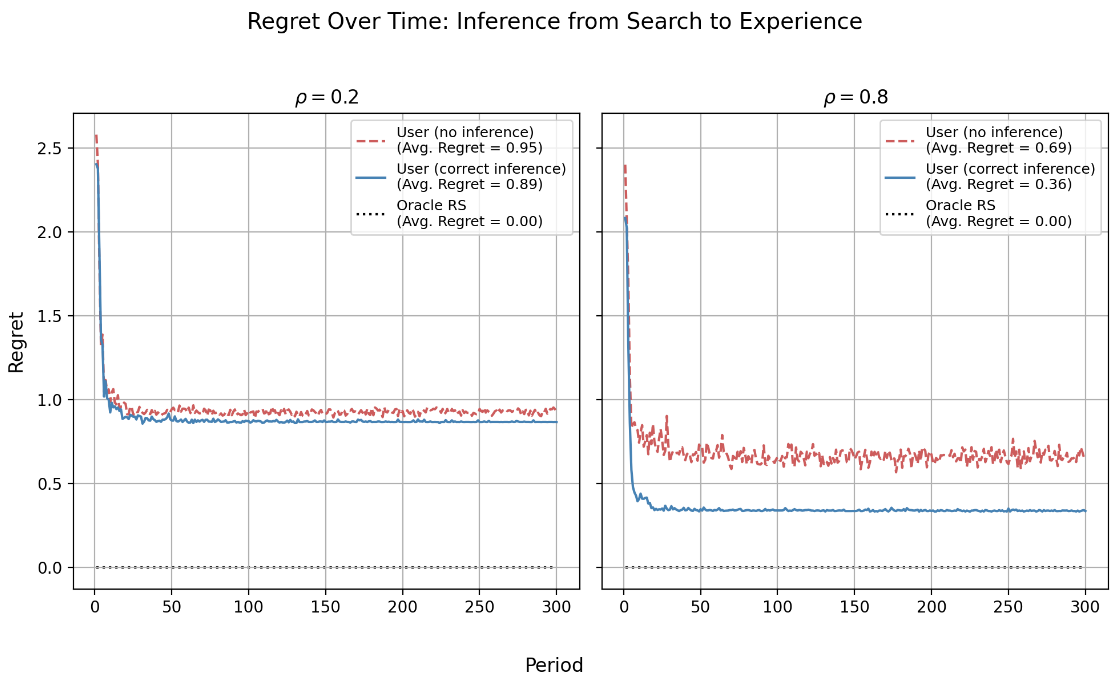
- **Recommender system.** The recommender system learns only from realized ratings. For each value of ρ, we train the recommender system on an external cohort and choose the matrix-factorization rank by cross-validation over

$$K = \{10, 20, \ldots, 100\}.$$

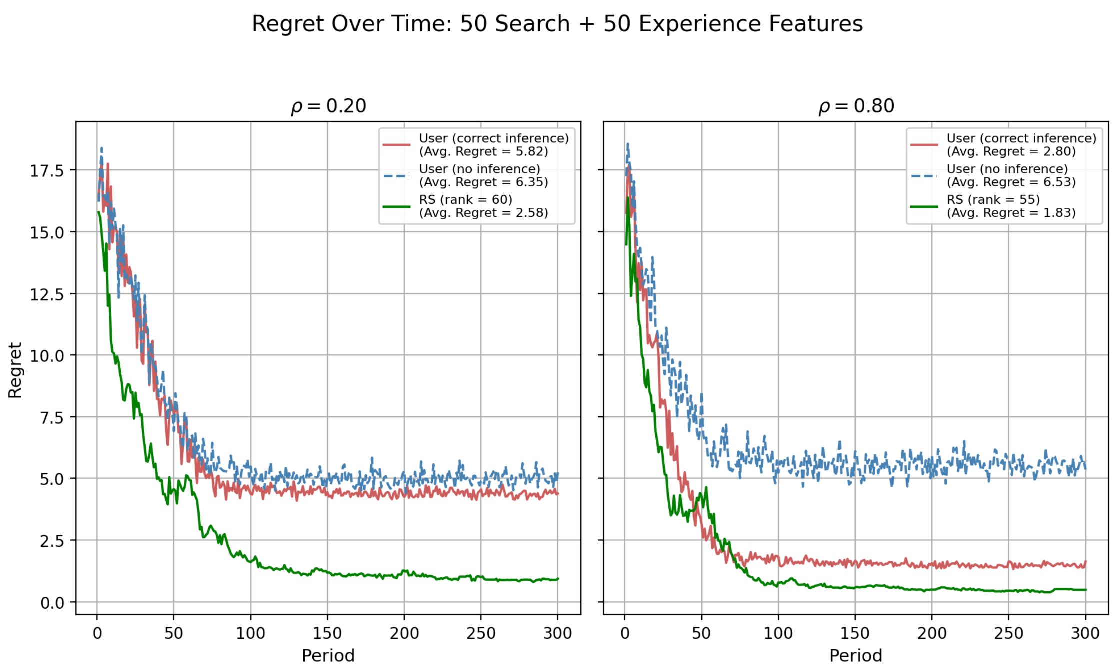
Panel (b) of Figure A12 reports the high-dimensional exercise with 50 search features and 50 experience features. We set $r_s = r_x = 40$ and again focus on $\rho = 0.2$ and $\rho = 0.8$.

When $\rho = 0.2$, search attributes are only moderately informative about latent experience attributes. Correct inference therefore improves over no inference, but the gain is limited: average regret is about 5.82 under correct inference and 6.35 under no inference. The RS performs substantially better, with average regret of about 2.58, because it starts from an item-factor representation learned from an external cohort. When $\rho = 0.8$, search attributes become much more informative, and the value of correct inference increases sharply. Average regret falls to about 2.80 under correct inference, compared with 6.53 under no inference. The RS still performs best, with average regret of about 1.83, but its advantage over the correct-inference user becomes much smaller.

Overall, the figure shows that the user's ability to compete with the recommender system depends on how informative observable search attributes are about latent experience utility. When the search–experience correlation is weak, users cannot infer much before consumption, and the RS retains a large informational advantage. When the correlation is strong, users can form better expectations from observable attributes, which substantially lowers regret and narrows the gap with the RS. Thus, consistent with our discussion under Proposition 2, recommendation systems are more valuable when a greater share of the variation in experience utility cannot be inferred by observable product information, i.e., residual variance of $\sigma^2_{i,x}$.



(a) Two-feature setting: regret over time for $\rho \in \{0.2, 0.8\}$. The Oracle RS observes both search and experience features and therefore achieves zero regret.



(b) High-dimensional setting: 50 search + 50 experience features, with $\rho \in \{0.2, 0.8\}$.

Figure A12: Inference performance in (a) a two-feature setting and (b) a high-dimensional setting.

G Alternative RS Algorithms with MF: Bayes-UCB, Bayes-EXP4 and Bayes-TS (γ-parameterization)

In the main text, the recommender system (RS) uses matrix factorization (MF) to represent items and users. The RS then updates a new user's low-dimensional latent weights as more movies are watched. This section examines whether our main results depend on the particular RS decision rule used in the baseline analysis. The goal is to separate the value of the RS's information structure from the value of any specific exploration algorithm.

Throughout this exercise, we keep the same MF representation and vary only the RS action rule. The comparison focuses on three Bayesian RS policies:

- MF + Bayes-UCB, which recommends optimistically using posterior uncertainty;
- MF + Bayes-EXP4, which samples probabilistically from exponential weights based on posterior-mean scores;
- MF + Bayes-TS, which applies Thompson sampling directly in the MF latent space.

These policies operate on the same low-dimensional MF state space and differ only in how they convert posterior beliefs into recommendations.

MF + Bayes-UCB. Under MF + Bayes-UCB, each candidate movie j is assigned an optimism-based score:

$$\mathrm{UCB}_{i,t}(j) = F_j^\top \mu_{i,t}^{(\gamma)} + \alpha\sqrt{2\log t}\,\sqrt{F_j^\top \Sigma_{i,t}^{(\gamma)} F_j}.$$

The first term is the posterior mean prediction, and the second term is an exploration bonus proportional to posterior uncertainty. As more data accumulate, posterior variance shrinks and the exploration bonus becomes less important. The parameter $\alpha > 0$ governs exploration intensity: larger values generate more exploration, while smaller values make the policy more exploitative. The RS recommends

$$j_{i,t}^{\mathrm{UCB}} = \arg\max_{j\in\mathcal{A}} \mathrm{UCB}_{i,t}(j).$$

MF + Bayes-EXP4. Under MF + Bayes-EXP4, the RS converts posterior-mean predictions into probabilistic choice weights. Let

$$\widehat{u}_{i,j,t}^{\mathrm{RS}} \equiv F_j^\top \mu_{i,t}^{(\gamma)}$$

denote the MF-based posterior-mean score for movie j. The unnormalized weight evolves according to

$$w_{i,t}(j) = w_{i,t-1}(j)\exp\big(\eta\,\widehat{u}_{i,j,t}^{\mathrm{RS}}\big),$$

and the RS samples from the normalized distribution

$$P_{i,t}(j) = \frac{w_{i,t}(j)}{\sum_{\ell\in\mathcal{A}} w_{i,t}(\ell)}.$$

With initial condition $w_{i,0}(j) = 1$, this implies

$$w_{i,t}(j) = \exp\left(\eta \sum_{\tau \leq t} \widehat{u}^{\text{RS}}_{i,j,\tau}\right).$$

Thus, the sampling rule is a Gibbs or Boltzmann distribution that places higher probability on movies with persistently favorable posterior-mean scores. The parameter η is an inverse-temperature parameter. When η is small, choice probabilities are relatively flat and the algorithm explores broadly. When η is large, the distribution concentrates on movies with the highest cumulative scores and behaves more greedily.

MF + Bayes-TS. Under MF + Bayes-TS, the RS applies Thompson sampling directly in the MF parameter space. At period t, it draws

$$\tilde{\gamma}_{i,t} \sim \mathcal{N}\big(\mu^{(\gamma)}_{i,t}, \Sigma^{(\gamma)}_{i,t}\big),$$

and recommends the movie with the highest sampled linear score:

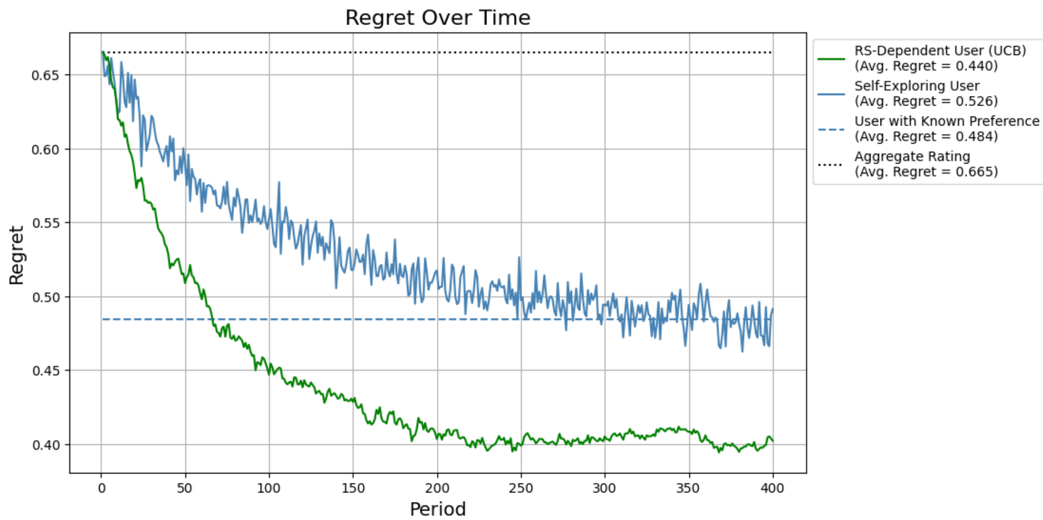
$$j^{\text{TS}}_{i,t} = \arg\max_{j \in \mathcal{A}} F_j^{\top} \tilde{\gamma}_{i,t}.$$

Unlike Bayes-UCB and Bayes-EXP4, Bayes-TS does not require a tuning parameter such as α or η. Exploration is generated entirely by posterior uncertainty. When the posterior covariance is large, sampled parameters vary more and the RS explores more. As the posterior contracts, the policy becomes increasingly exploitative.

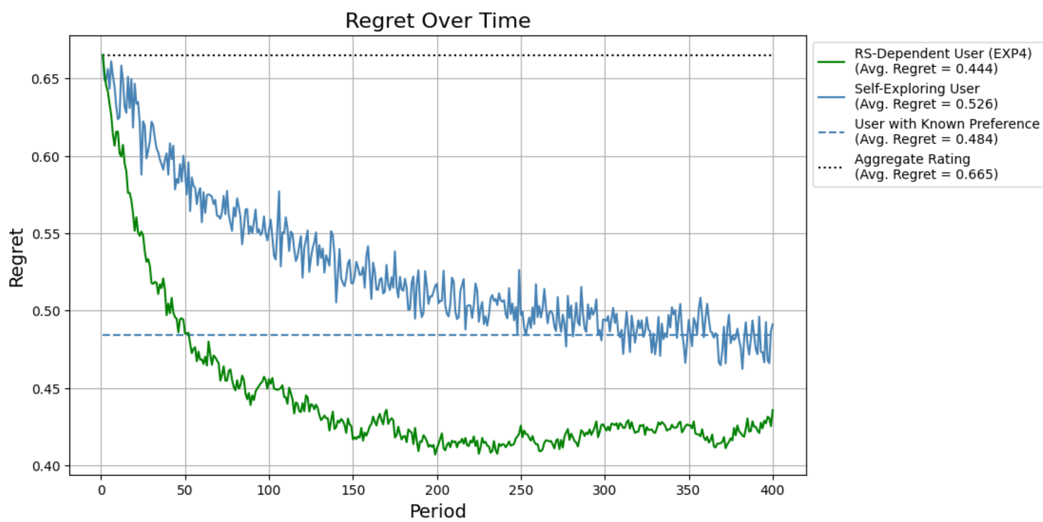
To compare these algorithms, we replicate the main regret analysis using the same MF representation but different Bayesian action rules. In the reported exercises, we set $\alpha = 0.01$ for Bayes-UCB and $\eta = 100$ for Bayes-EXP4. The comparison therefore holds fixed the RS information structure and varies only the rule used to convert posterior beliefs into recommendations.

Results Figure A13 summarizes the regret trajectories for the three alternative MF-based RS algorithms. Across all panels, Bayes-UCB, Bayes-EXP4, and Bayes-TS continue to outperform the self-exploring user benchmark and the Aggregate Rating baseline. Among the three alternatives, Bayes-TS performs best, with average regret of about 0.433, followed by Bayes-UCB at about 0.440 and Bayes-EXP4 at about 0.444.

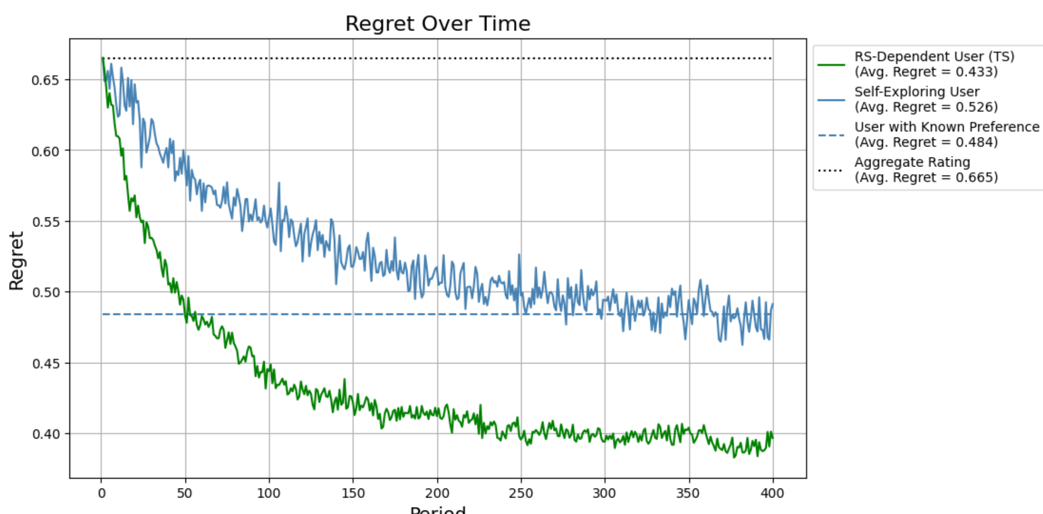
The differences across RS algorithms are modest. The main-text mean-greedy RS still performs slightly better, with average regret of about 0.431, but Bayes-TS comes very close. This suggests that the platform's advantage is not driven by a particular exploration rule. Rather, the key source of the RS advantage is the MF-based information structure: the platform can learn user preferences in a low-dimensional latent space constructed from cross-user data.



(a) MF + Bayes-UCB.



(b) MF + Bayes-EXP4.



(c) MF + Bayes-TS.

Figure A13: Regret performance under alternative MF-based RS algorithms.

H Objective Misalignment: RS vs. User

In the baseline model, the recommender system (RS) is assumed to maximize the same objective as users, namely expected utility or expected rating. In practice, however, platform objectives may differ from user welfare. For example, a platform may place weight on engagement, advertising exposure, retention, or other business-side outcomes that do not perfectly align with the user's utility. To study the implications of such misalignment for recommendation quality and user learning, we introduce reduced-form objective noise into the RS decision rule and compare the resulting outcomes with the aligned benchmark.

To keep notation consistent with the main framework, let F_j denote the RS-space item factor for movie j, and let $\gamma_{i,t}$ denote the RS posterior mean estimate of user i's latent weight in the MF space at period t. Under objective alignment, the RS recommends items according to the predicted utility

$$\gamma_{i,t}^{\top} F_j.$$

To introduce objective misalignment, we perturb this score with an item-period-specific noise term:

$$\text{score}_{i,j,t} = \gamma_{i,t}^{\top} F_j + \xi_{j,t}, \qquad \xi_{j,t} \sim \mathcal{N}\big(0, \sigma_{\text{obj}}^2\big).$$

The RS then recommends the item with the highest perturbed score:

$$j_{i,t}^{\text{RS}} = \arg\max_{j \in \mathcal{A}} \text{score}_{i,j,t}.$$

When $\sigma_{\text{obj}} = 0$, this specification reduces to the aligned benchmark in which the RS objective is perfectly aligned with user utility. As σ_{obj} increases, the RS recommendation rule becomes noisier relative to the user's true objective. This provides a simple reduced-form way to capture objective misalignment without changing the rest of the learning environment.

In the simulation, we set

$$\sigma_{\text{obj}} = 0.1.$$

This value introduces a moderate degree of misalignment while preserving the same MF representation and Bayesian updating structure as in the baseline model. The exercise therefore isolates the effect of changing the RS objective from the effect of changing the RS information structure.

Figure A14 shows that objective misalignment reduces the value of the RS along two margins. First, it weakens direct recommendation performance: the RS average regret rises to about 0.457, compared with about 0.431 in the aligned benchmark, although it still outperforms the Self-Exploring User, whose average regret is about 0.526. Second, it makes RS-generated histories less useful for user learning: the RS-Dependent User Counterfactual has average regret of about 0.612, compared with about 0.553 in the aligned benchmark, and performs worse than the Self-Exploring User. Thus, when recommendations are distorted by a noisy platform-side objective, the exposure path becomes less informative about the user's latent tastes. Objective misalignment therefore lowers both the immediate quality of recommendations and the informational value of the recommendation history, even though the RS retains some advantage in direct recommendation performance.

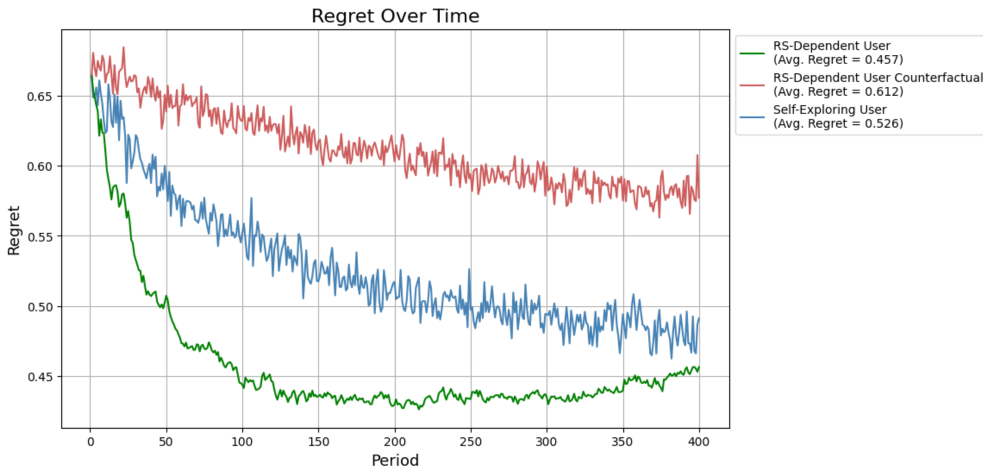


Figure A14: Regret and counterfactual regret under RS objective misalignment.

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